



**COMSATS University Islamabad,
Sahiwal Campus, Pakistan**

Health Hub

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COMSATS University Islamabad, Sahiwal Campus**

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of the requirement for the degree of**

Bachelor of Science in Computer Science (2022-2026)

By

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Executive Summary

The rapid digitization of healthcare highlights a gap between complex medical data and patient comprehension. Generic health apps often ignore local dietary habits and clinical safety. To address this, Health Hub is an advanced, AI-powered clinical nutrition ecosystem designed to automate personalized, medical-grade health management.

Unlike standard platforms, Health Hub relies on a robust hybrid AI architecture. It integrates a dedicated multi-model predictive framework, featuring a Native XGBoost Booster model for Iron Deficiency Anemia screening, a clinical hybrid WHO-guideline/XGBoost model for Vitamin D Deficiency assessment, and an engineered Scikit-Learn pipeline using 19 metabolic feature interactions for Type 2 Diabetes Risk estimation, alongside a multi-LLM orchestration layer (utilizing Llama-3.3 and Qwen via the Groq API) for generative meal planning. Engineered specifically for the Pakistani healthcare landscape, it guarantees Halal-compliant, culturally relevant recipes while strictly filtering allergens and disease-specific contraindications (such as high-sodium for Hypertension).

The platform features a Clinical Engine that interprets critical lab biomarkers (like HbA1c and LDL) to categorize health risks and provide actionable insights. The AI Diet Planner dynamically calculates BMI and activity-based caloric goals, ensuring clinical safety by blocking unsafe health objectives.

Health Hub's technical architecture includes a highly responsive React 18 frontend, a high-performance FastAPI (Python) backend, and a cloud-native, secure NoSQL data orchestration layer using Firebase Firestore and Firebase Authentication. Crucially, the system features a Nutritionist Portal with a "Human-in-the-Loop" workflow, allowing verified professionals to review, edit, and approve AI-generated plans.

By fusing custom Machine Learning, Generative AI, and strict clinical safety protocols, Health Hub delivers a secure, scalable, and highly accurate solution empowering users to take informed control of their well-being.

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Arooj Ahmad

Saira Bashir

Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
BMI	Body Mass Index
BMR	Basal Metabolic Rate
CORS	Cross-Origin Resource Sharing
CRUD	Create, Read, Update, Delete
FR	Functional Requirements
JSON	JavaScript Object Notation
JWT	JSON Web Token (Used implicitly via Firebase Authentication for session verification)
LLM	Large Language Model
ML	Machine Learning
XGB	Extreme Gradient Boosting (Used for Anemia and Vitamin D Native Booster Models)
NFR	Non-Functional Requirements
NoSQL	Not Only SQL (Document Database architecture used for Firebase Firestore)
UBJ	Universal Binary JSON (Serialization format for Native XGBoost models)
RBAC	Role-Based Access Control
REST	Representational State Transfer
SDK	Software Development Kit
SDLC	Software Development Life Cycle
SPA	Single Page Application
PKL	Pickle Format (Serialization format used for Scikit-Learn Diabetes pipeline)

TDEE	Total Daily Energy Expenditure
UI	User Interface
UID	Unique Identifier (Firebase Authentication User ID used across NoSQL Collections)
UX	User Experience
Vite	Next Generation Frontend Tooling

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Chapter 1

INTRODUCTION

1. Introduction

Healthcare in Pakistan faces long-standing challenges related to accessibility, coordination, and the intelligent handling of patient information. A significant portion of healthcare facilities, especially in smaller cities like Sahiwal and rural areas, continue to rely on manual record-keeping, handwritten prescriptions, and fragmented lab reporting. As a result, patient histories, laboratory reports, and nutritional data are often scattered or misplaced. These gaps create delays in diagnosis and make it difficult for patients to maintain a consistent health routine. With a rapidly growing population and overburdened public hospitals, the need for modern, AI-driven healthcare solutions in Pakistan has become increasingly urgent.

In recent years, Pakistan has seen a growing shift toward digital services in banking and e-commerce. However, the healthcare sector still lacks a unified "intelligent" ecosystem. While some private hospitals use electronic records, they remain isolated. The absence of centralized, smart health records means that when a patient visits different specialists, their data does not follow them. This leads to repeated lab tests and inconsistent dietary advice. A cloud-based, Machine Learning-integrated system like Health Hub has the potential to address these issues by not only storing information centrally but also providing automated analysis of that data to help users make better health decisions.

Health Hub is developed as a modern, AI and Machine Learning-powered solution to these challenges within the context of Pakistan's healthcare environment. It is a comprehensive digital health and nutrition management system designed to integrate medical data analysis and personalized diet planning into a single platform. The purpose of the system is to digitalize workflows, reduce the administrative burden on nutritionists, and make expert health insights accessible to people across Pakistan. Whether a patient is in a large metropolitan city or a smaller district, Health Hub enables them to analyze lab reports, maintain medical histories, and consult with professionals without being limited by physical distance.

The system consists of a comprehensive web application for users and nutritionists, supported by a robust backend infrastructure. It is built using modern technologies such as React 18 and Vite for a fast and responsive frontend, and FastAPI (Python 3.13) for high-performance backend orchestration. Unlike traditional systems, Health Hub implements a dedicated multi-model predictive framework executing local inference incorporating Native XGBoost Boosters in Universal Binary JSON format for Iron Deficiency Anemia and Vitamin D Deficiency tracks alongside a custom Scikit-Learn pipeline for Type 2 Diabetes Risk assessment and utilizes advanced Large Language Models (Llama

3.3 via Groq API) to ensure that AI-generated advice is strictly grounded in clinical guidelines and nutritional facts. Cloud-native NoSQL architectures like Firebase Firestore handle structured clinical records and real-time data streaming, while Firebase Authentication ensures granular session safety and highly secure client validation.

Health Hub enables patients in Pakistan to create digital accounts and manage their entire health journey. Users can log their weight, dynamically calculate their BMI and BMR, and use the Generative AI Diet Engine to generate calorie-precise meal plans that are specifically tailored to Pakistani cultural tastes and strict Halal standards. Since medical reports can be difficult to read, the platform's Clinical Engine allows users to upload their blood reports directly. The integrated script extracts data points seamlessly and interprets critical clinical biomarkers such as HbA1c, fasting blood glucose, and hemoglobin, providing immediate, easy-to-understand actionable insights on potential metabolic and physiological health risks based on adjusted decision thresholds.

Nutritionists and doctors in Pakistan benefit from a dedicated professional dashboard. They can review patient histories, track weekly calorie adherence and progress through interactive visual graphs (via Recharts), and seamlessly manage consultation appointments. A critical feature of this system is its "Human-in-the-Loop" approval workflow, which empowers verified nutritionists to review, edit, and approve AI-generated diet plans before clients adopt them. The system's underlying safety tier logic also acts as a second pair of eyes, proactively blocking allergens, filtering disease-specific contraindications (like high-sodium for hypertension), and flagging dangerous diet goals to ensure all recommendations are medically sound.

Overall, Health Hub aims to modernize Pakistan's healthcare landscape by introducing an intelligent, cloud-based environment. The system reduces dependence on manual record-keeping, minimizes the risk of lost medical data, and provides a platform that supports remote health monitoring and secure professional consultation. By addressing the unique challenges of data fragmentation and medical literacy in Pakistan, Health Hub represents a practical, scalable, and highly accurate AI solution that has the potential to enhance healthcare quality and transparency across the country.

1.1 Brief

The Health Hub project is designed as an intelligent, AI and Machine Learning-powered health and nutrition management ecosystem that bridges the gap between complex medical data and personalized patient care. The primary aim of the project is to overcome the critical limitations of traditional healthcare systems, where medical records, laboratory reports, and dietary advice are often

fragmented, manually handled, or provided without clinical context.

Unlike conventional platforms, Health Hub consolidates essential health services into a unified digital environment. The core of the system is built upon three native machine learning diagnostic components coupled with optimal decision threshold tuning, alongside a multi-LLM orchestration layer (utilizing Llama 3.3 via Groq API) to ensure that every recommendation from calorie-precise diet plans to automated lab analysis is accurate, secure, and grounded in clinical guidelines.

By integrating automated processing of clinical biomarkers (such as HbA1c, LDL, and serum glucose levels) and a "Human-in-the-Loop" approval workflow for nutritionists, the system ensures easy accessibility and efficient workflows for both patients and healthcare professionals. Health Hub utilizes modern cloud technologies and an asynchronous NoSQL document store architecture (Firestore/Firebase/FastAPI) to provide a reliable, scalable, and user-friendly experience, specifically tailored to meet the cultural standards (Halal compliance) and medical requirements of the Pakistani healthcare landscape.

1.1.1 Purpose of the Project

The primary purpose of Health Hub is to centralize and democratize healthcare intelligence by offering an all-in-one AI and Machine Learning-powered solution for personalized nutrition, medical data analysis, and professional consultation. The project aims to simplify the interaction between patients and healthcare providers by removing communication gaps and eliminating the dependency on manual, fragmented medical records.

Beyond simple record-keeping, Health Hub seeks to empower users by providing an automated Clinical Engine that parses complex lab results through regular expressions and custom predictive models such as a dual-path hybrid track for Vitamin D assessment and a mathematically engineered 19-feature matrix for Type 2 Diabetes tracking which forms the core of medically safe and culturally relevant (Halal) meal plans. The system ensures that users have direct, secure access to their longitudinal health history and progress logs, enabling them to share this data seamlessly with authorized nutritionists. Ultimately, the project is designed to enhance medical literacy and promote proactive health management within the Pakistani healthcare landscape by combining machine intelligence with professional human oversight.

1.1.2 Scope of the System

The scope of Health Hub extends across an integrated digital environment that includes a comprehensive web portal for users and nutritionists, a specialized ML-orchestration backend

(FastAPI), and an administrative layer for system-wide data management. The project handles a diverse range of core functionalities designed to automate the health journey, including:

- **Machine Learning Predictive Screening:** Utilizing custom-trained native booster and specialized pipelines (.ubj and .pkl formats) to accurately assess clinical risks for Iron Deficiency Anemia, Vitamin D Deficiency, and Type 2 Diabetes based on precise feature maps.
- **Intelligent Diet Planning:** Automated generation of calorie-precise and culturally relevant (Pakistani Halal) meal plans using Llama 3.3, based on real-time BMI and TDEE analysis.
- **Automated Lab Report PDF Parsing:** Streamlining data collection by extracting text from unstructured lab reports using an automated parser (pdf plumber + Regex) to instantly pre-fill diagnostic parameters.
- **Professional Consultation & Approval:** A centralized platform for nutritionist discovery, appointment scheduling, and a "Human-in-the-Loop" workflow where experts review and approve AI-generated diet plans.
- **Health Risk Assessment:** A built-in safety engine that cross-references user goals with medical conditions to provide critical warnings and block risky dietary choices (e.g., allergen filtering).
- **Longitudinal Progress Tracking:** Visualizing weight trends, calorie adherence, and health biomarkers through document-driven interactive dashboards using Recharts.

1.1.3 Technologies Used

The Health Hub system is developed using a modern and high-performance technology stack designed to handle complex ML computations and real-time data synchronization. The frontend is built using React 18 combined with Vite, providing a modular and lightning-fast user interface. For the styling and UI components, Tailwind CSS v4 and Lucide-React are utilized to ensure a responsive and professional medical-themed design.

The backend is powered by FastAPI (Python 3.13), which offers asynchronous and high-performance server capabilities to manage the local ML Orchestration Layer. This layer handles the ingestion and execution of standalone, serialized predictive models (anemia_xgb.ubj, vitd_xgb.ubj, and diabetes_pipeline_v2.pkl) and Large Language Models (Llama 3.3 via Groq API). For data persistence,

a robust, cloud-native NoSQL schema is implemented using Firebase Firestore for structured health profiles, messaging feeds, and biomarker metadata, while secure, decoupled user authentication and role validation are executed entirely via Firebase Authentication.

1.1.4 System Overview

Health Hub operates as a sophisticated, interconnected ecosystem that supports healthcare users and professionals through a multi-layered intelligent platform. At the user level, patients can create personalized accounts to manage their complete health journey, which includes dynamically calculating BMI/BMR, generating AI-driven diet plans, analyzing lab results through the clinical engine, and tracking their physical progress through visual logs. The system empowers users to take a proactive role in their health by providing easy-to-understand insights derived from complex clinical data.

Nutritionists and health professionals access a structured, professional dashboard designed to streamline patient management. They can review detailed patient profiles, approve or edit AI-generated diet plans, track their clients' weekly calorie adherence, and manage consultation appointments. This centralized view ensures that nutritionists have all the necessary data from ML-analyzed lab markers to historical trends to provide high-quality, evidence-based advice.

The administrative and orchestration layer supervises the entire system's integrity. This includes managing system-wide metrics, ensuring the security of sensitive medical data through role-based document access controls, and monitoring the Safety Engine that validates AI recommendations. By integrating automated ML diagnostics with professional human oversight, Health Hub ensures a seamless, secure, and highly accurate healthcare experience for all stakeholders.

1.1.5 Outcome of the Project

The project results in a complete, intelligent healthcare ecosystem that significantly improves the quality, accuracy, and efficiency of patient care in Pakistan. By centralizing disparate health data, Health Hub successfully saves time for both patients and nutritionists, reduces manual errors in dietary and clinical calculations, and prevents the loss of vital medical records through secure cloud storage and robust document collections.

One of the primary outcomes is the transformation of complex clinical biomarkers into actionable insights; the Clinical Engine parses PDF text dynamically to extract parameters instantly, while the native ML estimators evaluate specialized risk profiles to provide recommendations that are not only calorie-precise but also culturally and medically appropriate. The system ensures a secure, digital

environment where healthcare information flows through an organized pipeline, enabling nutritionists to make better-informed decisions based on longitudinal data. Ultimately, the outcome of Health Hub is a trusted digital platform that empowers users to manage their well-being and access professional medical services without geographical barriers.

1.2 Relevance to Course Modules

The Health Hub project is strongly connected with the core subjects taught throughout the Bachelor of Science in Computer Science program. Every stage of the system, from advanced AI orchestration to secure medical data handling, reflects concepts learned during the degree. The project demonstrates how theoretical knowledge from various semesters can be applied to develop an intelligent healthcare ecosystem that includes an AI-integrated web application and a professional nutritionist portal. The following subheadings explain the contribution of major course modules to the project.

1.2.1 Software Engineering

Software Engineering played a fundamental role in shaping the modular architecture of Health Hub. Concepts such as the Software Development Life Cycle (SDLC), requirement analysis, and system modeling were applied directly. The project followed an Agile-like approach to handle the iterative development of the multi-model predictive screening pipeline and backend parsing services. Structured practices, including architectural design diagrams and Use Case modeling, ensured that the functional and non-functional requirements such as the "Human-in-the-Loop" approval workflow for nutritionists were met efficiently.

1.2.2 Database Systems

Database Systems provided the knowledge required to design a scalable and secure cloud-native storage layer. Health Hub utilizes Firebase Firestore NoSQL document database collections for structured clinical records, ensuring real-time capabilities and schema flexibility for sensitive health logs. Concepts such as composite indexing (defined via `firestore.indexes`), hierarchical sub-collections, data integrity constraints via document design, and read/write optimization were vital in managing nested data records for diagnostic insights, lab markers, and user profiles, ensuring that clinical data is retrieved with minimal latency.

1.2.3 Artificial Intelligence and Machine Learning

This course was central to the development of the Health Hub intelligence engine. Beyond Large Language Models, the project implements three distinct clinical screening models executing local backend inference: a Native XGBoost Booster (`anemia_xgb.ubj`) for Iron Deficiency Anemia, a dual-

path clinical guideline and XGBoost model hybrid for Vitamin D Deficiency, and a Scikit-Learn pipeline (`diabetes_pipeline_v2.pkl`) leveraging 19 mathematically engineered feature interactions for Type 2 Diabetes Risk estimation. Theoretical understanding of supervised learning, split imputation via `joblib`, hyperparameter optimization, and custom decision threshold tuning allowed for the creation of a system that provides medically grounded and contextually accurate health advice, moving beyond static rule-based systems.

1.2.4 Web Engineering

The development of the Health Hub web portal is supported by the principles learned in Web Engineering. The use of React 18, component-based architecture, and modern build tools like Vite align with professional standards. The project applies concepts of responsive design using Tailwind CSS v4 to create a seamless experience across various devices. Furthermore, the integration of FastAPI (Python 3.13) as a high-performance, asynchronous web framework demonstrates advanced server-side engineering concepts to execute predictive scoring and handle file upload streams seamlessly.

1.2.5 Object-Oriented Programming (OOP)

OOP contributed significantly to the maintainability and scalability of the backend and frontend codebase. Concepts such as abstraction and encapsulation were used to build reusable service layers and structural Pydantic data schemas (e.g., `AnemiaInput`, `VitDInput`, `DiabetesInput`) to handle and validate client API requests. This modular approach allowed for clean business logic, making it easier to parse multi-model feature vectors and apply fallback safety protocols without compromising the system's stability.

1.2.6 Information Security

Since Health Hub handles highly sensitive medical information, Information Security principles were paramount. Concepts of Authentication and Authorization were implemented completely using Firebase Authentication. Secure server-side validation using Firestore Security Rules (`firestore.rules`), protection of document-level endpoints through strict cryptographic parameters, and role-based access control (RBAC) to distinguish between Patients, Nutritionists, and Admins reflect the core security countermeasures learned during the course.

1.2.7 Data Structures and Algorithms

Efficient data handling within the system relies on the principles of Data Structures and Algorithms. Features such as the string pattern mapping dictionary (`LAB_PATTERNS`) combined with Regular

Expressions (Regex) for text extraction inside the pdfplumber engine, alongside matrix transformations for native booster processing (xgb.DMatrix), utilize optimized algorithms. These topics ensured that the system remains performant even when calculating complex non-linear transformation arrays and mapping multi-variable metabolic risk parameters with minimal computing overhead.

1.2.8 Human-Computer Interaction (HCI)

HCI principles guided the design of the Health Hub dashboard. The focus was on creating a "User-Centric" design that simplifies complex medical and ML data. Principles of visual hierarchy, consistency, and cognitive load reduction were applied to the interactive Analytics charts (via Recharts) and Lab Result cards, ensuring that users can understand their health status at a glance without being overwhelmed by technical jargon.

1.2.9 Computer Networks

The cloud-based nature of Health Hub relies on server-client communication and API integrations, which were studied in Computer Networks. Understanding HTTP/HTTPS protocols, request-response cycles, asynchronous fetching, and network latency was essential for integrating the React frontend with the FastAPI backend endpoints (such as /predict/anemia and /parse-pdf) through secure Cross-Origin Resource Sharing (CORS) configurations, ensuring a reliable flow of information across the distributed network.

1.2.10 Final Year Project I and II

The FYP courses provided the structured framework needed to plan, document, and implement Health Hub. These courses ensured that the project was developed systematically through various phases from the initial proposal and ML training pipeline to the design review and final production-ready implementation. The skills gained here shaped the professional discipline and organization required to complete such a technologically advanced healthcare solution for graduation in June 2026.

1.3 Project Background

The idea behind the Health Hub project originates from the increasing need for a centralized, intelligent, and reliable digital system in the modern healthcare sector. Traditional healthcare practices in Pakistan rely heavily on manual processes, paper-based records, and disconnected communication channels between patients and nutritionists. In many local clinics, patient files are stored physically, medical histories are often incomplete, and complex lab reports are difficult for

patients to interpret. This creates significant delays, increases the chances of human error, and makes it difficult for both patients and health professionals to access accurate, data-driven insights when needed.

With the rapid growth of Machine Learning (ML) and the widespread use of cloud-based applications, healthcare environments now require systems that do more than just store data; they require systems that can "predict" and "analyze" it. The concept of Health Hub focuses on connecting patients and nutritionists through a unified, intelligent digital ecosystem. This idea aims to eliminate the gap between diagnostic data and actionable health outcomes by moving beyond static databases to a dynamic ML-driven Orchestration Layer. By implementing standalone native machine learning boosters and specialized serialized pipelines, the backend can now analyze distinct physiological and physical feature maps to accurately predict multi-condition clinical disease risk probabilities for a user.

The project idea further developed by analyzing the challenges faced by patients in managing their longitudinal medical information. Many individuals struggle to keep track of previous lab reports, weight trends, and nutritional advice. Nutritionists also face difficulty in making accurate dietary recommendations when a patient's medical history is incomplete or when lab markers are not clearly understood. By creating an automated system with a cloud-native Firebase Firestore NoSQL document architecture, Health Hub ensures that medical records are not only organized and secure but also intelligently analyzed. The system allows nutritionists to view patient progress instantly through interactive Recharts analytics and enables patients to extract and interpret their lab results through an integrated PDF parsing engine.

The idea was also influenced by the need for safer, medically grounded nutrition management. In many traditional settings, diet plans are generic and do not account for specific medical conditions like Diabetes or Hypertension. Health Hub addresses this by integrating Llama 3.3 and Qwen models via the Groq API, using strict prompt engineering. This ensures that every recommendation is cross-referenced with clinical guidelines and the user's specific health profile. Furthermore, the introduction of a "Human-in-the-Loop" approval workflow ensures that no AI-generated plan reaches the patient without the professional validation of a nutritionist, adding a critical layer of safety.

Another important factor behind the project idea is the growing demand for culturally relevant health solutions. Most international health platforms do not cater to the specific dietary habits and Halal standards of the Pakistani population. Health Hub addresses this by utilizing a specialized dataset of Pakistani Halal recipes and local nutritional sources. The backend, built on FastAPI, ensures that

these complex AI and ML computations are handled locally with high performance and low latency.

Overall, the idea for Health Hub is based on creating a modern, cloud-based healthcare ecosystem that addresses the limitations of traditional, fragmented systems. The project aims to bring convenience, security, and intelligence to the healthcare process by centralizing all essential features from Machine Learning diagnostics to personalized diet planning into one interconnected platform. It reflects how advanced AI and data science can support better medical care, improve professional communication, and enhance the patient experience through an organized, reliable, and scientifically sound digital system.

1.4 Literature Review

The reviewed literature shows that several healthcare applications offer features such as online appointments, doctor discovery, and basic medical record handling. However, each of these platforms also contains significant weaknesses, particularly in their lack of automated medical data analysis, predictive modeling, and clinical safety protocols. Most existing systems act as simple "directories" or "storage folders" without providing any intelligent insights into the data they store. They lack the ability to bridge the gap between diagnostic raw data and personalized nutritional outcomes.

To clearly highlight these gaps, the following table compares well-known healthcare applications with the proposed Health Hub system. Table 1.1 presents the application name, the major weakness identified through literature and practical usage, and the specific solution provided by Health Hub to overcome that weakness using modern AI and Machine Learning techniques. This comparison demonstrates why an ML-integrated, context-aware healthcare platform is necessary for modern medical management. The synthesis of this literature review indicates that while digital health is evolving, there is a profound need for systems that provide "actionable intelligence" rather than just data storage. Health Hub addresses the identified gaps by implementing an automated diagnostic and nutritional ecosystem that prioritizes clinical accuracy, Machine Learning-driven prediction, and cultural relevance. By moving from a static file storage model to a localized AI Orchestration Framework, Health Hub provides a trusted digital solution for complex healthcare needs.

Table 1.1: Competitive Analysis & Proposed Project Solutions

Application Name	Identified Weakness	Proposed Project Solution (Health Hub)
Zocdoc	Focuses mainly on appointment booking	Health Hub integrates

	and does not provide intelligent medical record analysis or longitudinal tracking.	appointments with an automated Clinical Engine that parses unstructured lab reports via backend Regex pattern algorithms to provide immediate actionable insights.
Practo	Provides generic diet and health tips that do not account for a user's specific clinical biomarkers or existing medical conditions.	Health Hub uses local, custom-trained native machine learning estimators (XGBoost Boosters and Scikit-Learn Pipelines) to screen multi-track physiological risks (Anemia, Vitamin D, Diabetes) based on explicit biomarkers.
Marham	Does not offer automated nutrition planning or calorie-precise meal generation tailored for the local Pakistani food context.	Health Hub includes a dynamic Generative AI Diet Engine (via Groq API) designed specifically for Pakistani Halal dietary standards and local ingredients.
MyFitnessPal	Primarily relies on manual tracking and lacks the ability to adjust goals based on actual medical lab reports (e.g., Blood Glucose).	Health Hub creates an automated data pipeline linking extracted PDF lab parameters directly to physiological risk prediction loops using a high-performance FastAPI Multi-Model Orchestration framework.
Manual / Paper Based Clinics	High chances of human error, misplaced reports, and no safety checks for dangerous or conflicting dietary goals.	Health Hub secures and automates all records within cloud-native Firebase Firestore document collections protected by granular security rules that strictly validate and flag clinical contraindications.

Basic Health Apps	Do not combine AI diagnostics, predictive ML planning, and a nutritionist approval workflow into one unified ecosystem.	Health Hub unifies Predictive Health Screening, Multi-LLM Meal Planning, and a professional Human-in-the-Loop Approval workflow within a single integrated platform.
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1.5 Analysis from Literature Review

The analysis of the reviewed literature demonstrates how the Health Hub project aligns with global trends in digital healthcare transformation and how it specifically addresses the existing gaps within Pakistan’s healthcare environment through the integration of Artificial Intelligence and Machine Learning. The literature emphasizes that modern healthcare systems must move beyond simple record-keeping and toward "Intelligent Decision Support Systems" that offer predictive modeling, secure medical storage, and automated clinical analysis. These findings strongly support the core features implemented in Health Hub, such as the localized machine learning risk screening modules and the automated Clinical Engine, validating the relevance of developing such a system in a region where medical literacy and digital health adoption are still emerging.

The comparison between existing research and the functionalities of Health Hub shows that the project is not only aligned with modern healthcare solutions but is specifically designed to overcome the limitations highlighted in previous studies such as the "hallucination" risks of general Large Language Models and the lack of cultural contextualization in international health apps. By grounding AI responses in verified clinical data and focusing on Pakistani Halal dietary standards, Health Hub effectively fills the technological void identified in current literature, providing a scalable and medically safe digital health ecosystem.

1.5.1 Alignment with Digital Healthcare Adoption Trends

The literature identifies a major shift from traditional, paper-based healthcare systems toward intelligent, cloud-based platforms worldwide. Research highlights that digital health solutions, specifically those integrating Machine Learning, significantly reduce medical errors, improve longitudinal access to records, and enhance the quality of communication between patients and providers. The Health Hub project aligns strongly with these global trends by offering more than just a centralized database; it provides an Intelligent Data Layer for analyzing medical histories and

nutritional needs.

This alignment shows that the project is built upon well-established research regarding the benefits of digital healthcare transformation, such as improved predictive accuracy and long-term data accessibility. By incorporating custom-trained native XGBoost boosters and engineered classification pipelines, Health Hub follows the same principles discussed in modern medical literature: reducing the dependency on manual interpretation, ensuring evidence-based advice, and providing secure, real-time access to health insights.

1.5.2 Comparison with Cloud-Based Healthcare Research

Studies on cloud-based healthcare solutions highlight critical advantages such as massive scalability, remote accessibility, and significantly reduced infrastructure costs. Recent literature also stresses the paramount importance of creating "Secure Cloud Environments" specifically for sensitive clinical and nutritional data. Health Hub reflects these research findings by utilizing a modern cloud-native architecture with a FastAPI backend and a Firebase Firestore NoSQL document store that supports a wide diversity of medical information ranging from high-resolution lab markers to dynamic health metrics.

The system allows nutritionists and users to access longitudinal patient histories, real-time progress logs, and AI-generated diet plans regardless of their physical location, which is entirely consistent with global research on remote healthcare access and telemedicine. The analysis further shows that Health Hub not only adopts the fundamental strengths of cloud computing discussed in the literature but also applies them to a practical system that handles complex local ML-inference orchestration natively within the backend deployment.

1.5.3 Relevance to Existing Appointment and Patient Management Systems

Research on platforms like Zocdoc and Marham emphasizes the importance of simplifying doctor discovery and appointment booking. These systems mainly focus on scheduling processes and reducing waiting times. However, the literature also mentions significant limitations, such as the lack of deep medical record integration and automated health data interpretation.

Health Hub extends far beyond the functionality of these existing systems by integrating detailed patient histories, ML-predicted diet types, and automated lab biomarkers within the same platform. Unlike many systems mentioned in the literature, Health Hub also introduces a "Human-in-the-Loop" approval workflow, allowing nutritionists to review and approve AI-generated diet plans. This

comparison indicates that Health Hub significantly expands upon existing capabilities by offering a more holistic, secure, and data-intelligent healthcare experience.

1.5.4 Integration of Electronic Medical Records (EMR)

Studies show that electronic medical records significantly improve diagnosis accuracy and ensure reliable, long-term storage of sensitive patient data. The literature also identifies recurring issues with traditional handwritten prescriptions and manual lab reports, such as misinterpretation and loss of historical data. Health Hub directly addresses these concerns through a sophisticated NoSQL document storage architecture and standardized interpretation of medical markers.

The project incorporates structured health histories, BMR/TDEE calculations, and easy multi-device accessibility. The analysis highlights that Health Hub builds upon established EMR research but adds an additional layer of Intelligent Processing through its Clinical Engine, which is tailored specifically to bridge the medical literacy gap in Pakistan's healthcare context. By transforming raw data into structured digital insights, the system ensures a higher standard of care and data reliability.

1.5.5 Application of Security and Privacy Standards

The literature strongly focuses on security risks and the need for strict authentication and authorization. Health Hub implements security mechanisms that match these global recommendations, such as Role-Based Access Control (RBAC), secure Firebase Authentication, and document-level security rules (Firestore.rules). The analysis shows that the project is developed in line with recognized healthcare security guidelines, ensuring that the system remains trustworthy and reliable while protecting the integrity of sensitive health insights.

1.5.6 Involvement of Mobile Health Concepts

Research on mobile health applications suggests that mobile platforms significantly increase user engagement and support better real-time communication. Health Hub incorporates these findings by offering a highly responsive web interface (React 18 + Vite) that allows users to track their progress on the go. Features such as real-time calorie adherence tracking and mobile-friendly nutritionist consultation reflect the benefits discussed in mobile health literature.

1.5.7 Addressing Limitations of Existing Healthcare Technologies

The literature identifies several shortcomings in current digital healthcare solutions, such as decentralized records, "hallucinating" AI models, and poor integration between diagnostic data and

nutritional advice. Health Hub is designed specifically to overcome these issues by centralizing data through a high-performance FastAPI Orchestration framework, which ensures that AI advice is grounded in verified clinical facts.

Unlike traditional systems, Health Hub integrates automated ML-based multi-track disease risk screening with personalized generative planning, ensuring secure record sharing and providing a consistent, intelligent interface. The analysis confirms that the project aligns with international healthcare standards while offering features tailored to the unique challenges faced in Pakistan, specifically regarding Halal dietary standards and clinical safety tiers.

1.6 Methodology and Software Lifecycle

The development of the Health Hub project follows a structured approach that combines modern software engineering practices with a suitable Software Development Life Cycle (SDLC) model. Considering the complexity of the system which integrates a multi-track Machine Learning risk screening pipeline, a high-performance FastAPI backend, and a professional nutritionist portal, a specialized methodology that supports iterative development and continuous validation was deemed essential. This approach ensures that all components, from automated clinical biomarker extraction via backend regular expressions to personalized machine learning assessments, are rigorously tested and effectively integrated into a cohesive digital healthcare ecosystem.

1.6.1 Methodology

The Health Hub project adopts the Agile methodology, which emphasizes iterative development, incremental delivery, and close collaboration. Agile is ideal for AI and ML-integrated healthcare applications where requirements for model accuracy, clinical safety protocols, and prompt engineering may evolve based on feedback from health professionals or advancements in Large Language Models (LLMs).

1.6.2 Agile Practices Applied

In Health Hub, Agile practices are implemented through short development cycles called Sprints. Each sprint focuses on a specific set of high-priority modules, such as the serialized XGBoost Booster training loops, the rule-based Clinical Engine for biomarker interpretation, or the secure Firebase Authentication flow. Iterative development allows the team to implement, test, and refine Machine Learning parameters and AI safety guardrails progressively. Feedback from supervisors and health experts is integrated after each sprint, ensuring that the system's predictive suggestions remain

medically sound and culturally relevant.

1.6.3 Software Development Life Cycle (SDLC) Model

The project adopts the Scrum model of Agile SDLC due to its suitability for multi-component software development involving data science and web engineering. Scrum divides the project into time-boxed sprints (typically 2 weeks), during which specific technical tasks like cloud-native Firebase NoSQL collection design, local backend model inference integration, or UI/UX enhancements for the nutritionist dashboard are planned, developed, and reviewed.

1.6.4 Product Backlog Management

Scrum methodology relies on a prioritized product backlog. In Health Hub, the backlog includes critical technical modules such as:

- **Machine Learning Training:** Serving and managing the specific disease screening and biochemical risk models (Anemia, Vitamin D, and Diabetes tracking pathways) natively inside the FastAPI local directory.
- **Clinical Engine Logic:** Rule-based interpretation of raw parsed lab biomarkers (HbA1c, LDL, Glucose, etc.) into contextualized risk tags.
- **Generative AI Pipeline:** Orchestrating Llama 3.3 and Qwen via Groq API to produce structured, calorie-precise meal plan arrays.
- **Role-Based Access Control (RBAC):** Restricting data permissions for Patients, Nutritionists, and Admins utilizing Firebase backend authentication schemas.
- **Data Visualization:** Implementing continuous calorie adherence tracking charts via Recharts tied to NoSQL user datasets. The backlog is updated continuously to prioritize clinical safety and the accuracy of the ML-driven recommendations.

1.6.5 Incremental Delivery

Health Hub is developed in stages to manage complexity. The first sprint focused on the core infrastructure (FastAPI environment configuration and cloud-native Firebase Firestore mapping), while subsequent sprints implemented the local Machine Learning validation pipelines and the Generative Diet Engine. This modular delivery allowed for each AI component to be evaluated for "hallucinations" or logical errors before moving to the integration phase.

1.6.6 Rationale behind Selected Methodology

Agile and Scrum were chosen for Health Hub because of their adaptability to AI-driven systems. Unlike traditional software, Machine Learning models require frequent hyperparameter tuning and data validation. Agile provides the flexibility to refine these ML interactions based on testing results, while Scrum ensures structured progress tracking for the overall full-stack architecture.

1.6.7 Development Tools and Environment

The system is developed using React 18 and Vite for the frontend and FastAPI (Python 3.13) for the backend microservices. The database layer implements a document-driven environment served entirely via Firebase Firestore for multi-device real-time sync, logs, and user metadata. The development environment also utilizes Git for version control, Postman for API endpoint optimization, and local Python scripts for managing serialized model dependencies natively inside the project repository.

1.6.8 Testing and Quality Assurance

Testing is integrated into every phase through a systemic validation module. Special attention is given to AI Safety Testing, ensuring that the system blocks risky health goals or allergens. Unit tests for BMI/BMR calculations and integration tests for local FastAPI microservice execution routes are performed iteratively. Quality assurance also includes "Prompt Validation" to ensure the Generative AI remains within medical ethical boundaries.

1.6.9 Advantages of the Selected Methodology and SDLC

The Agile-Scrum approach provides the flexibility needed to handle complex ML logic while ensuring early detection of errors. It promotes collaboration between technical development and healthcare safety standards, enhancing the overall reliability, scalability, and usability of the Health Hub ecosystem.

Chapter 2

PROBLEM DEFINITION

2. Problem Definition

This chapter discusses the precise problem to be solved by the Health Hub project. The main issue is the lack of a unified, intelligent, and predictive digital platform for managing health and nutrition data in Pakistan. Currently, patient information is scattered across multiple locations, and most health tracking is either manual or relies on generic applications that do not account for a user's specific medical conditions or clinical biomarkers. This fragmented system creates significant inefficiencies, increases the risk of medical errors, and hinders the ability of users to take proactive, data-driven control of their well-being.

Healthcare in Pakistan faces numerous challenges that affect both patients and nutritionists. The absence of a centralized digital ecosystem results in fragmented patient records, delayed interpretations of complex diagnostic reports, and significant gaps in coordination between users and health professionals. Many clinics and individuals continue to rely on manual logging or paper-based reports, which often lead to lost medical histories, repeated lab tests, and a lack of continuity in longitudinal dietary management. These challenges are particularly pronounced when users try to follow complex medical advice without understanding the underlying clinical markers in their lab reports.

In addition, the growing burden of lifestyle diseases like Type-2 Diabetes and Hypertension has intensified the need for an organized system that can handle nutritional data effectively. Patients frequently have to carry physical lab reports and struggle to explain their medical history to new nutritionists, increasing the risk of misinterpretation. Doctors and nutritionists struggle to access complete, longitudinal patient histories in a timely manner, which negatively impacts the accuracy of diet plans and treatment decisions. Without a centralized view, tracking a patient's weight trends and calorie adherence over time through manual methods becomes nearly impossible.

Furthermore, a critical problem is the "Clinical Literacy Gap." Most patients cannot interpret their own lab results (such as HbA1c, LDL Cholesterol, or Serum Creatinine levels), leading to a total dependency on professionals for even basic health understanding. Generic diet plans found online often contribute to health risks because they do not consider these clinical markers or the user's BMI-specific safety limits.

The most significant technological gap is the lack of Predictive Intelligence. Existing systems act as

static storage folders rather than active health assistants. There is no automated mechanism that can take raw clinical data and predict the most suitable medical diet while filtering out contraindications. The absence of an intelligent system prevents healthcare providers from analyzing patient data efficiently or implementing evidence-based nutritional practices tailored to the Pakistani Halal context. Health Hub is designed to solve this by implementing a Machine Learning-driven Clinical Engine that transforms fragmented raw data into actionable, safe, and culturally relevant health intelligence.

2.1 Problem Statement

The healthcare and nutrition system in Pakistan faces significant challenges due to fragmented patient records, the inability of users to interpret complex lab results, and the reliance on generic dietary advice that lacks clinical context and predictive accuracy. Currently, patients struggle to maintain and share their longitudinal medical history, leading to repeated laboratory tests, delayed health interventions, and a total lack of continuity in personalized care. Users often possess critical diagnostic data in the form of lab reports but lack the medical literacy to understand clinical biomarkers like HbA1c, Lipid profiles, or Creatinine levels, which are essential for proactive health management.

Nutritionists and health professionals face equal difficulties in accessing complete, visualized patient histories, managing consultation workflows efficiently, and generating medically safe diet plans that account for specific conditions such as Type-2 Diabetes or Hypertension. Furthermore, existing digital solutions in Pakistan are often static and do not cater to regional dietary habits or Halal standards, forcing users to follow culturally irrelevant nutrition plans that lack scientific validation.

The Health Hub project addresses these critical issues by providing an intelligent, cloud-based healthcare ecosystem. It replaces fragmented, manual processes with a high-performance FastAPI-driven architecture and cloud-native Firebase Firestore NoSQL document collections. The system moves beyond static tracking by implementing a multi-track native Machine Learning screening pipeline specifically deploying standalone XGBoost boosters for Anemia and Vitamin D screening alongside an engineered Scikit-Learn pipeline for Diabetes risk profiling to assess critical clinical conditions based on a user's unique biomarkers. By integrating an automated backend PDF report parser for seamless data ingestion and a Generative AI Diet Planner (orchestrating Llama 3.3 via Groq API), the system empowers users to interpret their health data and follow medically validated, Halal-compliant nutrition plans. Through a centralized platform that features a "Human-in-the-Loop" approval workflow, Health Hub aims to improve diagnostic efficiency, reduce dietary errors, and

deliver a reliable, AI-driven healthcare experience across Pakistan.

2.2 Deliverables and Development Requirements

The Health Hub project is designed to create an intelligent, AI and Machine Learning-powered digital health and nutrition management system. This section outlines the main deliverables of the project and the technical and functional requirements necessary for its successful development and deployment. Deliverables define the expected outputs of the project, while development requirements specify the tools, technologies, and system capabilities needed to achieve these outcomes.

2.2.1 Deliverables

The primary deliverables of the Health Hub project include:

2.2.1.1 Intelligent Web Portal for Users

A responsive web application built with React 18 and Vite that enables users to manage their complete health journey. This includes a personalized dashboard for BMI/BMR tracking, an AI Diet Planner for generating calorie-precise meal plans, and a Clinical Engine for interpreting diagnostic markers. The portal provides a seamless interface for users to visualize their health progress through interactive charts.

2.2.1.2 Professional Nutritionist Dashboard

A dedicated web interface for health professionals to manage patient consultations and diet validation. Nutritionists can view structured patient profiles, analyze multi-track machine learning health risk logs, track client calorie adherence, and manage appointments. The dashboard is designed to simplify complex clinical data and improve the accuracy of professional medical advice.

2.2.1.3 Machine Learning Prediction Engine

A core backend deliverable featuring localized, custom-trained predictive health tracks served natively through FastAPI. This engine hosts standalone, high-performance estimators specifically an XGBoost booster for Iron Deficiency Anemia screening, a hybrid WHO-guideline/XGBoost tracking setup for Vitamin D Deficiency, and an engineered Scikit-Learn model processing 19 metadata features for Type 2 Diabetes Risk classification serving as a reliable diagnostic foundation for the platform.

2.2.1.4 Clinical Biomarker Analysis System

An automated backend script that parses uploaded PDF lab reports rather than processing ungrounded textual structures. This system implements the pdfplumber core stream parser alongside structured Regular Expression (Regex) pattern mapping to extract key medical markers such as HbA1c, LDL, and Fasting Blood Glucose, providing immediate health risk assessments and insights based on clinical thresholds.

2.2.1.5 Human-in-the-Loop Approval Workflow

A specialized deliverable that bridges AI and human expertise. This module allows nutritionists to review, modify, and approve AI-generated diet plans before they are finalized, ensuring 100% clinical safety and professional oversight.

2.2.1.6 Secure Cloud NoSQL Infrastructure

A robust backend architecture built with FastAPI and a cloud-native Firebase Firestore NoSQL document store to manage clinical biomarkers, metadata logs, and messaging streams with absolute transactional safety. This includes the configuration of Firebase Authentication and secure server-side authorization patterns to ensure that sensitive user health records are strictly protected.

2.2.2 Development Requirements

The development requirements define the technical and infrastructural needs for building the Health Hub system, ensuring it can handle complex ML computations and secure real-time data orchestration.

2.2.2.1 Technical Requirements

The technical requirements define the specific tools and technologies selected for Health Hub. The system utilizes a high-performance Python-based backend and a component-based frontend.

- **Frontend Development:** React 18 with Vite for a high-performance web interface. Tailwind CSS v4 and Lucide-React for responsive, medical-themed styling and iconography.
- **Backend Development:** FastAPI (Python 3.13) for high-speed asynchronous server-side orchestration, ML model serving, and API endpoint management.
- **Database & Storage:** Cloud-native Firebase Firestore document store utilized for structuring

health data tracks, user profile metadata, and real-time consultation records.

- **Machine Learning & AI:** Natively hosted serialized models (.ubj booster files and Scikit-Learn .pkl structures) for localized backend health classification, and the Groq API (Llama 3.3 & Qwen) for generative diet planning workflows.
- **Authentication & Security:** Firebase Authentication for secure multi-role login infrastructure combined with strict database-level validation scopes (firestore.rules) for enclosing sensitive clinical markers.
- **Cloud Hosting:** Deployment of the frontend web application on Vercel and the Python backend on a high-performance asynchronous cloud environment (such as Render) to enable 24/7 accessibility across Pakistan.

2.2.2.2 Hardware Requirements

To ensure smooth development and deployment of the Health Hub ecosystem, the following hardware specifications are recommended:

- **Processor:** Intel Core i5 (10th Gen) or higher / Apple M1 or higher to handle concurrent ML local processing and async backend requests.
- **RAM:** Minimum 8GB (16GB recommended for running backend multi-model server environments).
- **Storage:** 256GB SSD for fast development workflows and asset rendering.
- **Internet:** High-speed stable connection (min 10 Mbps) for real-time document syncing, localized backend execution loops, and Groq Cloud AI API calls.

2.3 Current System

The healthcare and nutrition landscape in Pakistan is still largely dependent on traditional, manual, and fragmented processes that severely limit efficiency, accuracy, and patient empowerment. Although a few private hospitals and premium fitness centers have begun adopting basic digital tools for record-keeping, these solutions remain isolated and lack "predictive intelligence." There is currently no nationwide or centralized digital platform that combines longitudinal patient health records, automated clinical biomarker analysis, and Machine Learning-driven personalized diet

planning into a single integrated ecosystem.

Due to the absence of such a unified and intelligent platform, both patients and nutritionists face immense challenges in maintaining complete medical histories and ensuring continuity of care. Many clinics continue to rely on handwritten prescriptions, physical paper files, and manual entries for weight and calorie tracking, which are not only time-consuming but also prone to human error, loss, and duplication. Furthermore, even the existing digital systems in some private institutions are restricted to internal administrative use and do not offer any diagnostic assistance or predictive modeling, leaving patients unable to understand their own medical markers without professional intervention.

These gaps become even more evident when considering the Clinical Literacy of the general population. In the current system, a patient receives a laboratory report (e.g., Lipid Profile or HbA1c) but has no immediate way to interpret the results or understand how those results should affect their daily nutrition. Rural and semi-urban areas face an even greater disadvantage, where specialized nutritionists are scarce. Patients often travel long distances for consultations while carrying piles of physical documents, and health professionals have no reliable, centralized way of accessing previous medical records or trend-lines from other facilities.

The most critical failure of the current system is the lack of Data-to-Action conversion. Even if data is digitized, it is not "analyzed" to predict health outcomes. Current health apps in the market operate on static, hard-coded rules that fail to account for the complex interactions between multiple clinical markers. Overall, Pakistan's current healthcare system lacks the high-performance integration, automated machine intelligence, and digital accessibility required to support modern medical workflows, highlighting the urgent need for a comprehensive, AI-driven solution like Health Hub.

2.3.1 Manual Record-Keeping

A significant portion of healthcare and nutrition facilities in Pakistan including small diet clinics, private practices, and rural healthcare centers continue to rely heavily on paper-based record-keeping. Patient files, nutritional histories, physical lab reports, and BMI progress logs are maintained in physical folders. Nutritionists commonly write diet plans manually, and laboratories generate printed reports that are stored in physical files.

This manual system creates severe operational and clinical challenges. Paper records are highly vulnerable to human errors such as incomplete documentation, illegible handwriting on meal plans, and accidental misplacement of critical blood reports. This lack of reliable documentation increases

the risk of recommending medically unsafe diets. Health Hub addresses this by replacing paper with a cloud-native Firebase Firestore NoSQL document database architecture, ensuring that every clinical marker is digitally indexed, searchable, and safe from physical loss.

2.3.2 Fragmented Appointment Management

Appointment scheduling in the majority of Pakistani healthcare facilities remains inconsistent and manual. Patients typically book consultations through phone calls or walk-in registrations using handwritten logbooks. While some premium centers use standalone software, these systems lack integration with the patient's medical history or nutritional progress.

Common issues such as double booking and long waiting times persist. Nutritionists often experience bottlenecks because they cannot view a patient's health status before the meeting. Health Hub solves this by providing a centralized Appointment Management module that links schedules directly to the user's AI-analyzed health profile, allowing nutritionists to prepare for consultations in advance.

2.3.3 Limited Accessibility and Communication

In the existing system, the accessibility of nutritional history is restricted because data is stored in isolated physical files. This makes it nearly impossible for patients to seek second opinions or change dietitians without losing their history. Continuity of care is disrupted, leading to inappropriate nutritional recommendations based on incomplete verbal information.

Communication between professionals and patients is mostly through unorganized messaging apps that are not integrated with the patient's profile. Health Hub bridges this gap by providing a Real-time Consultation Portal and secure Role-Based Access Control (RBAC), allowing patients to share their longitudinal health trends and AI-analyzed reports with authorized experts instantly.

2.3.4 Security and Privacy Concerns

The current infrastructure does not adequately safeguard sensitive patient information. Physical files and unprotected standalone computers are vulnerable to unauthorized access and environmental damage. There is no mechanism to ensure that a patient's lab reports are only seen by authorized personnel.

Health Hub introduces robust data security through secure Firebase Authentication and server-side authorization patterns governed by custom Firestore Security Rules (firestore.rules). This ensures that personal health metrics and chronic disease histories are protected by modern encryption standards,

enhancing patient trust in the digital healthcare ecosystem.

2.3.5 Inadequate Integration of Digital Tools

Although some institutions use basic digital systems, these tools are typically "Data Silos" designed for a single office. There is no unified infrastructure to bridge the gap between diagnostic laboratories and nutritional consultants. A patient's report generated at one lab cannot be automatically shared with an AI analysis engine in another clinic.

Health Hub overcomes this through a Unified API Orchestration layer built on FastAPI. This allows for the smooth flow of information across clinical diagnostics and nutritional planning modules, ensuring that a user's diagnostic biomarkers (like HbA1c or LDL) automatically trigger adjustments in their multi-track machine learning risk assessments and adjusted dietary guidelines.

2.3.6 Regional Disparities

Regional disparities create a noticeable divide between major urban centers and rural areas in Pakistan. Rural patients face travel burdens to consult specialized nutritionists, and when they do, new providers have no access to their previous medical records.

Health Hub is a cloud-native platform that provides expert-level nutritional guidance regardless of geographical location. By deploying on a high-performance cloud infrastructure, Health Hub enables a patient in a remote village to receive the same Llama 3.3-driven diet planning and ML-based diagnostics as someone in a metropolitan city, effectively narrowing the urban-rural healthcare gap.

2.3.7 Summary of Limitations

In summary, Pakistan's current healthcare system is hindered by fragmented records, inconsistent data storage, and a total lack of Predictive Intelligence. Nutritionists lack the longitudinal data required for evidence-based decisions, and appointment management is plagued by poor coordination.

Health Hub addresses these collective limitations by providing a secure, centralized, and Machine Learning-driven ecosystem. By digitizing and integrating clinical markers with intelligent tools, Health Hub elevates the quality and reliability of nutritional services, replacing human error with data-driven accuracy and professional human oversight.

2.4 Summary

The first chapter of this report presents a detailed exploration of the challenges within Pakistan's current healthcare and nutritional landscape and establishes the necessity for a unified, intelligent, and predictive digital solution such as Health Hub. The chapter begins by defining the core problem: Pakistan lacks a centralized, ML-driven healthcare platform capable of integrating longitudinal patient records, automated clinical biomarker analysis, and personalized, context-aware diet planning. This fragmented environment creates systemic inefficiencies that negatively impact patients, nutritionists, and healthcare administrators alike.

The chapter highlights how the absence of digital and intelligent integration results in scattered patient histories, delayed interpretations of medical markers, and poor continuity of care. Many facilities still rely on handwritten meal plans and manual registers that operate in "Data Silos". As a result, critical health information becomes difficult to track, easy to misplace, and impossible to analyze for long-term health trends. These limitations are further intensified by the Clinical Literacy Gap, where patients struggle to understand their own diagnostic data, and the lack of culturally relevant (Halal) nutritional tools trained on local datasets.

The chapter concludes that these challenges manual processes, fragmented systems, and the total lack of Predictive Diagnostics collectively demonstrate the urgent need for a centralized, AI-integrated healthcare management system. Health Hub is presented as a comprehensive solution designed to digitize and harmonize healthcare operations through a high-performance FastAPI-driven architecture. By offering a secure cloud-native Firebase Firestore NoSQL document architecture, localized standalone Machine Learning risk screening tracks (Anemia, Vitamin D, and Type 2 Diabetes risk pathways), an automated backend PDF report parser, and a "Human-in-the-Loop" nutritionist approval workflow, Health Hub aims to enhance healthcare efficiency and ensure continuity of care across all regions. Through this intelligent ecosystem, the project strives to transform Pakistan's nutritional landscape from a fragmented system into a unified, modern, and data-driven digital framework.

Chapter 3

REQUIREMENT ANALYSIS

3. Requirement Analysis

The Requirement Analysis chapter provides a comprehensive and structured understanding of what the proposed system, Health Hub, must accomplish in order to function effectively as an intelligent healthcare and nutritional management platform. It serves as a foundation for translating user expectations into clear technical requirements by identifying system behavior, Machine Learning-driven interactions, and the specific tasks the system must perform such as automated clinical biomarker interpretation and context-aware diet classification.

This chapter outlines how the Health Hub system is expected to operate, detailing both functional and non-functional requirements while defining the roles and responsibilities of each actor involved, including Patients (Users), verified Nutritionists, and the AI-Orchestration Layer. Using the requirements as a reference, this chapter organizes the system's capabilities into well-defined sections that guide the database schema design, ML model integration, and the final validation processes.

Through this analysis, the development team gains a precise understanding of the system's objectives such as bridging the clinical literacy gap through AI and its constraints, ensuring that the final implementation aligns with the needs of all stakeholders. This structured approach ensures that the final product effectively addresses the limitations of current fragmented healthcare systems in Pakistan by providing a data-driven, secure, and user-centric experience.

3.1 Introduction to Requirement Analysis

Requirement Analysis is a critical phase in the software development lifecycle, serving as the foundation upon which the entire system is designed, implemented, and evaluated. For the Health Hub system, this phase ensures that all intelligent functionalities such as automated clinical biomarker extraction and multi-track Machine Learning health screening performance needs, and operational constraints are clearly understood before development begins.

The purpose of conducting requirement analysis is to bridge the gap between user needs for personalized, culturally relevant nutrition and the technical implementation of high-performance FastAPI services and Large Language Models (LLMs).

This ensures the final system effectively resolves existing healthcare challenges, such as the clinical literacy gap, manual record handling, and the lack of clinically grounded nutritional advice in Pakistan. This chapter outlines the detailed requirements of the Health Hub platform, describing how each AI component interacts with users, how the data flows through the Machine Learning Pipeline,

and what outcomes are expected from every interaction.

3.2 System Overview

The Health Hub system is designed as an integrated, AI and Machine Learning-powered digital platform that connects patients (users) and nutritionists within a unified, data-driven environment. The system enables users to manage their nutritional health by dynamically calculating BMI and BMR, generating personalized meal plans via a Generative AI Engine, and inputting medical lab results for automated clinical interpretation. Nutritionists, on the other hand, can access a specialized dashboard to view longitudinal patient progress through interactive charts (via Recharts), analyze multi-track machine learning health risk parameters, and provide expert consultation through an integrated professional portal.

The overall architecture reflects the system's role as a centralized hub that replaces manual dietary logging and fragmented paper reports with a streamlined, cloud-based solution. The system is built upon a Multi-Model Orchestration Layer (integrating Llama 3.3 and Qwen via Groq API), ensuring that the AI engine does not operate in isolation but is grounded in a verified clinical knowledge base and custom-trained localized machine learning pipelines served natively within the backend deployment.

The relationship between these actors Users, Nutritionists, and the AI Orchestration Layer shows how functionalities such as biomarker parsing, calorie adherence tracking, and professional appointment scheduling are interconnected. By consolidating these processes, Health Hub addresses the key issues of data fragmentation, the inability to interpret complex medical markers, and the lack of culturally relevant (Halal) health advice. This overview validates the system's objective to transform raw medical data into actionable, life-improving health outcomes through a cloud-native Firebase Firestore NoSQL framework.

3.3 Actors in the System

The Health Hub system interacts with multiple categories of users and internal system components, each performing distinct roles that contribute to the overall functionality of the platform. These users, also known as actors, engage with the system through specific use cases that reflect their responsibilities, permissions, and operational needs. Understanding these actors and their interactions is crucial for defining accurate system requirements and ensuring that the platform delivers a seamless, Machine Learning-integrated, and secure experience. The following sections describe each actor in detail, outlining their characteristics, goals, and the actions they perform within the Health

Hub ecosystem.

3.3.1 Patient (User)

The patient or general user is the primary beneficiary of the Health Hub system. Users interact with the platform to manage their nutritional journey by providing personal health metrics such as age, weight, height, and activity levels. Through the system, patients can upload laboratory report PDFs for automated backend text extraction and clinical risk screening, generate personalized Halal-compliant diet plans via the Generative AI Engine, and track their BMI and calorie adherence over time. The system ensures that patients can conveniently manage their health without needing advanced medical literacy, as the AI translates complex markers into actionable, easy-to-understand advice.

3.3.2 Nutritionist (Healthcare Professional)

Nutritionists are essential users who utilize the platform to provide expert oversight and professional consultations. The system enables nutritionists to access a comprehensive dashboard where they can review longitudinal patient health trends (via Recharts), analyze multi-track machine learning health risk metrics, and monitor user adherence to suggested plans. A critical responsibility of this actor is the "Human-in-the-Loop" validation, where they review, modify, and approve AI-generated diet plans before the patient adopts them, ensuring the highest standard of clinical safety.

3.3.3 AI and ML Orchestration Layer (System Actor)

Unlike traditional systems, Health Hub includes an AI & ML Orchestration Layer as a core system actor. This actor is responsible for processing raw data through a high-performance FastAPI pipeline. Its role is to execute serialized native XGBoost boosters and Scikit-Learn pipelines to predict multi-condition physiological risk scores, orchestrate multi-LLM calls (Llama 3.3 and Qwen via Groq API), and cross-reference clinical biomarkers with a verified medical knowledge base. This actor ensures that all recommendations are context-aware, clinically grounded, and free from hallucinations, providing the data-driven intelligence that supports both the user and the professional.

3.3.4 Administrator

The administrator oversees the entire Health Hub platform's operational integrity. Their responsibilities include nutritionist verification (KYC), managing system-wide configurations, and ensuring data security across the Firebase Firestore and Authentication infrastructure. Administrators handle the management of user roles, monitor system health, and ensure the reliability of the local

ML Model inference environment. Their control panel provides tools for monitoring API usage, managing the Halal recipe database, and ensuring that clinical safety guardrails are maintained across the platform.

3.4 Use Case Diagrams

Use case diagrams provide a clear and structured representation of how different users interact with the Health Hub system. They help define the system's functional boundaries and illustrate the various intelligent operations performed by each actor. Figure 3.1 shows the primary use case diagram for the Health Hub platform, based on the Machine Learning and AI-integrated requirements identified for the system.

In Figure 3.1, four main actors Patient (User), Nutritionist, the AI/ML Orchestration Layer, and Admin interact with the system through a set of defined use cases:

- **The Patient (User):** Uses the system to register an account via Firebase Authentication, set up a comprehensive health profile (BMI, BMR, chronic conditions), upload laboratory report PDFs for automated backend text parsing, execute localized multi-track Machine Learning health risk screening (Anemia, Vitamin D, and Type 2 Diabetes tracks), and track daily calorie adherence and physical activity logs.
- **The Nutritionist:** Interacts with the system by managing professional consultation schedules, reviewing AI-interpreted patient biomarkers and multi-track machine learning risk profiles, and monitoring longitudinal progress charts. A core use case for this actor is the "Review and Approve AI Diet Plan" function, ensuring that all generative plans are clinically validated before use.
- **The AI/ML Orchestration Layer (System Actor):** Performs back-end use cases such as executing localized inference for native XGBoost boosters and engineered Scikit-Learn pipelines, orchestrating Multi-LLM Generative logic (Llama 3.3/Qwen), and driving automated laboratory PDF text extraction using Regular Expression pattern algorithms.
- **The Admin:** Oversees system-level functionalities such as verifying nutritionist credentials (KYC), managing the Halal recipe database, monitoring localized ML model asset directories inside the FastAPI deployment, and ensuring data security and Firebase Firestore NoSQL database integrity.

The diagram also highlights how these actors connect with key system functionalities, such as automated biomarker parsing, secure NoSQL document store record management, and integrated professional consultation workflows. Each use case represents a specific function from predictive ML logic to human-led professional oversight that contributes to the overall healthcare workflow. By visualizing these interactions, the use case diagram ensures that all user needs, functional requirements, and intelligent system behaviors are correctly captured.

Overall, Figure 3.1 serves as a foundational model for understanding the operational structure of the Health Hub system. It guides developers and stakeholders by clearly depicting what the system is expected to perform and how different users will interact with its AI-driven features. This helps eliminate ambiguity, supports accurate requirement validation, and ensures that the system design aligns with the intended objectives of providing culturally relevant and medically grounded health solutions in the Pakistani context.

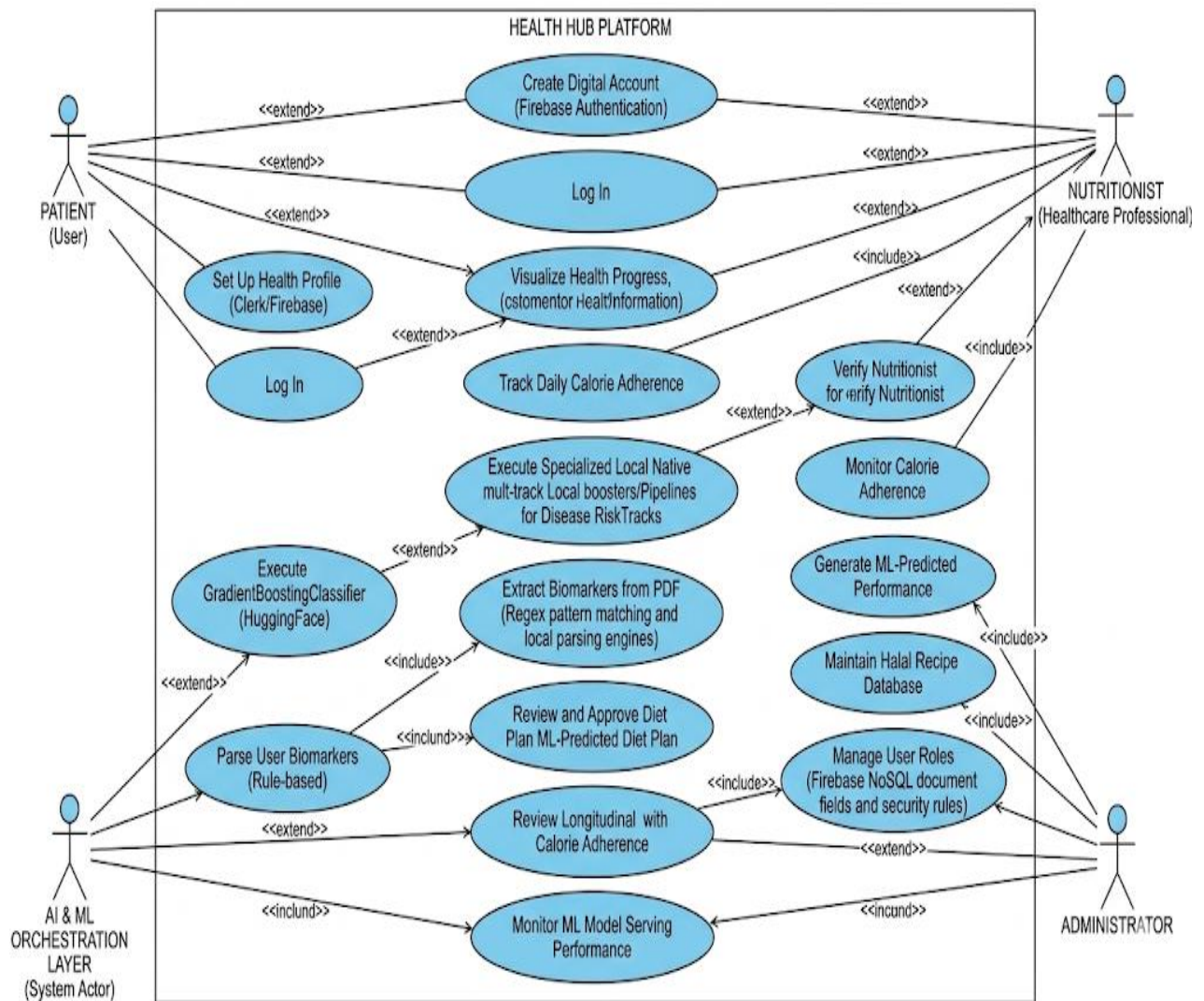


Figure 3.1: Use case diagram

3.5 Use Case Descriptions

This section provides detailed descriptions of all 17 major use cases identified within the Health Hub ecosystem, reflecting the new system architecture and actors as depicted in the revised Use Case Diagram. Each use case description outlines the specific actors involved, including the Patient (User), Nutritionist (Healthcare Professional), Administrator, and the custom AI & ML Orchestration Layer (System Actor). These descriptions detail standard triggers, preconditions, postconditions, normal and alternative flows, exceptions, and critical business rules such as strict clinical biomarker thresholds, document privacy logic via Firestore Security Rules, and local model inference validation tailored to an AI and Machine Learning-driven healthcare environment.

The following tables present the detailed descriptions of these use cases, meticulously updated to align with the technical implementations, such as multi-role authentication via Firebase Authentication, cloud-native document record management via Firebase Firestore NoSQL collections, and complex system interactions like "Human-in-the-Loop" validation (Review and Approve AI Diet Plan). Each table strictly follows a standardized template, ensuring clarity, technical consistency, and complete alignment with professional healthcare documentation standards, effectively bridging the gap between raw health data and intelligent nutritional insights.

3.5.1 Login/Registration Use Case

The Table 3.1 “Login/Registration” use case defines the fundamental process through which Patients, Nutritionists, and Administrators gain secure access to the Health Hub platform. Authentication is handled completely via Firebase Authentication, ensuring granular role-based access control (RBAC) mapping directly onto Firestore user documents.

Table 3.1: Login/Registration Use Case

Use Case Attribute	Details
Use Case Name	User Login & Registration
Primary Actors	User (Patient), Nutritionist, Admin
Secondary Actors	Firebase Authentication, Firebase Firestore NoSQL Store
Description	Enables users to create an account or log in securely. The system grants dashboard access permissions based on explicit user role properties stored inside the corresponding Firestore document.
Trigger	User attempts to sign up or sign in through the React frontend interface.

Pre-Conditions	Stable internet connection and active client session initiation.
Post-Conditions	User is authenticated successfully via token verification and redirected to their role-specific dashboard view.
Normal Flow	1. User selects "Sign In" or "Sign Up" on the interface. 2. User enters credentials which are verified securely by Firebase Authentication. 3. Backend validates the authenticated token payload against the database role map.v 4. User is dynamically routed to their specific interface module.
Business Rules	Nutritionists must maintain a verified KYC status flag within their profile document before full professional platform dashboard features are unlocked by the Admin.

3.5.2 Set Health Profile

The Table 3.2 “Set Health Profile” use case allows Patients to establish their baseline physiological data (Age, Height, Weight, Gender, and Activity Level), which serves as the metadata mapping input layer for local predictive screening algorithms and dietary estimation loops.

Table 3.2: Set Health Profile use case.

Use Case Attribute	Details
Use Case Name	Set Health Profile
Primary Actors	User (Patient)
Description	Users input or modify their key physical metrics and baseline lifestyle parameters to drive automated, real-time calculation models.
Trigger	User completes the initial app onboarding configuration or manually updates their physical parameters.
Pre-Conditions	User is authenticated and securely logged into a client account mapped under the 'user' role.
Post-Conditions	Updated metrics are written to the user's health profile document map

	inside Firestore; BMI, BMR, and TDEE values are recalculated automatically.
Normal Flow	1. Patient navigates to the Health Profile setting tab. 2. Enters updated values for weight, height, and age parameters. 3. Confirms modification by clicking "Save Changes". 4. The FastAPI backend triggers execution formulas, updating metric vectors chronologically.
Business Rules	Active lifestyle activity coefficients must be mapped explicitly to execute precise Total Daily Energy Expenditure estimations.

3.5.3 Input Clinical Biomarkers

The Table 3.3 “Input Clinical Biomarkers” use case enables users to bridge the clinical literacy gap by uploading digital medical reports from local labs into the automated parsing pipeline for seamless parameter ingestion.

Table 3.3: Input Clinical Biomarkers use case.

Use Case Attribute	Details
Use Case Name	Input Clinical Biomarkers (Lab Report Ingestion)
Primary Actors	User (Patient)
Secondary Actors	FastAPI PDF Parser Engine (pdfplumber)
Description	Users upload their unstructured laboratory report PDFs directly to the application to extract and compile blood parameters.
Trigger	User requests a digital interpretation of their latest laboratory results.
Pre-Conditions	User session is active; user possesses a valid, readable digital medical report file.
Post-Conditions	Biomarker data fields are extracted, cross-referenced, and saved into a structured sub-collection within the user's Firestore path.
Normal Flow	1. User opens the Clinical Lab section and uploads a report PDF. 2. FastAPI receives the file stream and passes it to the pdfplumber parsing module.

	3. Regular expression arrays map and extract numerical values for required fields. 4. Pre-filled parameters are presented to the user to confirm database entry.
Business Rules	Values must correspond to strict physiological boundaries; duplicate report processing instances targeting the same calendar date are automatically flagged.

3.5.4 Interpret Lab Reports (Clinical Engine)

The Table 3.4 “Interpret Lab Reports” use case transforms raw parsed biomarker parameters into highly structured health risk indices using automated clinical threshold evaluations.

Table 3.4: Interpret Lab Reports use case.

Use Case Attribute	Details
Use Case Name	Interpret Lab Reports (Clinical Analysis)
Primary Actors	Clinical Engine (System Actor)
Secondary Actors	User (Patient)
Description	Processes clinical biomarkers to evaluate specific metabolic levels and assign clear diagnostic risk tiers.
Trigger	Fresh biomarker parameters are compiled and verified inside the user's sub-collection path.
Pre-Conditions	Extracted numerical biomarker inputs must exist inside the database path.
Post-Conditions	Categorized risk evaluation flags are mapped onto the document and stored to condition the generative food engine.
Normal Flow	1. System pulls parameters from the database collection path. 2. Compares numerical bounds against reference clinical decision thresholds. 3. Assigns corresponding risk markers to the user's record profile. 4. Displays simplified diagnostic insights on the user dashboard.

Business Rules	Every automated interpretation interface output must include a clear legal medical disclaimer indicating the system is not a replacement for professional human diagnostics.
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3.5.5 Generate ML-Predicted Diet Plan

The Table 3.5 “Generate ML-Predicted Diet Plan” use case This is the core computational layer of Health Hub, utilizing localized machine learning classification tracks coupled with Llama 3.3 orchestration to produce customized nutritional schedules.

Table 3.5: Generate ML-Predicted Diet Plan use case.

Use Case Attribute	Details
Use Case Name	Generate ML-Predicted Diet Plan
Primary Actors	AI & ML Orchestration Layer (System Actor), User
Description	Executes localized diagnostic risk screenings to identify contraindications before drafting an iterative meal profile via the Groq cloud gateway.
Trigger	User clicks the "Request New Diet Plan" option from their interface panel.
Pre-Conditions	User baseline health profile dimensions and recent clinical biomarker matrices are fully updated inside Firestore.
Post-Conditions	A structured 7-day meal layout document is saved under a "Pending Approval" state collection path.
Normal Flow	1. FastAPI invokes localized estimators (.ubj / .pkl models) to capture physiological risk vectors. 2. Probability metrics are wrapped into an engineered prompt template. 3. The payload is sent to the Llama 3.3 model instance via the Groq cloud gateway. 4. The validated Halal meal structure is created and posted to the database.

Business Rules	Generated profiles must strictly adhere to calculated calorie constraints, accommodate cultural criteria, and exclude foods flagged by the Clinical Engine.
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3.5.6 Review and Approve AI Diet Plan

The Table 3.6 “Review and Approve AI Diet Plan” use case focuses the vital Human-in-the-Loop safety tier, providing verification capabilities to human experts before plans go live.

Table 3.6: Review and Approve AI Diet Plan use case.

Use Case Attribute	Details
Use Case Name	Review and Approve AI Diet Plan
Primary Actors	Nutritionist
Description	Enables professional nutritionists to review, refine, and sign off on AI-generated dietary profiles to ensure safe application.
Trigger	A new meal profile document reaches a "Pending Approval" flag state for an active client.
Pre-Conditions	Target meal layout document exists in the workspace collection with a pending flag state.
Post-Conditions	Document state advances to an "Approved" flag, immediately updating the synchronized client user dashboard layout.
Normal Flow	1. Nutritionist accesses the professional expert workspace tab. 2. Selects a pending client dietary plan folder from the interface list. 3. Evaluates structural items and modifies meal nodes directly if needed. 4. Clicks "Approve Plan" to release the file update to the user.
Business Rules	AI-generated plan components remain locked and invisible on the patient interface until a verified nutritionist document signature is appended.

3.5.7 Visualize Health Progress

The Table 3.7 “Visualize Health Progress” use case handles the processing and presentation of historical tracking fields through clear graphical modules.

Table 3.7: Visualize Health Progress use case.

Use Case Attribute	Details
Use Case Name	Visualize Health Progress
Primary Actors	User (Patient), Nutritionist
Secondary Actors	Recharts Component Library
Description	Processes tracking history fields to render scalable line and bar tracking visualizers covering nutritional behaviors.
Trigger	Actor opens the progress tracking dashboard panel via the user or expert client view.
Pre-Conditions	Multiple chronological tracking log updates must exist within the client's database collection path.
Post-Conditions	Responsive, interactive progress chart metrics are compiled and displayed on screen.
Normal Flow	1. Client app makes an asynchronous query request to the sub-collection tracking database stream. 2. Data vectors are structuralized sequentially on the client side. 3. The system maps values onto Recharts container configurations. 4. Interactive data visualization vectors display weight anomalies and metric logs.
Business Rules	Log values must be aligned and mapped chronologically based on Unix creation timestamp keys to prevent data plotting conflicts.

3.5.8 Manage User Roles & Permissions

The Table 3.8 “Manage User Roles & Permissions” use case allows administrators to control operational boundaries and enforce data isolation guidelines across the ecosystem.

Table 3.8: Manage User Roles use case.

Use Case Attribute	Details
Use Case Name	Manage User Roles & Permissions

Primary Actors	Admin
Secondary Actors	Firestore Authentication, Firestore
Description	Admin initializes role transformations or modifies specific account access capabilities inside database records.
Trigger	System audit request or new credential processing submission by a registered specialist.
Pre-Conditions	Target user record must exist; Admin session must satisfy absolute authorization validation.
Post-Conditions	Account role configuration modifications are written to the document map and enforced via security filters.
Normal Flow	1. Admin logs into the administrative console space. 2. Locates target account entry via user registry lookup. 3. Modifies the structural profile permission value field. 4. Saves structural alterations, forcing immediate platform token validation updates.
Business Rules	Account elevation to 'nutritionist' requires successful authentication verification alongside structural professional proof verification records.

3.5.9 Monitor ML Model Performance

The Table 3.9 “Monitor ML Model Performance” use case keep track of system response parameters, processing overhead, and inference delays across all localized processing pathways.

Table 3.9: Monitor ML Model Performance use case.

Use Case Attribute	Details
Use Case Name	Monitor ML Model Performance
Primary Actors	Admin
Description	Records monitoring telemetry, logging execution times for local .ubj and .pkl file processing layers alongside Groq platform endpoints.

Trigger	Scheduled routine performance query loop or server-side latency exception flag.
Pre-Conditions	Server logging metrics and telemetry collection pipelines are operational.
Post-Conditions	Performance records are structuralized and reviewed to manage local backend computing allocations.
Normal Flow	1. Telemetry monitors record compute times across active pipeline channels. 2. Execution logs are stored inside an internal system monitoring space. 3. Admin reviews execution trends through the configuration panel layout. 4. Optimizes file streaming configurations to keep execution pathways clear.
Business Rules	Internal computing latency constraints for localized booster classification steps are capped at a maximum threshold of 2.0 seconds.

3.5.10 Consult via Real-Time Chat

The Table 3.10 “Consult via Real-Time Chat” use case provides communication tools to facilitate immediate feedback and consultation loops between active platform actors.

Table 3.10: Consult via Real-Time Chat Use Case

Use Case Attribute	Details
Use Case Name	Consult via Real-Time Chat
Primary Actors	User (Patient), Nutritionist
Secondary Actors	Firebase Firestore Realtime Stream Sync
Description	Establishes instant message syncing to handle direct consultation requirements securely.
Trigger	An actor types an entry and submits it within the consultation chat

	window component.
Pre-Conditions	A active, verified consultation appointment linking both parties must be active in the system.
Post-Conditions	Messages sync instantly on screen and append directly to the conversation sub-collection stream.
Normal Flow	1. User inputs textual updates inside the conversation text component. 2. React interface pushes document object to the chat collection pathway. 3. Database listener intercepts the payload, syncing the viewport of the connected recipient. 4. Document updates are appended to the historical messaging path.
Business Rules	Message entries must be cryptographically bound to explicit patient-nutritionist relation maps, blocking all external document read attempts.

3.5.11 Book Consultation Appointment

The Table 3.11 “Book Consultation Appointment” use case bridges automated guidance with specialized human care by enabling patients to schedule consultation slots with verified clinical professionals.

Table 3.11: Book Consultation Appointment Use Case

Use Case Attribute	Details
Use Case Name	Book Consultation Appointment
Primary Actors	User (Patient)
Secondary Actors	Nutritionist
Description	Allows a user to view professional openings and book a consultation appointment inside the provider's active calendar collection.
Trigger	User initializes a booking request from the selected specialist's

	professional directory card.
Pre-Conditions	User account role is verified; nutritionist has set up available time block intervals within their profile map.
Post-Conditions	A unique appointment document is instantiated under a 'pending' state flag attribute.
Normal Flow	1. User navigates through the professional directory view. 2. Chooses an open slot from the target provider's schedule component. 3. Confirms selection details and submits the registration payload. 4. System instantiates a record document inside the global appointments collection.
Business Rules	The system automatically checks timestamps to block overlapping appointment entries for the same professional profile.

3.5.12 Manage Consultation Schedules

The Table 3.12 “Manage Consultation Schedules” use case structure their operational calendars, defining open slots and processing customer booking submissions.

Table 3.12: Manage Consultation Schedules Use Case

Use Case Attribute	Details
Use Case Name	Manage Consultation Schedules
Primary Actors	Nutritionist
Description	Enables experts to define work hour parameters and accept, decline, or update incoming client consultation logs.
Trigger	Reception of an automated booking event or manual update action inside the dashboard view.
Post-Conditions	Appointment status flags update across collections, instantly triggering a notification state reload on the patient client app.
Normal Flow	1. Expert signs in and pulls up the consultation manager calendar view. 2. Reviews unresolved client booking request document records.

	3. Exercises status alteration actions (Confirm / Decline / Re-schedule). 4. The backend writes modifications to the target appointment record.
Business Rules	Rescheduling updates must be logged at least 24 hours prior to the original appointment time to protect workflow stability.

3.5.13 Verify Nutritionist Credentials

The Table 3.13 “Verify Nutritionist Credentials” use case provides verification protocols to confirm the background and validation credentials of professional platform sign-ups.

Table 3.13: Verify Nutritionist Credentials Use Case

Use Case Attribute	Details
Use Case Name	Verify Nutritionist Credentials (KYC)
Primary Actors	Admin, Nutritionist
Secondary Actors	Firebase Storage
Description	Admin evaluates background files to authorize account transitions into the authenticated provider workspace tier.
Trigger	A professional registration submission file reaches the admin review queue.
Pre-Conditions	Professional documentation files (PDF/Images) are successfully loaded into the protected Firebase Storage segment.
Post-Conditions	Profile document access role values change to 'nutritionist', opening associated feature paths.
Normal Flow	1. Admin accesses the credential confirmation panel list. 2. Opens associated verification folders and checks uploaded certificate files. 3. Confirms legitimacy against regional healthcare registry records. 4. Updates account status properties to complete profile validation.
Business Rules	Account validation parameters require matching confirmation entries from recognized national or medical regulatory associations.

3.5.14 Maintain Halal Recipe Database

The Table 3.14 “Maintain Halal Recipe Database” use case allows admins update the core nutritional reference listings, ensuring the generative advice layer matches regional food metrics.

Table 3.14: Maintain Halal Recipe Database Use Case

Use Case Attribute	Details
Use Case Name	Maintain Halal Recipe Database
Primary Actors	Admin
Description	Allows administrators to create, update, or remove entries from the verified collection of local, Halal-compliant food records.
Trigger	Nutritional updates are published or an asset item modification request is processed.
Post-Conditions	Updates are applied across global collections, immediately shifting the semantic matching context used by generative loops.
Normal Flow	1. Admin opens the recipe indexing platform tool view. 2. Inputs raw recipe attributes, macro properties, ingredient indices, and local designations. 3. Confirms compliance constraints and pushes the entry to live execution status. 4. System commits the document parameters directly into the recipes collection path.
Business Rules	Every food entry must compile explicit, non-zero macronutrient measurements per baseline consumer serving dimension to preserve tracking accuracy.

3.5.15 Track Daily Calorie Adherence

The Table 3.15 “Track Daily Calorie Adherence” use case enables users to record daily nutritional intake values, compiling performance data to feed analytical visualization charts.

Table 3.15: Track Daily Calorie Adherence Use Case

Use Case Attribute	Details
Use Case Name	Track Daily Calorie Adherence
Primary Actors	User (Patient)
Description	Users log their actual food intake components, comparing cumulative counts against target daily energy profiles.
Trigger	User logs a meal or ingredient choice inside the interface timeline component.
Post-Conditions	The adherence score parameter is calculated and stored inside the user's progress tracking document.
Normal Flow	1. User taps the tracking icon, choosing a meal category node. 2. Specifies target food item parameters or enters explicit calorie counts. 3. Confirms submission to log values to their profile history path. 4. Backend computes total macro balances against target values.
Business Rules	Adherence percentages are determined using the structured calculation: $\text{Adherence} = \left(\frac{\text{Actual Calorie Intake}}{\text{Target TDEE}} \right) \times 100\%$

3.5.16 Review Longitudinal Patient Histories

The Table 3.16 “Review Longitudinal Patient Histories” use case allows specialists to evaluate customer metrics over multi-month tracking cycles, facilitating safe, long-term intervention adjustments.

Table 3.16: Review Longitudinal Patient Histories

Use Case Attribute	Details
Use Case Name	Review Longitudinal Patient Histories
Primary Actors	Nutritionist

Description	Professionals pull up chronological tracking logs covering user weights, metabolic shifts, and extracted lab markers over 3–6 month timelines.
Trigger	Specialist initializes an analytical inspection ahead of a scheduling event or plan modification action.
Pre-Conditions	Active consultation linkages exist; user profile document access clearances are valid.
Post-Conditions	Longitudinal health vectors are decrypted and structured into analytical report templates.
Normal Flow	1. Expert navigates to the customer profile sub-directory interface map. 2. Initializes a history log compilation query loop. 3. FastAPI queries document paths, calculating rolling average metrics. 4. Progress tracking indicators are rendered to the expert dashboard interface.
Business Rules	Access keys are transient; document authorization validations automatically expire once active consultation timelines finish.

3.5.17 Orchestrate Multi-LLM Diet Generation

The Table 3.17 “Orchestrate Multi-LLM Diet Generation” use case manages internal LLM operations, coordinating API streams to produce cleanly structured nutritional plans.

Table 3.17: Orchestrate Multi-LLM Diet Generation Use Case

Use Case Attribute	Details
Use Case Name	Orchestrate Multi-LLM Diet Generation
Primary Actors	AI & ML Orchestration Layer (System Actor)
Description	Governs programmatic streaming requests to Groq platforms, evaluating JSON formats to guarantee well-structured data.
Trigger	The execution pipeline receives a valid execution call from the

	parent meal generation module.
Post-Conditions	A validated JSON object map containing complete meal data is generated and saved to the database.
Exceptions	Network timeout occurrences or rate-limit violations trigger local exponential backoff retry policies.
Business Rules	Generative string outputs must validate cleanly against system Pydantic schema maps before being committed into Firestore collections.

3.6 Functional Requirements

Functional requirements define the specific behaviors and operations that the Health Hub system must provide to satisfy the needs of its users. These requirements outline the essential capabilities that the system must possess to manage personalized nutritional data, Machine Learning-driven health risk screening, and seamless communication between patients and verified nutritionists. The Health Hub system is designed to offer a secure, AI-integrated, and high-performance environment that ensures data integrity through cloud-native Firebase Firestore NoSQL architecture and real-time responsiveness through FastAPI.

Table 3.18: Functional Requirements

Module ID	Module Name	Key Features & Functional Requirements
1.	Secure Identity and Role Management	Authentication: Implementation of Firebase Authentication for secure multi-role login infrastructure. Role-Based Access Control (RBAC): Distinct operational environments and document-level view boundaries for Patients, Nutritionists, and Admins. Dynamic Health Profiling: Automated calculation of BMI, BMR, and TDEE metrics based on user-provided physical data tracks. KYC Verification: Professional credential submission and verification folder workflow for incoming nutritionists.

2.	Predictive and Clinical Intelligence	Multi-Track ML Risk Screening: Execution of local standalone native XGBoost boosters (.ubj) and specialized Scikit-Learn pipelines (.pkl) to screen multi-condition physiological profiles (Anemia, Vitamin D, and Type 2 Diabetes) based on extracted biomarkers. Automated Lab PDF Parsing: Streamlining data collection by utilizing backend Regular Expression pattern algorithms inside a pdfplumber execution loop to instantly pre-fill diagnostic parameters. Actionable Insights: Translation of complex clinical risk probabilities into simplified summaries for enhanced user medical literacy.
3.	Nutritional Engine and AI Generation	Generative AI Diet Planner: Automated generation of 7-day, calorie-precise, and Pakistani Halal-compliant meal plan arrays using Llama 3.3 via the Groq cloud gateway. Clinical Safety Tier: Core validation logic that automatically filters and excludes contraindicated food components or allergens based on underlying clinical screening engine flags. Interactive Analytics: Chronological visualization of patient weight changes and daily calorie adherence tracking using the Recharts library.
4.	Professional Portal and Governance	Human-in-the-Loop Approval: A dedicated verification workflow allowing nutritionists to review, edit, and approve AI-generated plans before they become visible to the patient. Unified Appointment Manager: Centralized calendar scheduling, slot allocation management, status updates (Accept/Decline), and history tracking.

		Real-time Consultation: Secure, document-driven messaging feed powered by real-time Firestore stream sync for direct patient-professional interaction.
5.	Backend Orchestration and Ops	FastAPI Integration: Asynchronous server-side orchestration handling incoming report streams, local ML model inference blocks, and external Groq API processing routes. Document Store Management: Structuring real-time collection streams and user tracking indices using decoupled Firebase Firestore NoSQL schemas. Knowledge Base Tuning: Administrative controls to adjust local Halal recipe datasets and clinical threshold rules.

3.7 Non-Functional Requirements

Non-functional requirements define the quality attributes, performance standards, and operational constraints that the Health Hub system must meet. While functional requirements describe what the system does, non-functional requirements specify how the system performs and the standards it must uphold in terms of reliability, security, scalability, and clinical accuracy. These requirements are critical for ensuring that the Health Hub system operates effectively under real-world conditions and maintains the trust of both patients and healthcare professionals. The non-functional requirements for the Health Hub platform are detailed below:

3.7.1 Security & Privacy

The system must ensure high-level security as it handles sensitive clinical data. Authentication and authorization must be managed entirely through Firebase Authentication, implementing Role-Based Access Control (RBAC) to ensure that a user's medical markers are only accessible to themselves and their authorized nutritionist. All document entries stored within cloud-native Firebase Firestore collections must be protected using granular server-side authorization patterns governed by custom Firestore Security Rules (firestore.rules). API endpoints configured inside FastAPI must implement strict Cross-Origin Resource Sharing (CORS) policies and secure headers to prevent unauthorized cross-origin requests and data leakage.

3.7.2 Performance & Response Time

As an AI-driven platform, low latency is critical for user satisfaction. The system shall execute localized native machine learning screening booster tracks (anemia_xgb.ubj, vitd_xgb.ubj, and diabetes_pipeline_v2.pkl) in less than 2 seconds. Due to the complex nature of Large Language Models, the generation of a full 7-day personalized diet plan array via the Groq cloud gateway (Llama 3.3) should be optimized to deliver results within 5–8 seconds. The frontend, built with Vite, must ensure a "First Contentful Paint" (FCP) of under 1.5 seconds to provide a lightning-fast user experience.

3.7.3 Accuracy & Clinical Integrity

The primary non-functional requirement for the AI engine is the elimination of "hallucinations." All generated nutritional advice must be strictly grounded in the verified Clinical Knowledge Base. The Clinical Engine must maintain 100% accuracy in comparing biomarker values (like HbA1c) against established medical thresholds. Furthermore, the Human-in-the-Loop approval workflow serves as a critical reliability layer, ensuring that no AI-generated plan is finalized without professional validation.

3.7.4 Reliability & Availability

The Health Hub system shall aim for 99.9% availability, leveraging cloud-native deployment. The backend, built on FastAPI, must handle asynchronous request streams efficiently to prevent system bottlenecks during peak hours. In the event of an external API connection failure (e.g., Groq cloud endpoint downtime), the system must implement a local exponential backoff retry policy or fail gracefully by pulling previously verified data records from the Firebase Firestore local cache storage.

3.7.5 Scalability

The architecture must be designed for fluid horizontal scaling. By utilizing an asynchronous FastAPI backend and a cloud-native Firebase Firestore NoSQL document database, the system should be capable of seamlessly handling an increasing volume of concurrent users and nested medical logging feeds without computing resource exhaustion. The ML model serving layer, executing local inference directories natively inside the FastAPI project deployment, must maintain consistent prediction times as the request volume grows.

3.7.6 Usability & HCI Standards

The interface must adhere to Human-Computer Interaction (HCI) principles, ensuring that complex medical data is visualized through simple, intuitive interactive charts (via Recharts). The system shall provide a consistent experience across various screen sizes (Responsive Design) and use clear typography to ensure that elderly users or those with limited medical literacy can easily navigate their health dashboard.

Chapter 4

DESIGN AND ARCHITECTURE

4. Design and Architecture

The Design and Architecture chapter of the Health Hub system explains how the AI and Machine Learning-driven software is structured, how its intelligent components interact, and how clinical data flows throughout the platform. This chapter provides a detailed overview of the system's architecture specifically the integration between the React-Vite frontend, the high-performance FastAPI backend, and the localized Machine Learning inference pipeline.

It covers the data representation, document collection schemas, process flows, and design models that form the backbone of the application. The chapter highlights the technical decisions made to ensure high-speed, localized ML predictions, secure document-oriented data handling in Firebase Firestore, and a scalable cloud infrastructure. By presenting the system design clearly, this chapter demonstrates how Health Hub meets both functional and non-functional requirements transforming raw health metrics and parsed PDF report files into clinically grounded nutritional advice while delivering a reliable and secure healthcare experience for users and nutritionists alike.

4.1 System Architecture

The system architecture defines the overall structure of Health Hub and illustrates how its intelligent components work together to deliver AI-driven nutritional and healthcare services efficiently. It ensures that each module from automated biomarker parsing to localized native machine learning disease risk screening performs its specific tasks while interacting seamlessly via an asynchronous API layer to maintain smooth operations and clinical data integrity.

4.1.1 Architecture Style

Health Hub is developed using a Layered Client-Server Architecture integrated with a specialized ML Orchestration Layer.

- The Client Layer: Consists of a responsive Web interface built with React 18, allowing patients and nutritionists to interact with the system.
- The Backend Layer: Powered by FastAPI (Python 3.13), managing asynchronous business logic, local ML model inference serving, and secure communication with the Firebase Firestore database collections.
- The Data & AI Layer: Utilizes cloud-native Firebase Firestore document storage for structured records, and the Groq API (Llama 3.3/Qwen) combined with serialized native model binaries for

executing predictive diagnostic tracks. This separation of concerns enhances maintainability, supports independent scaling of AI processing, and ensures secure handling of sensitive medical records.

4.1.2 Components

The Health Hub system is composed of several interconnected components, each designed to perform a specific function within the intelligent architecture. By dividing the system into modular parts, Health Hub maintains clarity in functionality and allows Machine Learning modules to evolve without affecting the entire system.

4.1.2.1 User Interface (UI)

The UI provides access via web to all user types: patients, nutritionists, and administrators. Built with React 18 and Tailwind CSS, it is designed to be intuitive and responsive, ensuring users can navigate features such as clinical marker input, ML-predicted diet plans, and health tracking charts.

4.1.2.2 Identity & Role Management (Security Module)

Powered entirely by Firebase Authentication, this service manages user registration, secure login, and Role-Based Access Control (RBAC). It ensures that only authorized users can access sensitive health metrics, implementing security measures such as token-based validation and cryptographic rule-enforced document constraints (firestore.rules) to safeguard patient privacy.

4.1.2.3 Clinical Engine (Biomarker Analyzer)

This specialized module is responsible for the structured extraction and analysis of laboratory data fields. It utilizes an automated PDF stream parser (pdfplumber combined with Regular Expression pattern maps) to interpret raw values like HbA1c, fasting glucose, and hemoglobin, identifying risk levels based on clinical thresholds. It translates raw, unstructured laboratory files into structured digital document records stored in Firebase Firestore for further nutritional analysis.

4.1.2.4 ML & Nutritional Engine (Core Intelligence)

This is the core intelligence component of Health Hub. It utilizes localized machine learning risk tracks (native XGBoost boosters in Universal Binary JSON format alongside a 19-feature Scikit-Learn validation pipeline) to evaluate multi-condition physiological profiles, and a Generative AI pipeline (Llama 3.3 via Groq API) to generate daily meal plans. It ensures that the generated plans are medically grounded, Halal-compliant, and tailored to the user's specific health status, accounting

for calculated TDEE, BMI, and metabolic risk flags.

4.1.2.5 Analytics & Visualization Module

This module manages the storage and visualization of longitudinal health data. It calculates BMI and BMR, displaying progress through interactive charts using Recharts. It allows patients and nutritionists to track the impact of diet plans over time, ensuring data-driven health monitoring based on structured document history streams.

4.1.2.6 Professional Consultation & Chat Module

The consultation module facilitates the booking and management of appointments. It includes a Human-in-the-Loop approval system where nutritionists validate AI plans and a Real-time Chat service powered by real-time Firestore collection stream synchronization for instant communication between patients and professionals.

4.1.2.7 Admin Control & Model Ops Dashboard

The admin dashboard provides tools for administrators to verify nutritionist credentials (KYC), monitor local ML model inference latency, and update the Halal recipe database. It allows oversight of system health and provides logs for API usage and security audits.

4.1.3 Interactions

The interactions between the various components of Health Hub are managed through a combination of high-speed RESTful API calls and real-time document sync patterns, ensuring efficient and standardized data exchange across the cloud ecosystem. The React-Vite UI sends asynchronous requests to the FastAPI backend, which serves as the central orchestration layer. This layer processes raw health data, executes Machine Learning models natively via serialized binary structures, and interacts with the cloud-native Firebase Firestore NoSQL document store.

For instance, when a user uploads a clinical report, the UI triggers a secure POST request containing the file stream to the FastAPI Clinical Engine. This event initiates an interaction where the system extracts and validates parameters against clinical thresholds and updates the user's health profile in the Firestore database collection path. Simultaneously, the AI & ML Orchestration Layer interacts with these updated markers to trigger localized machine learning pipelines, which instantly evaluate if a change in physiological risk profiles or adjusted dietary contraindications is necessary.

All communication within the system is secured using HTTPS protocols, SSL encryption, and

Firebase Authentication token states to protect sensitive medical data and ensure robust user privacy. Furthermore, the integration of Firebase Firestore real-time snapshot listeners ensures that specific updates such as a nutritionist's approval of a diet plan or new chat messages are reflected on the user's dashboard instantly. This decoupled interaction model ensures that complex local AI computations do not block the user interface, providing a smooth and responsive experience even during heavy data processing.

4.1.4 System Architecture Diagram

The system architecture diagram (Figure 4.1) illustrates the high-performance, layered structure and interaction of the Health Hub components. It shows how users including Patients, Nutritionists, and Administrators interact with the platform through responsive, web-based dashboards built with React 18 and Vite. The diagram highlights the multi-layered flow of information between the user interface, the FastAPI Backend Orchestration, the cloud-native Firebase Firestore NoSQL Document Database, and the localized Machine Learning and Generative AI Layer (native binary models and Groq API).

The architecture is designed to ensure asynchronous, non-blocking communication between modules and efficient data management for high-stakes medical insights. When a user performs an action, such as uploading laboratory report PDFs for automated biomarker parsing or requesting a personalized diet plan, the request is sent via secure RESTful API calls to the FastAPI backend.

The server-side logic processes the request by executing the localized native machine learning estimators such as standalone XGBoost boosters in Universal Binary JSON format (anemia_xgb.ubj, vitd_xgb.ubj) and an engineered Scikit-Learn pipeline (diabetes_pipeline_v2.pkl) or the Generative AI Diet Engine (orchestrated via Groq Llama 3.3/Qwen). These services interact with the Firebase Firestore database through structured Pydantic validation schemas to ensure secure, transaction-safe document storage of clinical profiles. The processed results are then stored and returned to the UI in real-time, utilizing Firebase Firestore snapshot streams for instant nutritionist-patient dashboard synchronization. This structured approach provides a clear separation of concerns, improves system scalability for a growing user base, and ensures that the local AI orchestration is natively optimized within the backend tier for maximum maintainability.

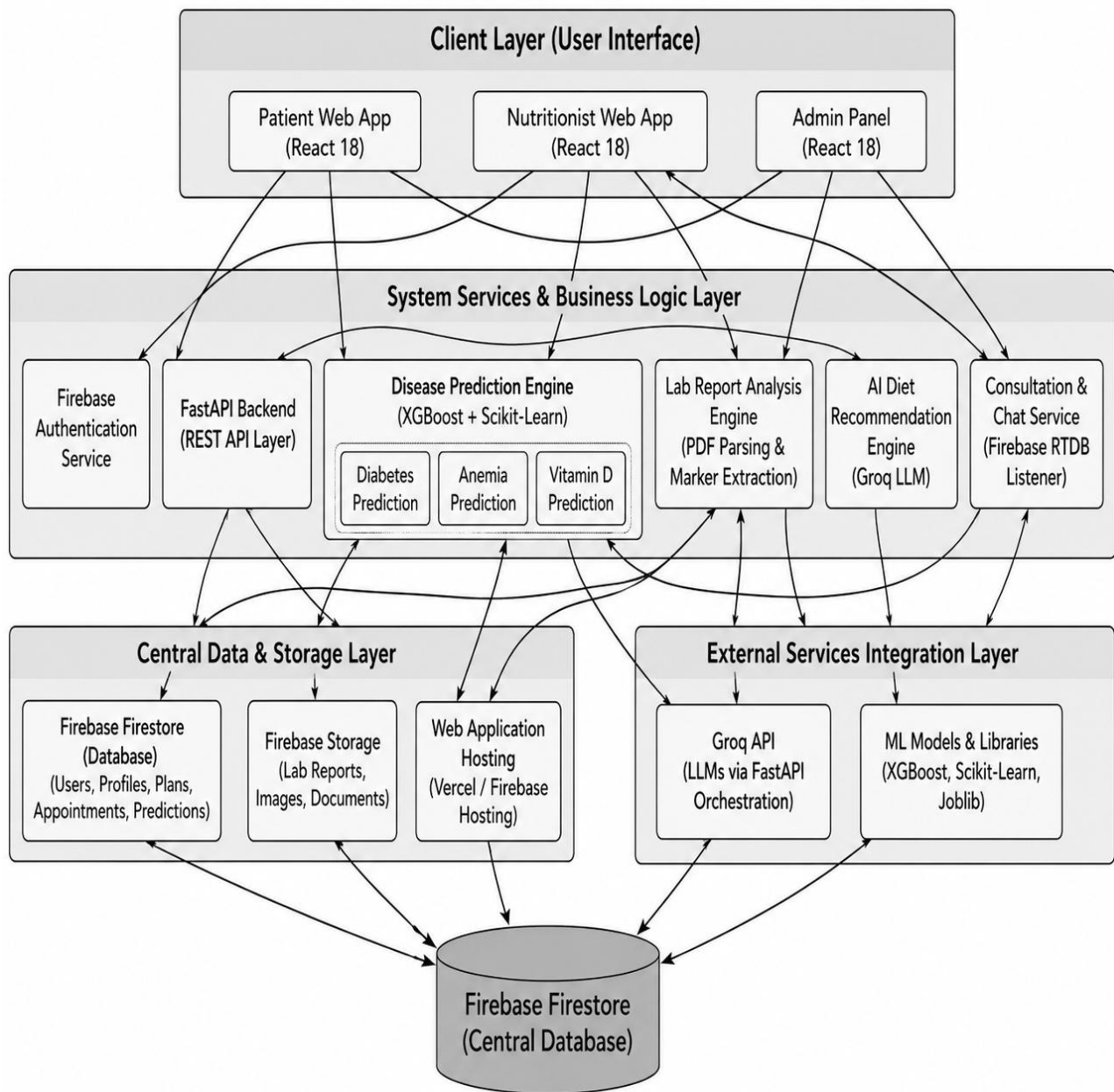


Figure 4.1: System Architecture Diagram

4.2 Data Representation

The Data Representation section explains how information is structured, stored, and managed within the Health Hub system. It covers the database design approach, entity-relationship mappings for a relational environment, and the logic behind how clinical data flows between the FastAPI orchestration layer and the Machine Learning modules. This section ensures that all system components including the Gradient Boosting Predictor and the Generative AI Diet Engine can efficiently access and process the information they need while maintaining clinical accuracy, data consistency, and patient privacy through structured relational integrity.

4.2.1 Database Design

Health Hub utilizes Firebase Firestore as its primary cloud-based NoSQL database for storing and managing health-related information. Firestore was selected because it provides real-time synchronization, flexible document structures, high scalability, and seamless integration with Firebase Authentication and Firebase Storage. Since the platform manages diverse healthcare information such as user profiles, disease prediction results, diet plans, appointment records, chat messages, and laboratory reports, a document-oriented database offers greater flexibility for handling dynamic and evolving health data.

The database design is organized into multiple collections including Users, User Profiles, Lab Reports, Predictions, Diet Plans, Appointments, Nutritionists, and Chat History. This structure allows the FastAPI backend and AI-powered modules to efficiently retrieve and update user information whenever disease predictions, diet recommendations, or consultation activities are performed. Firestore's real-time capabilities ensure that changes in health metrics, appointment status, prediction outcomes, and nutrition plans are immediately reflected across the system without requiring manual synchronization.

The database architecture also supports secure access control through Firebase Authentication and role-based authorization mechanisms. Each user's health records, prediction history, uploaded laboratory reports, and consultation data are securely associated with their unique account. This approach maintains data consistency while allowing healthcare professionals and administrators to access authorized information efficiently. By utilizing Firebase Firestore and cloud-based storage services, Health Hub ensures reliable performance, secure healthcare data management, and scalability for future expansion while supporting real-time health monitoring and intelligent clinical analysis.

4.2.2 Entity–Relationship Representation

Health Hub utilizes a document-oriented data model implemented through Firebase Firestore to manage healthcare information efficiently. Unlike traditional relational database systems, Firestore organizes data into collections and documents, allowing flexible storage of user profiles, laboratory reports, disease prediction results, diet plans, appointments, and consultation records. This structure enables the platform to handle dynamic healthcare information while maintaining scalability, security, and real-time synchronization.

As illustrated in Figure 4.2, the User entity serves as the central component of the system. Each

registered user maintains a unique profile containing personal information, health metrics, medical history, and lifestyle data. A user can be associated with multiple Lab Reports, Disease Predictions, Diet Plans, Appointments, Progress Records, and Chat Messages. These relationships allow the system to maintain a complete longitudinal health history and support personalized healthcare recommendations.

The Disease Prediction module stores prediction results generated by the Machine Learning models, including Diabetes Risk Prediction, Anemia Prediction, and Vitamin D Deficiency Prediction. These prediction records are linked to the corresponding user profile and are utilized by the AI Diet Recommendation Engine to generate personalized nutritional guidance. Similarly, uploaded laboratory reports are stored and analyzed through the Lab Report Analysis Module, providing additional clinical information that supports prediction accuracy and dietary planning.

The Appointment and Consultation entities establish relationships between users and nutritionists, enabling professional healthcare guidance through appointment scheduling and real-time communication. This document-based representation allows the FastAPI backend, Machine Learning services, and AI-powered modules to efficiently access related healthcare information while ensuring data consistency, security, and scalability across the platform.

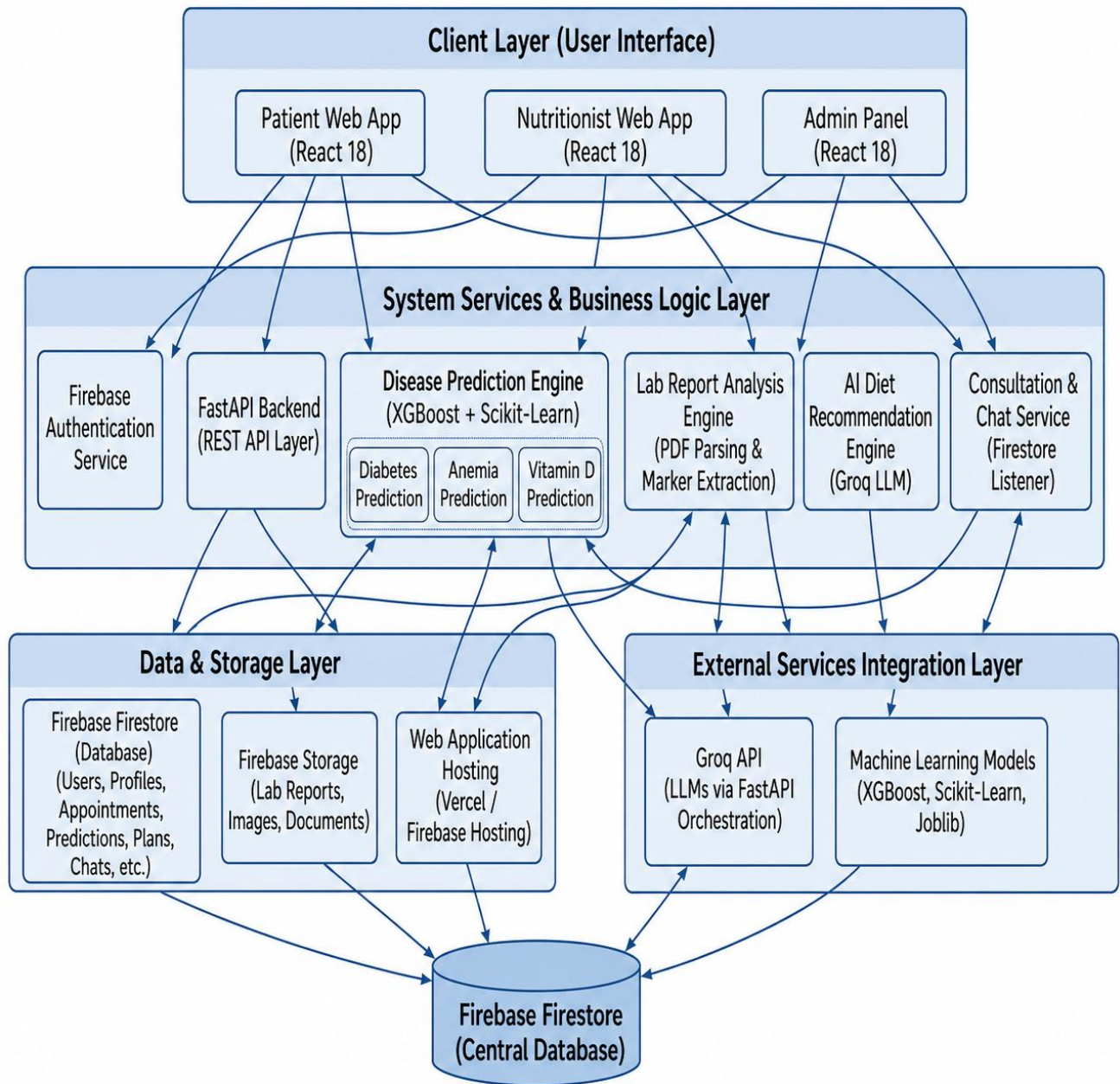


Figure 4.2: Entity-Relationship Diagram of the System

4.2.3 Detailed Document Relationship Structure

The Detailed Document Relationship Structure illustrates the specialized document-oriented architecture designed for Health Hub, prioritizing high-speed queries, horizontal scalability, and flexible data payloads for real-time Machine Learning processing. Moving away from rigid relational tables, this architecture utilizes a Normalized NoSQL Schema modeled inside cloud-native Firebase Firestore collections to maintain a cohesive, secure link between a patient's identity and their clinically sensitive datasets. The primary users collection acts as the foundational root entity, which aggregates user states and is cleanly separated from professional provider accounts contained within a top-level nutritionists collection.

The most critical architectural pattern is the logical indexing path linking a Patient Document to their multi-track `clinical_biomarkers` sub-collection logs and their active `diet_plans`. Because the FastAPI orchestration engine requires simultaneous access to physical metrics (Age, Weight, Height) and clinical markers (such as parsed blood data fields), these datasets are co-located or reference-linked using unique Firebase Auth UIDs (`auth_uid`). This allows the system to execute low-latency document fetches across nested paths without resource-intensive multi-table SQL JOIN operations, ensuring that the localized Python machine learning pipeline receives a complete, synchronized data payload for accurate risk screening.

Furthermore, the appointments and real-time chat collections are modeled as decoupled top-level directories linked via unique reference ID string attributes (`patient_uid` and `nutritionist_uid`), establishing a secure, highly auditable communication trail. This structure ensures that even as the volume of medical logging feeds grows exponentially, the system maintains ultra-low fetch times and native schema flexibility. By leveraging Firestore's built-in single-field and composite indexing mechanisms, Health Hub delivers clinically grounded nutritional insights in a highly responsive, cloud-native ecosystem.

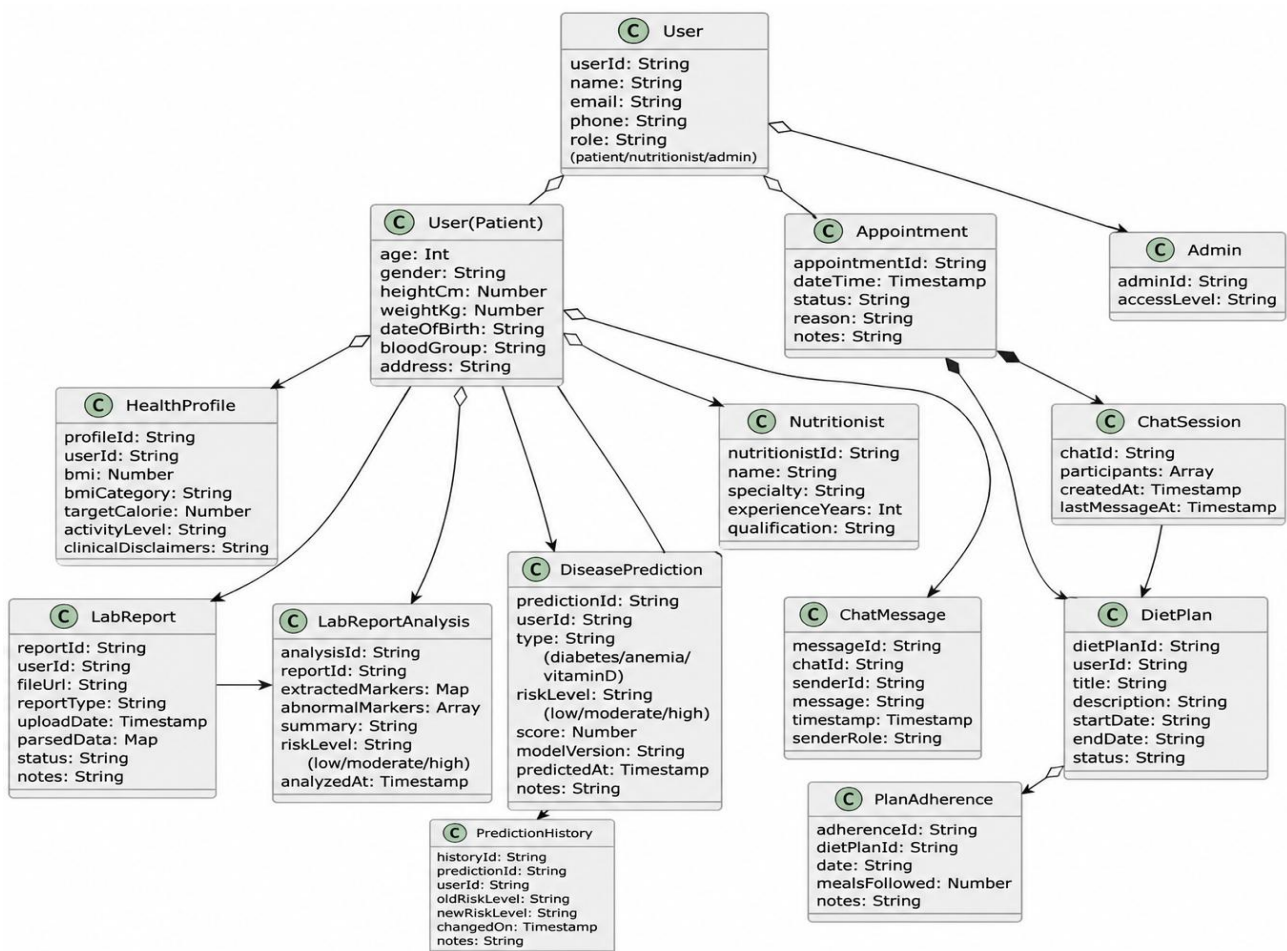


Figure 4.3:Firestore Document Relationship structure

4.2.4 Data Flow

The Data Flow section describes the dynamic movement of information throughout the Health Hub ecosystem, illustrating how raw inputs are transformed into actionable health insights through a series of specialized computational layers. The flow initiates at the Client Layer (React-Vite UI), where the user provides physical metrics, uploads lab report PDFs, or adds food entries. This data is transmitted via secure HTTPS protocols and verified Firebase Authentication tokens to the FastAPI Backend, triggering a sequence of high-performance internal processes.

For clinical biomarker data, the flow enters the backend PDF parser module where text values are extracted using `pdfplumber` and mapped via Regular Expression pattern matching. These structured parameters are then committed to a dedicated sub-collection inside the Firebase Firestore document store, ensuring transaction-safe storage and longitudinal tracking consistency. The most critical interaction occurs during the nutritional classification and multi-track disease risk screening phase, where data flows directly from the Firestore collection state into the FastAPI ML Orchestration Layer. Here, localized native estimators specifically standalone XGBoost boosters and optimized Scikit-Learn pipelines process the clinical markers and physical metrics natively within the backend system memory to predict multi-condition risk states.

Once the health category and risk factors are identified, the data flows into the Generative AI Engine, where parameters are merged with verified nutritional standards and local, Halal-compliant recipes retrieved asynchronously from the Firestore recipes collection. The engine coordinates a prompt orchestration loop using Llama 3.3 via the Groq cloud gateway to generate a highly personalized, structured meal plan array. Finally, this processed information flows back to the presentation layer, updating the patient's dashboard and interactive Recharts components in real-time. This continuous, bidirectional loop ensures that every piece of data whether it is a new weight update, an extracted blood biomarker, or a nutritionist's approval flag is accurately routed, processed, and reflected across all system modules to maintain a consistent and reliable user experience.

4.2.4.1 Sequence Diagram (Data Flow Across Layers)

The sequence diagram (Figure 4.4) illustrates the dynamic interaction and chronological data flow between the four primary layers of the Health Hub architecture: Presentation Layer (React-Vite UI), Business Logic Layer (FastAPI & Local ML Orchestration), Data Access Layer (Firebase Firestore NoSQL Store), and the External Services Layer (Groq Cloud Gateway API). It provides a detailed breakdown of how various system components collaborate to execute core functional scenarios such

as local ML-driven health risk screening and clinical biomarker interpretation while maintaining strict logical separation and data integrity.

The interaction begins when the Presentation Layer initiates a secure asynchronous request containing a valid Firebase Auth ID token header. The FastAPI Orchestration Layer then manages the transaction handshake between internal clinical engines and the data layers. For instance, in a diet plan generation scenario, the backend first retrieves authenticated user metrics from the Firestore user document, feeds them directly into the locally deployed XGBoost and Scikit-Learn binaries for rapid health risk categorization, and then orchestrates a structured payload call with the Llama 3.3 model via the Groq API. Once the response is validated against target Pydantic schemas, it is committed back to the Firestore data collection and served down to the user interface. This sequence ensures that high-stakes medical computations are performed in a completely non-blocking, asynchronous manner, providing a high-performance experience while ensuring that every step of the data transformation is securely logged.

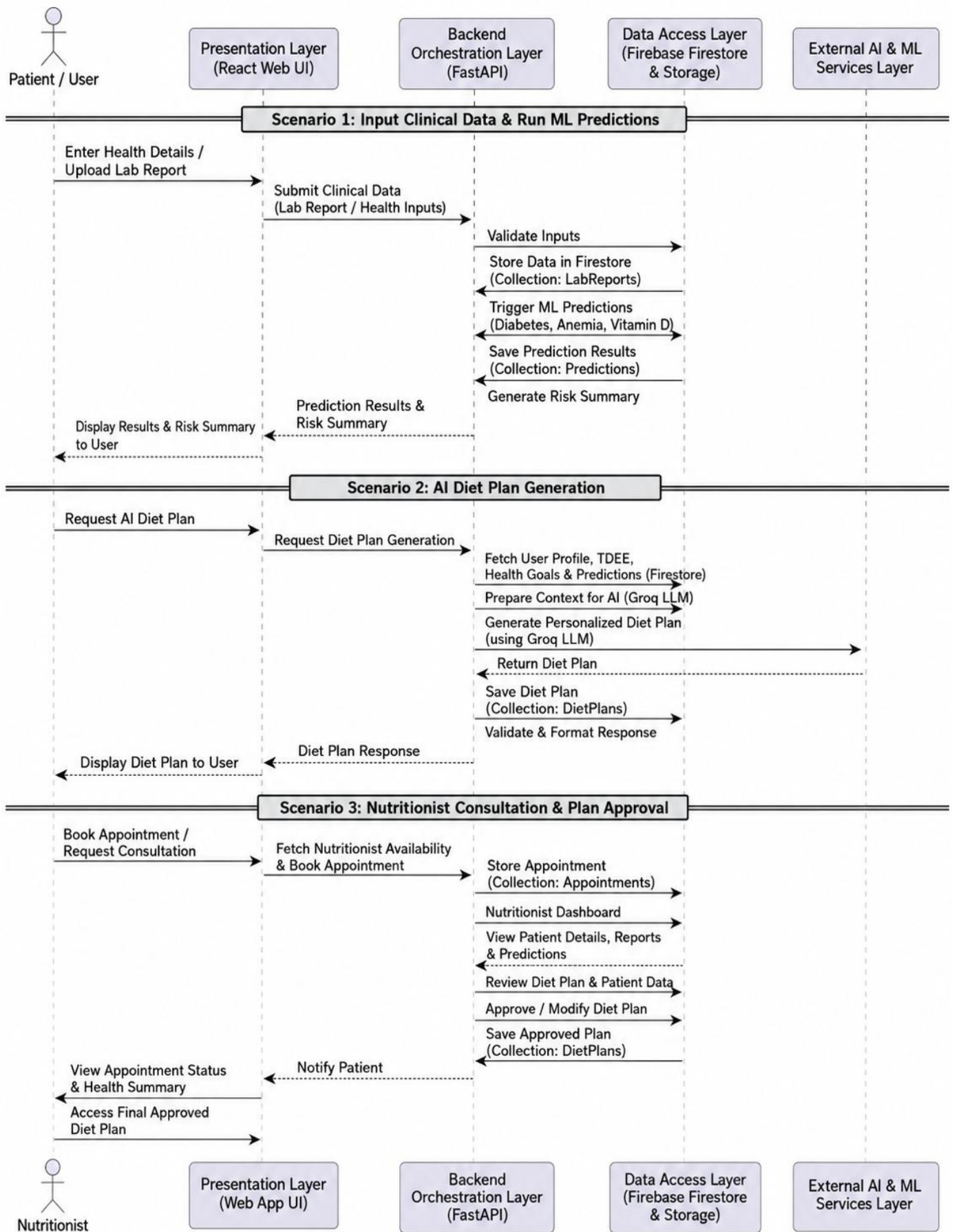


Figure 4.4: Layered sequence diagram

4.2.4.2 Activity Diagram (Overall System Workflow)

The Activity Diagram (Figure 4.5) illustrates the sequential and concurrent activities that form the overall operational workflow of the Health Hub system. It provides a visual guide on how various users including Patients and Nutritionists interact with both the automated Machine Learning logic and the professional consultation modules. The workflow initiates with the patient's secure authentication via Firebase Authentication, followed by the dynamic backend calculation of BMR and TDEE metrics based on the established health profile document fields.

The path then focuses on the critical Clinical Engine process: the automated upload and parsing of biomarker data. A decision diamond in the workflow handles data validation at the FastAPI schema level, ensuring that extracted clinical markers (e.g., Fasting Blood Glucose, HbA1c, Hemoglobin) fall within realistic numerical ranges before they are committed to the Firestore document store. Following successful storage, the workflow transitions to the ML Orchestration Layer, which executes the local multi-track XGBoost and Scikit-Learn pipelines to screen for physiological conditions (Anemia, Vitamin D, and Type 2 Diabetes risks).

A concurrent branch in the diagram represents the Human-in-the-Loop safety mechanism. Nutritionists, upon logging into their dedicated dashboard, can access assigned patient profiles to review both the rule-based clinical interpretations and the local ML-predicted risk vectors. The diagram includes a decision node where the professional can either "Approve" or "Modify" the generated diet plan array. Only after this professional validation does the processed information flow back to the patient's dashboard for visualization. Finally, the diagram illustrates the patient engaging in longitudinal tracking logging daily calorie adherence data and interacting via the real-time chat module effectively closing the continuous health improvement loop.

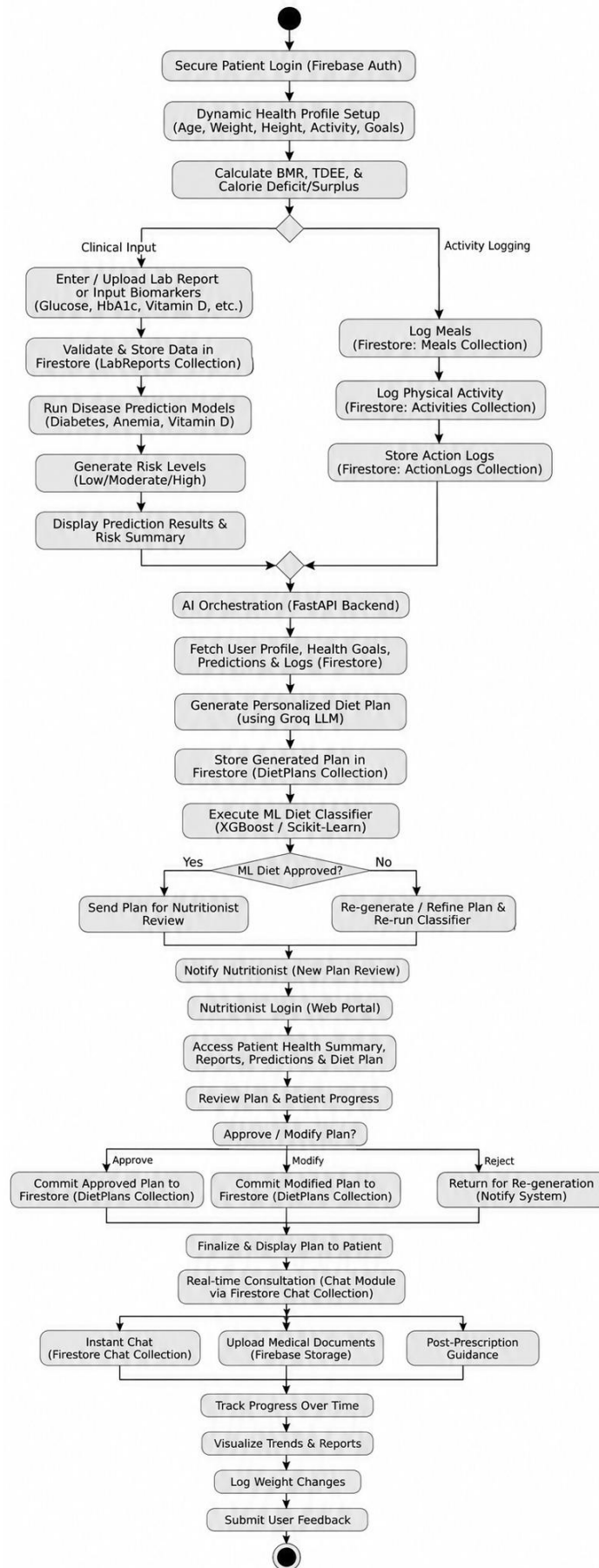


Figure 4.5: Vertical patient-journey activity diagram

4.2.5 Entity Descriptions and Constraints

Each entity within the Health Hub NoSQL database structure serves a specific purpose, meticulously designed to support high-stakes, AI-driven nutritional guidance and maintain clinical accuracy.

- **User (Patient) Document:** Stores core physiological metrics (age, weight, height, gender) and calculated baseline numbers such as BMI, BMR, and TDEE inside a dedicated nested profile map. This document encapsulates a sequential history log of physical changes to facilitate long-term longitudinal tracking.
- **Nutritionist Document:** Maintains professional profiles, including verified medical license IDs, clinical specializations, professional experience years, and active consultation availability status flags. It is linked to active patients through unique string reference pointer arrays.
- **Clinical Biomarkers Sub-collection:** Acts as the digital repository for structured medical markers. It contains verified document fields for glucose levels, lipid profiles, and hemoglobin counts extracted via Regex, with each document containing a risk-level flag generated by the automated validation engine.
- **ML-Diet Plan Collection:** Records the native ML risk classifications alongside the Llama 3.3-generated meal plan arrays. It stores caloric targets, macro-nutrient breakdowns, and Halal-compliance flags, linked directly via metadata pointers to the specific biomarker snapshot used during its compilation.
- **Consultation & Audit Documents:** Manage global appointments, real-time message text arrays, and plan approval statuses, ensuring every professional interaction is natively documented.

4.2.6 Data Constraints and Validation Rules

Each entity includes specific relational constraints such as Primary Keys (UUIDs), Foreign Key relationships, and NOT NULL constraints on critical clinical fields. Business logic validation rules are strictly enforced at the FastAPI level; for instance, a diet plan generation request is rejected if current weight metrics or interpreted biomarkers are missing.

Role-based access control (RBAC) is implemented via Clerk JWT scopes, ensuring that patients can only access their personal health records, while nutritionists are restricted to viewing data only for patients with active, authorized consultation IDs. Data integrity is maintained through ACID-compliant transactions in PostgreSQL, ensuring that AI-generated plans and clinical summaries are

fully validated and committed without risk of partial data entry. Security is further reinforced through database-level encryption, comprehensive audit logs for every data modification, and secure API endpoints that prevent unauthorized SQL injection or data exposure.

4.3 Process Flow / Representation

The Process Flow section explains how key operations are carried out within the Health Hub system. Each workflow is described through structured explanations to help readers understand the triggers, specific steps, and outcomes involved in the major system processes, ensuring a seamless integration between Machine Learning automation and human professional interaction.

4.3.1 Login and Authentication Process

The authentication process begins when a user enters their credentials through the Health Hub web interface. The system, utilizing Firebase Authentication, receives the input and verifies it against secure cloud identity records. Upon successful matching, the FastAPI backend evaluates the user's specific role (Patient, Nutritionist, or Admin) linked to their unique Firebase Auth UID (`auth_uid`) within the Firebase Firestore NoSQL document store to load the appropriate dashboard. A secure session token is generated, granting role-based access to sensitive clinical metrics and local ML tools. In case of incorrect credentials, the system enforces security protocols to prevent unauthorized access, ensuring the safety of private medical data.

4.3.2 Clinical Biomarker Interpretation Process

This process is triggered when a patient uploads a laboratory report PDF containing their physical medical metrics. The FastAPI Clinical Engine receives this file stream, utilizes a `pdfplumber` execution loop, and applies Regular Expression (Regex) pattern algorithms to parse out key values (e.g., HbA1c, Fasting Blood Glucose, Hemoglobin levels). Once these values are validated against structural data constraints, the system stores these markers as highly structured entries in a nested `clinical_biomarkers` sub-collection path within Firebase Firestore and triggers the automated interpretation logic. This logic generates a simplified risk summary (Low, Medium, or High risk), which is then displayed to the user, ensuring their health profile is always grounded in the latest medical data.

4.3.3 ML-Driven Diet Generation Process

The diet generation process is the core intelligence workflow, initiated when a user requests a personalized nutritional plan. The FastAPI Orchestration Layer retrieves the user's physical metrics

(BMR, TDEE) and the most recent clinical biomarkers from the Firestore document collections. This context is immediately fed into the locally deployed native Machine Learning estimators specifically standalone XGBoost boosters and optimized Scikit-Learn pipelines to execute multi-track physiological risk screening (Anemia, Vitamin D, and Type 2 Diabetes risk profiles). Subsequently, the system merges this identified clinical context with verified nutritional standards and sends a structured prompt to the Llama 3.3 model via the Groq Cloud API. The generated iterative, calorie-precise meal plan array is then stored in a "Pending Approval" state flag attribute within the Firestore database, awaiting professional validation.

4.3.4 Nutritionist Consultation and "Human-in-the-Loop" Workflow

This workflow bridges the gap between automated insights and human expertise. It is triggered when a nutritionist accesses their dedicated dashboard to review pending plans. The system facilitates a "Human-in-the-Loop" validation step where the professional reviews the ML-generated diet plan array alongside the patient's clinical history and risk factors. Nutritionists can modify caloric targets or specific food recommendations. Once they "Approve" the plan, the document state flag is finalized in the Firestore collection and reflected on the patient's interface in real-time. Secure, real-time messaging driven by a live Firestore real-time snapshot subscription stream allows for continuous follow-up and guidance between both actors.

4.3.5 Progress Tracking & Health Analytics

The tracking process starts when a user logs daily physical metrics, such as weight or calorie adherence. The system automatically recalculates the BMI and progress towards established TDEE goals using standard clinical formulas. These updates are recorded as transaction-safe, time-stamped log entries in the Firebase Firestore health tracking collections and reflected through interactive Recharts on the dashboard. This continuous feedback loop provides longitudinal visualization of the user's health journey, encouraging long-term adherence to the nutritional plan.

4.4 Design Models

Design models describe the structural and behavioral aspects of the Health Hub system through conceptual representations. Each model is thoroughly explained to reflect the intelligent system behavior, module interaction, and architectural decisions made to support Machine Learning-driven healthcare.

4.4.1 Class Model Description

The class model defines the main structural elements of Health Hub by outlining the central data schemas and their object dependencies. Document schemas such as User (Patient), Nutritionist, Clinical Biomarkers, ML-Diet Plan, and Appointment contain relevant attributes and operations that describe their behavior. The Patient schema maintains personal metrics (Age, Weight, Height) and calculates BMR/TDEE targets, while the Nutritionist schema stores professional credentials and verification status. The Biomarkers sub-collection is specifically designed to store numerical clinical data (e.g., HbA1c, Fasting Blood Glucose, Hemoglobin), which act as feature inputs for the local Machine Learning risk screening pipelines. These object representations demonstrate how components interact and how clinically grounded data is exchanged between the FastAPI logic and the Firebase Firestore NoSQL document store.

4.4.2 Sequence Model Description

The sequence model illustrates the chronological communication between system components during vital operations. For instance, during the ML Diet Generation process, the sequence begins with the React-Vite UI sending an asynchronous request containing a verified Firebase Authentication token. The FastAPI Orchestration Layer retrieves the medical context from the Firestore document store and coordinates natively with the locally deployed machine learning binary files (.ubj and .pkl) for physiological risk classification and the Groq LLM API for Halal-compliant recipe generation. After the system validates the response JSON against target Pydantic schemas, it commits the plan to the document store and notifies the user via real-time snapshot listeners. This sequence highlights the order of interactions and clarifies how messages flow securely between the web client and the high-performance backend services.

4.4.3 Activity Model Description

The activity model explains the operational workflows within the system by describing the actions, decisions, and outcomes that occur during various processes. In the Clinical Analysis activity, the user uploads their lab report PDF, and the system triggers an automated pdfplumber parsing routine and regular expression extraction loop. A decision diamond evaluates if the data falls within safe physiological ranges; if so, the markers are committed to the Firestore database, otherwise, the user is prompted for manual correction. This representation demonstrates how Health Hub handles high-stakes scenarios, such as Multi-Track Machine Learning Risk Screening and the Human-in-the-loop approval process, ensuring that all steps follow predefined clinical protocols before a diet plan is

finalized.

4.4.4 Component Model Description

The component model describes how the major modules of Health Hub are organized and interconnected. Components such as the Firebase Authentication Service, FastAPI Orchestration Layer, Clinical Engine, and Real-time Chat Processor operate independently while communicating through secure RESTful APIs and live collection sync pipelines. External services, including the Groq Cloud Gateway (Llama 3.3/Qwen), are seamlessly integrated into the ecosystem, while the localized machine learning estimators run natively within the backend deployment environment. This modularity allows individual components, especially the Machine Learning pipelines, to be updated or scaled without affecting the entire system's stability, ensuring a maintainable and high-performance healthcare platform.

Chapter 5

IMPLEMENTATION

5. Implementation

The Implementation chapter explains how the Health Hub system was transformed from a conceptual design into a functional, high-performance software solution at the technical level. This chapter focuses on the practical execution of the project's requirements, detailing the selection of the modern technology stack, the deployment of intelligent algorithms, and the configuration of cloud-native services. While it does not include extensive raw source code, it provides a comprehensive description of the localized MLOps pipeline, the orchestration of external LLM APIs via the Groq gateway, and the user interface decisions made using the React-Vite framework.

The goal of this chapter is to demonstrate how the system's modular architecture was implemented to ensure clinical accuracy in biomarker interpretation and efficiency in personalized diet generation. It highlights the technical steps taken to maintain data security through Firebase Authentication and the optimization of localized machine learning pipelines for high-accuracy health risk screening. By presenting the implementation journey, this chapter confirms that the final software product is a reliable, evidence-based health management platform that satisfies all predefined functional and non-functional goals through a robust integration of FastAPI and the Firebase Firestore NoSQL document store.

5.1 Technology Stack Selection

The implementation phase began with selecting tools that prioritize processing speed, horizontal scalability, and flexible data payloads.

- **Frontend:** React 18 with Vite for a lightning-fast user interface and Tailwind CSS for responsive design workflows.
- **Backend:** FastAPI (Python 3.13) for its asynchronous capabilities, allowing the system to handle multiple machine learning model requests concurrently.
- **Database:** Firebase Firestore NoSQL Document Database was implemented to manage flexible, low-latency collection relationships between users, biomarker sub-collections, and approved diet plans.
- **Machine Learning:** Serialized local model binaries specifically native XGBoost boosters (.ubj) and an optimized Scikit-Learn pipeline (.pkl) handle localized health risk screening, while the Groq Cloud Gateway serves Llama 3.3 models with ultra-low latency.

5.2 AI & ML Pipeline Implementation

A core part of the implementation was the orchestration of multiple AI models. The system was configured to first run an automated PDF parsing pipeline utilizing a pdfplumber loop and regular expressions to extract text and pre-fill biomarker properties into Firestore. Once saved, the system executes a localized, multi-track predictive ML Layer (XGBoost and Scikit-Learn) to screen for conditions such as Anemia, Vitamin D deficiency, and Type 2 Diabetes risks. Finally, the Generative AI Layer utilizes prompt engineering to compile the identified clinical risk factors and produce structured, Pakistani Halal-compliant meal plans in JSON format, ensuring easy parsing and visualization on the frontend.

5.3 Security & Data Governance

Implementation of data privacy involved setting up Firebase Authentication for secure user sessions and configuring server-side data validation schemas. Access control was strictly enforced at the database level using custom Firestore Security Rules (firestore.rules), ensuring that sensitive medical data remains isolated between patients and their explicitly assigned nutritionists.

5.4 Algorithms

This section describes the main algorithms used in the Health Hub system. Each algorithm represents the internal logic behind system features and explains how different functions operate. These descriptions are presented in natural language steps to clearly communicate the workflow behind major modules, particularly focusing on the integration of Machine Learning and structured clinical data processing.

5.4.1 Secure Authentication & RBAC Algorithm

The authentication algorithm ensures that only legitimate users can access the system. It protects sensitive health metrics by verifying credentials via Firebase Authentication and granting Role-Based Access Control (RBAC).

Steps:

- User enters credentials through the React-Vite web interface.
- The system transmits credentials to Firebase Authentication for JSON Web Token (JWT) verification.

- If credentials match:
 - Generate a secure session token with specific user scopes mapped to their unique account ID (auth_uid).
 - FastAPI backend retrieves the user role (Patient/Nutritionist/Admin) from the corresponding Firestore document.
 - Redirect the user to their specific dashboard with authorized API access.

If credentials do not match:

- Display an error message and enforce a rate-limiting delay for security.

5.4.2 Clinical Biomarker Interpretation Algorithm

This algorithm handles the transformation of unstructured medical report files into structured clinical risk assessments.

Steps:

- Patient uploads their laboratory report PDF file through the interface.
- FastAPI invokes a pdfplumber parsing execution loop and applies Regular Expression (Regex) pattern algorithms to extract key numerical values (e.g., HbA1c, Fasting Blood Glucose, Hemoglobin).
- FastAPI validates that the extracted inputs are within realistic physiological numerical ranges using structural validation schemas.
- Trigger the rule-based clinical validation logic to compare values against medical thresholds.
- If validation is successful:
 - Map values to the user's clinical_biomarkers sub-collection path in Firebase Firestore.
 - Categorize risk levels (Low/Medium/High) based on the target medical thresholds.
 - Notify the user and trigger a potential "Diet Plan Refresh" prompt.
- If validation fails:

- Prompt the user to re-verify or manually correct the numerical entries from their physical report.

5.4.3 ML-Driven Diet Generation Algorithm

This is the core "Intelligence" algorithm that utilizes localized multi-track machine learning screening models and Generative AI to create personalized plans.

Steps:

- User triggers the "Generate ML Diet Plan" request through the app dashboard.
- Context Retrieval: Fetch physical metrics (age, weight, height, calculated TDEE) and recent parsed biomarkers from the user's Firestore document path.
- Classification Phase: Execute the localized native ML estimators specifically standalone XGBoost boosters (anemia_xgb.ubj, vitd_xgb.ubj) and an optimized Scikit-Learn pipeline (diabetes_pipeline_v2.pkl) to screen for physiological conditions and predict required nutritional constraints.
- Augmentation Phase: Merge the predicted risk categories and the patient's physical context into a structured system prompt template.
- Generation Phase: Send the prompt payload to Llama 3.3 via the Groq Cloud Gateway API for iterative, calorie-precise meal plan generation.
- Validation Phase: Parse the generative JSON response against backend validation schemas, cross-check for Halal and cultural compliance, and store as a "Pending Approval" plan in Firestore for nutritionist review.

5.4.4 Health Progress & BMI Tracking Algorithm

This algorithm manages the continuous monitoring of the user's physical status and visualizes their journey.

Steps:

- User enters their latest weight and height metrics.
- The system executes the BMI formula:

$$BMI = \frac{weight(kg)}{height(m)^2}$$

- Compare the resulting numerical value against the World Health Organization (WHO) standard health categories.
- Update the Firestore tracking collection with a new timestamped log record.
- Render a visualized trend chart dynamically using the frontend Recharts library.
- If the calculated BMI category shifts significantly, trigger an automated notification flag for a nutritionist review.

5.5 External APIs

The following table describes the external Application Programming Interfaces (APIs) and integrated service frameworks utilized within Health Hub to facilitate intelligent data processing, local machine learning execution, and secure cloud management.

Table 5.1: Details of APIs used in the project

Name of API / Service	Description	Purpose of Usage	Function / Class Name
Groq Cloud AI API	High-speed, low-latency Large Language Model inference infrastructure.	To generate iterative, calorie-precise, and culturally tailored Pakistani Halal-compliant meal plan arrays using Llama 3.3.	DietOrchestrator, generate_meal_plan()
Local ML Native Inference	Serialized native binary machine learning models executing within the FastAPI environment .	To execute localized multi-track predictive risk screenings for Anemia (anemia_xgb.ubj), Vitamin D (vitd_xgb.ubj), and Type 2 Diabetes (diabetes_pipeline_v2.pkl).	MLService, predict_health_risks()
Firebase Auth API	Cloud-native identity and session management infrastructure.	To manage secure user sign-up, absolute JWT login verification, and role-based routing control across dashboards.	AuthMiddleware, verify_session()
Firebase Firestore NoSQL	Real-time, scalable cloud-native NoSQL document database (replacing	To store multi-track clinical biomarkers, health tracking metrics, app	FirestoreManager, commit_atomic_write()

API	PostgreSQL/SQLAlchemy).	configurations, and historical chat logs with strong transaction-safe execution.	
Firebase Firestore Sync API	Real-time document-level snapshot streaming pipeline.	To facilitate immediate, bidirectional text synchronization for the patient-nutritionist professional chat module.	ChatService, listen_for_messages()
Vercel / Google Cloud Platform	Managed cloud hosting and modular deployment infrastructure.	To host the compiled React frontend application and the high-performance containerized FastAPI backend for global availability.	DeploymentConfig, main_entry_point

5.6 User Interface

The User Interface (UI) of Health Hub is designed with a focus on minimalism, clinical clarity, and intuitive navigation. Since the platform manages complex medical data and AI-generated insights, the design philosophy prioritizes a "Mobile-First" approach using the web framework. The interface utilizes a clean color palette, predominantly medical blue and white, to instill trust and reduce cognitive load for users who may be stressed about their health results.

Key UI decisions were made to ensure that the application is accessible to users of all ages. Large, legible typography and high-contrast action buttons are used throughout the app to facilitate ease of use. The dashboard is structured hierarchically, placing the most critical information, such as the BMI Progress Tracker and Latest AI-Diet Plan, at the center of the user's view. Furthermore, the integration of real-time loops and progress visualizations ensures that complex clinical markers extracted from lab reports are presented in an easy-to-understand, graphical format. This strategic design approach transforms data-heavy health management into a simplified, engaging, and user-centric experience.

Figure 5.1 presents the User Dashboard interface of the Health Hub platform, designed to provide patients with a comprehensive overview of their health status. The interface utilizes a structured, grid-based layout that prioritizes a minimalist and intuitive user experience. At the top of the panel, a personalized greeting dynamically welcomes the patient by name while instantly displaying high-visibility status badges, such as the current BMI classification and the actively assigned nutritional path. Directly underneath, the dashboard aggregates the user's primary physical metrics into independent, high-contrast analytics cards that track daily calories burned, body weight, calculated

BMI, and consistency streaks to encourage regular logging.

The lower portion of the screen features an interactive data visualization canvas, which graphs the user's weight trends over time to assist in long-term health monitoring. Navigation across the ecosystem is facilitated by a persistent, clean sidebar menu that provides one-click access to core application features including the Diet Plan, Recipes, NutriBot, History, Progress logs as well as the specialized, local multi-track machine learning screening portals for Diabetes, Iron Deficiency, and Vitamin D checks. The entire interface employs a unified green, blue, and white color palette to minimize cognitive strain while upholding an accessible, professional aesthetic suitable for healthcare management.

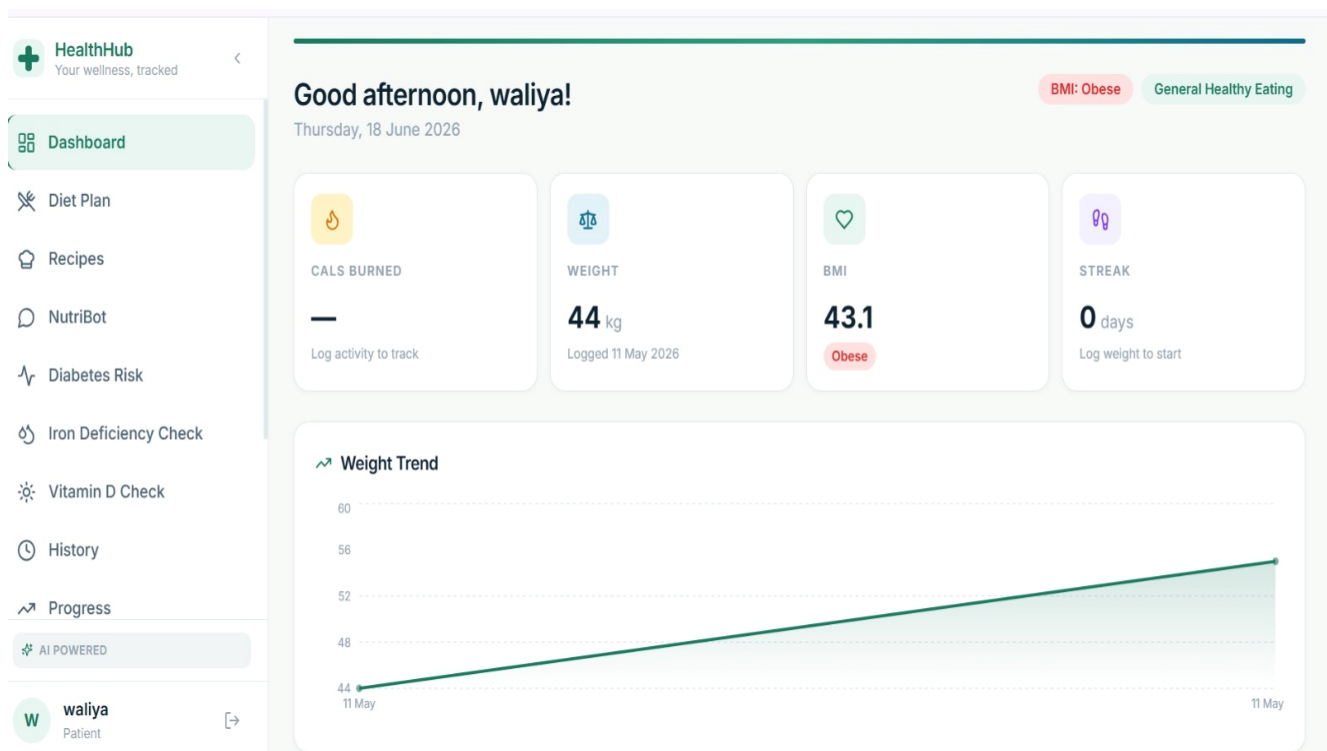


Figure 5.1:User Dashboard

Figure 5.2 presents the AI Diet Plan interface of the Health Hub platform, designed to provide patients with an automated, disease-aware personalized meal plan module. The interface is structured to guide the user seamlessly through plan generation and daily breakdown steps. At the top of the interface, under the main header, the user is presented with horizontal selection buttons to define the "Plan Duration," offering predefined options for 7 Days, 15 Days, or 30 Days. Directly beneath these duration options, the system explicitly highlights the current diagnostic context derived from the local machine learning risk pipelines displaying dynamic, high-visibility status tags such as "Diabetes High" and "Iron Deficiency High" notifying the patient exactly how their plan is being personalized. A prominent, high-contrast "Generate AI Plan" action button triggers the backend orchestration logic

to compile or update the plan payload.

The lower sections of the interface display the active structured meal plan details. A dedicated panel summarizes the current "7-Day AI Plan" parameters, incorporating localized data badges for target daily caloric intake (e.g., ~1330 kcal/day), the primary classification category (e.g., General Healthy Eating), and the timestamp of creation. To ensure general wellness guidelines are maintained alongside specific macro targets, a secondary blue-toned "Hydration Plan" notification box provides baseline daily water consumption goals. Finally, a collapsible accordion log system initializes at the bottom (starting with Day 1 showing a targeted 1310 kcal limit), allowing patients to expand daily views to review calorie-precise, culturally tailored recipe distributions.

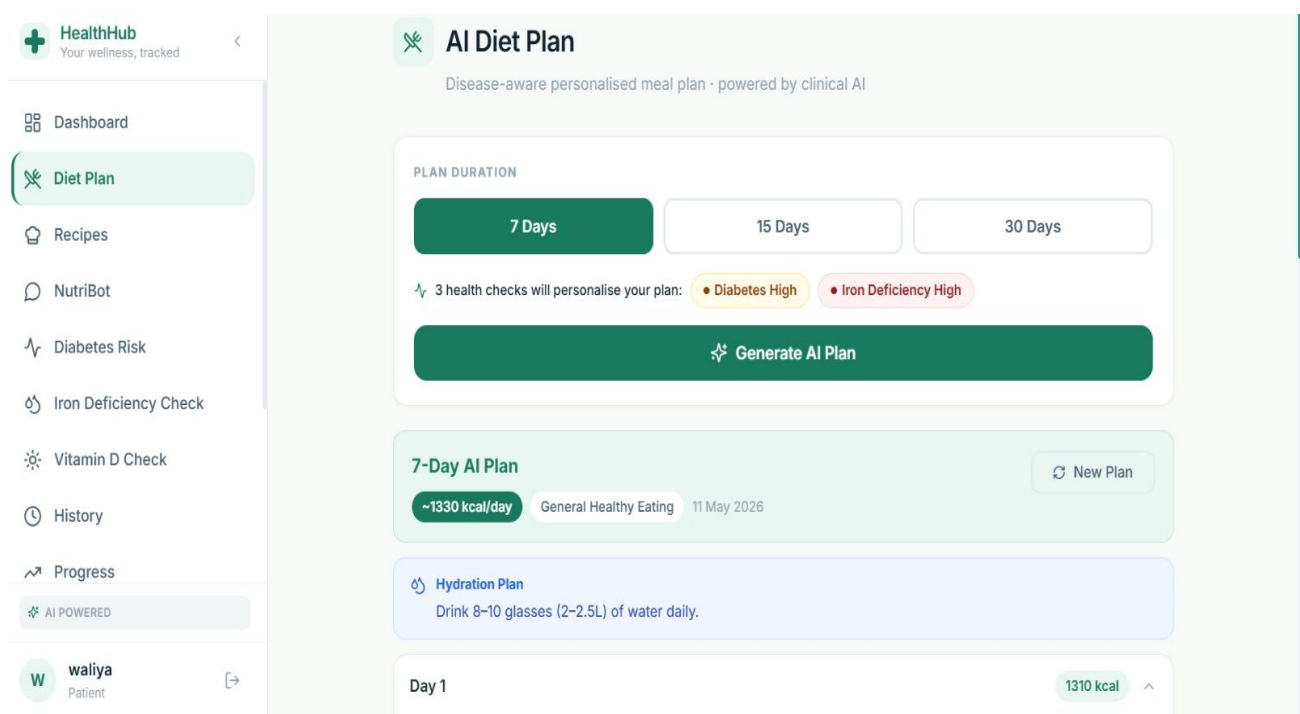


Figure 5.2: Diet Plan

Figure 5.3 presents the Recipes interface of the Health Hub platform, designed to provide patients with an interactive portal for disease-aware, personalized Pakistani recipe recommendations. The screen is structured around a central generation control card that allows users to seamlessly toggle between different generation modes. Under the "Generation Mode" header, the UI features two distinct selection boxes: a "Health Recommended" mode where the system pre-selects dietary assets based on the user's localized medical risk conditions, and a "By Ingredients" mode which allows users to select items they currently have available in their kitchen.

Directly below the mode selection, a status alert box communicates the underlying multi-track medical screening activity, indicating that health checks have been detected and that the AI will

automatically personalize the output recipes. A prominent, high-contrast "Generate Health Recipes" call-to-action button is positioned at the bottom of the card to trigger the backend orchestration loop and fetch tailored culinary options from the Firestore database. This dashboard layout maintains visual consistency with the rest of the system, matching the minimalist design language, explicit structural text alignment, and green-accented color palette seen throughout the platform.

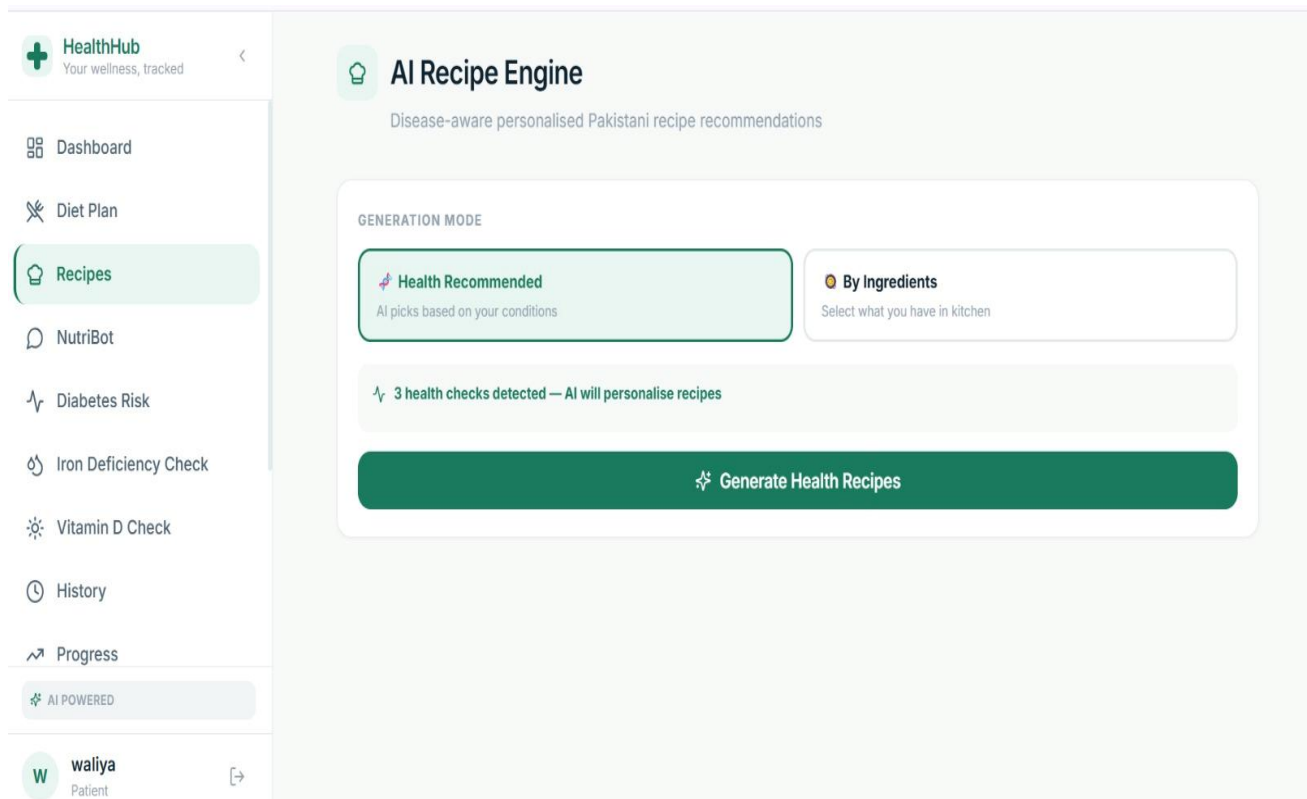


Figure 5.3: Recipes

Figure 5.4 presents the NutriBot interface of the Health Hub platform, designed to provide patients with an interactive, AI-powered conversational nutrition assistant. The interface is structured as a modern messaging workspace with a focus on real-time dialogue and clinical accessibility. At the top of the chat area, a dedicated header bar displays the assistant's name, "NutriBot," alongside an active "Online" status indicator and a utility button to clear the conversation history. The central chat canvas sequences messages chronologically using distinct, high-contrast text bubbles accompanied by specific time stamps; user inquiries appear wrapped in green bubbles on the right, while the AI's tailored responses are housed in white bubbles on the left, each paired with their respective profile avatars. At the bottom of the interface, a persistent input field with a clear placeholder text ("Ask about food, diet, nutrition...") and an adjacent send action button enables users to seamlessly type and dispatch their dietary queries. The conversational text illustrates the bot's specialized capability to deliver culturally contextualized, Pakistani-centric nutritional advice while maintaining the platform's

clean, minimalist aesthetic and green-accented visual theme.

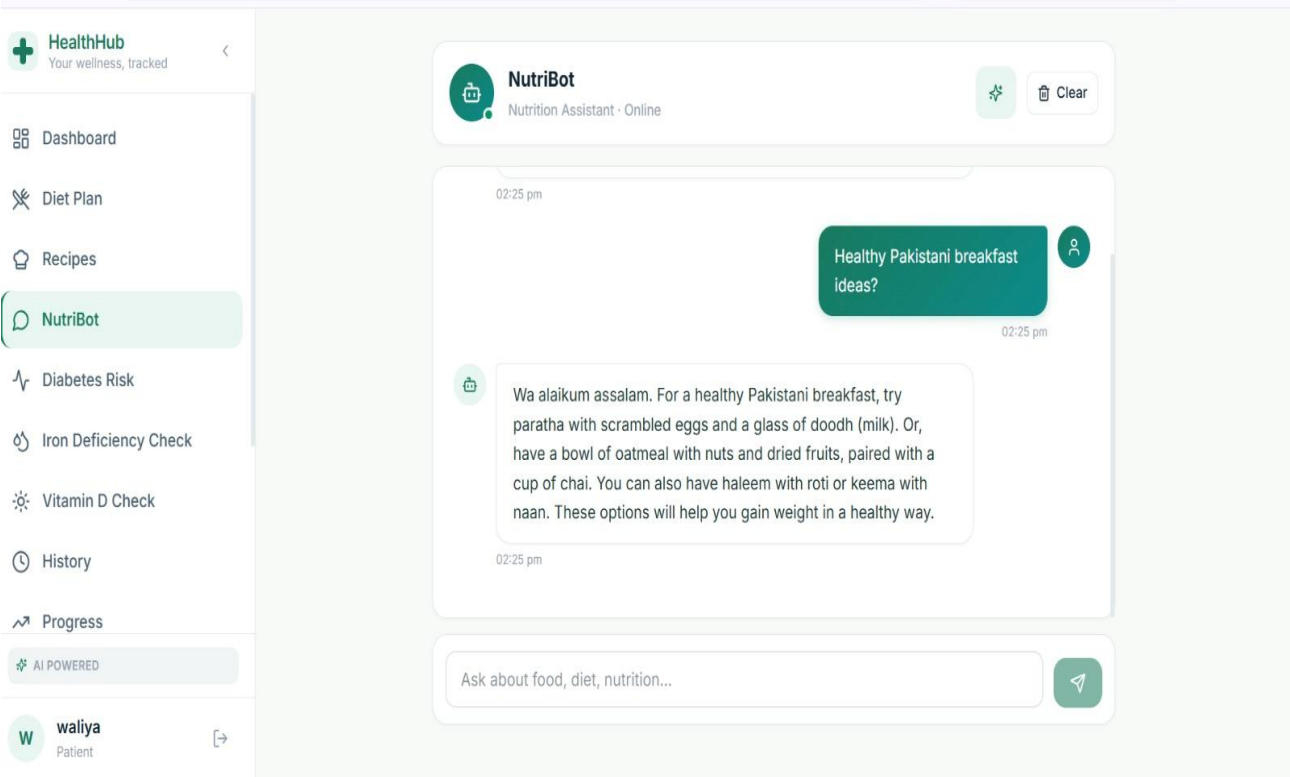


Figure 5.4: NutriBot

Figure 5.5 presents the Diabetes Risk Assessment interface of the Health Hub platform, designed to provide patients with an interactive portal to screen for type 2 diabetes risk metrics. The screen is structured around a centralized medical intake form backed by the system's machine learning infrastructure, explicitly highlighted under the main header as a "Stacking Ensemble Model" achieving an AUC of 0.83. At the top of the workspace, the interface displays personal physiological data cards for "Age" and "BMI (from profile)" that are dynamically pre-filled using the patient's existing health metrics to reduce manual entry load.

The central segments of the form require the patient to input specific clinical and familial parameters necessary for the underlying predictive logic. A dedicated numerical input field for "Fasting Blood Glucose" features descriptive placeholder styling alongside an informative subtext note indicating the normal physiological range of 70–99 mg/dL. Below this, the interface introduces high-visibility, binary action selection boxes for documenting a "Family History of Diabetes," followed by segmented horizontal button groups to log the patient's current "Physical Activity Level" across predefined categories ranging from Sedentary to Active. The entire assessment module maintains structural alignment with the platform's minimalist design philosophy, utilizing a persistent sidebar menu, clear typography, and a cohesive green-and-white visual aesthetic.

HealthHub
Your wellness, tracked

Dashboard
Diet Plan
Recipes
NutriBot
Diabetes Risk
Iron Deficiency Check
Vitamin D Check
History
Progress
AI POWERED

waliya
Patient

Diabetes Risk Assessment

Stacking Ensemble Model · AUC: 0.83

AGE: 22 yrs

BMI (from profile): 43.1

FASTING BLOOD GLUCOSE *
e.g. 95 mg/dL
Normal fasting range: 70–99 mg/dL

FAMILY HISTORY OF DIABETES? *
Yes No

PHYSICAL ACTIVITY LEVEL
Sedentary Light Moderate Active

Figure 5.5: Diabetes Risk Assessment

Figure 5.6 presents the Iron Deficiency Check interface of the Health Hub platform, designed to provide patients with an interactive portal to evaluate their risk for Iron Deficiency Anemia using a specialized XGBoost machine learning model. At the top of the interface, a primary header block incorporates an auto-filled user profile panel displaying core biological context parameters including Age, Gender, and BMI. These values are automatically populated from the user's main application profile to ensure seamless data integration, with a clean helper prompt instructing the patient on how to modify them if necessary.

The middle tier of the interface contains a dedicated clinical numerical entry field for logging the patient's "Hemoglobin Level" in g/dL. This critical field features explicit medical reference guidelines positioned directly beneath the input box, informing users of the normal standard baseline ranges for both women (12–16 g/dL) and men (13.5–17.5 g/dL). Below the clinical marker entry, the screen presents a segmented "Dietary Iron Intake" selection module categorized into Low, Moderate, and High tiers relative to the Recommended Dietary Allowance (RDA), supported by localized, contextual subtext prompts covering typical regional dietary staples such as meat, lentils (daal), spinach, and eggs. The entire interface maintains visual and structural harmony with the rest of the application modules through its minimalist layout, clear data hierarchy, and the persistent green-accented navigation sidebar.

Figure 5.6: Iron Deficiency Check

Figure 5.7 presents the Vitamin D Check interface of the Health Hub platform, designed to provide patients with an interactive portal to evaluate their vitamin deficiency risks. The module is engineered around a hybrid analytical approach, explicitly highlighted in the header as a deficiency assessment combining standard WHO guidelines with a localized XGBoost machine learning model. At the top of the form, an auto-filled user profile block dynamically pre-populates the patient's current Age and BMI data from the database to ensure rapid execution and minimize manual typing errors.

The primary input section features an optional data field for adding "Serum Vitamin D" levels in ng/mL, allowing users to supply direct values from a recent 25-OH Vitamin D blood test. A dynamic, purple-toned information badge directly beneath the input field acts as a safety routing indicator, clarifying that if the serum input is left blank, the system's backend will seamlessly fallback to using the alternative ML symptom model to predict risk based entirely on the user's logged symptoms and physical profile. Below this medical section, the interface displays a horizontal selector segment for capturing "Daily Sun Exposure" tiers classified into Low, Moderate, and High options. This interface maintains design continuity with the platform's core aesthetic through clean card constraints, legible typography, and a persistent application sidebar for straightforward navigation.

HealthHub
Your wellness, tracked

- Dashboard
- Diet Plan
- Recipes
- NutriBot
- Diabetes Risk
- Iron Deficiency Check
- Vitamin D Check**
- History
- Progress
- AI POWERED

waliya
Patient

Vitamin D Check

Deficiency Assessment · WHO Guidelines + XGBoost Model

YOUR PROFILE (auto-filled)

AGE	BMI
22 yrs	43.1

SERUM VITAMIN D (optional) Enables WHO path

e.g. 18.5 ng/mL

From your blood test (25-OH Vitamin D). Leave blank to use AI symptom model.

No serum value → ML model will predict based on your symptoms and profile.

DAILY SUN EXPOSURE

Low	Moderate	High

Figure 5.7: Vitamin D Check

Figure 5.8 presents the My History interface of the Health Hub platform, designed to provide patients with a comprehensive repository to review past logs. Under a clean module subtitle, the screen introduces a horizontal tab navigation row that segments past historical records into distinct categories: Appointments, Chat, Diet Plans, Health Checks, and Recipes. Each tab features a dynamic numerical indicator badge that explicitly shows the total number of logged transactions stored for that specific domain within the Firebase Firestore database.

The active workspace displays the list container for the selected category. In this specific view, an individual record card summarizes a consultant meeting, showcasing details such as the practitioner's name, a clear calendar date and time stamp, and a "Completed" transaction status badge. To facilitate direct user interaction and downstream workflows, each history record card embeds explicit action elements, allowing patients to trigger a modal to view deeper details or initiate a secure chat session stream. The screen fully aligns with the application's clean design philosophy, matching the persistent green-accented layout, crisp typography hierarchy, and minimalist aesthetic used throughout the system.

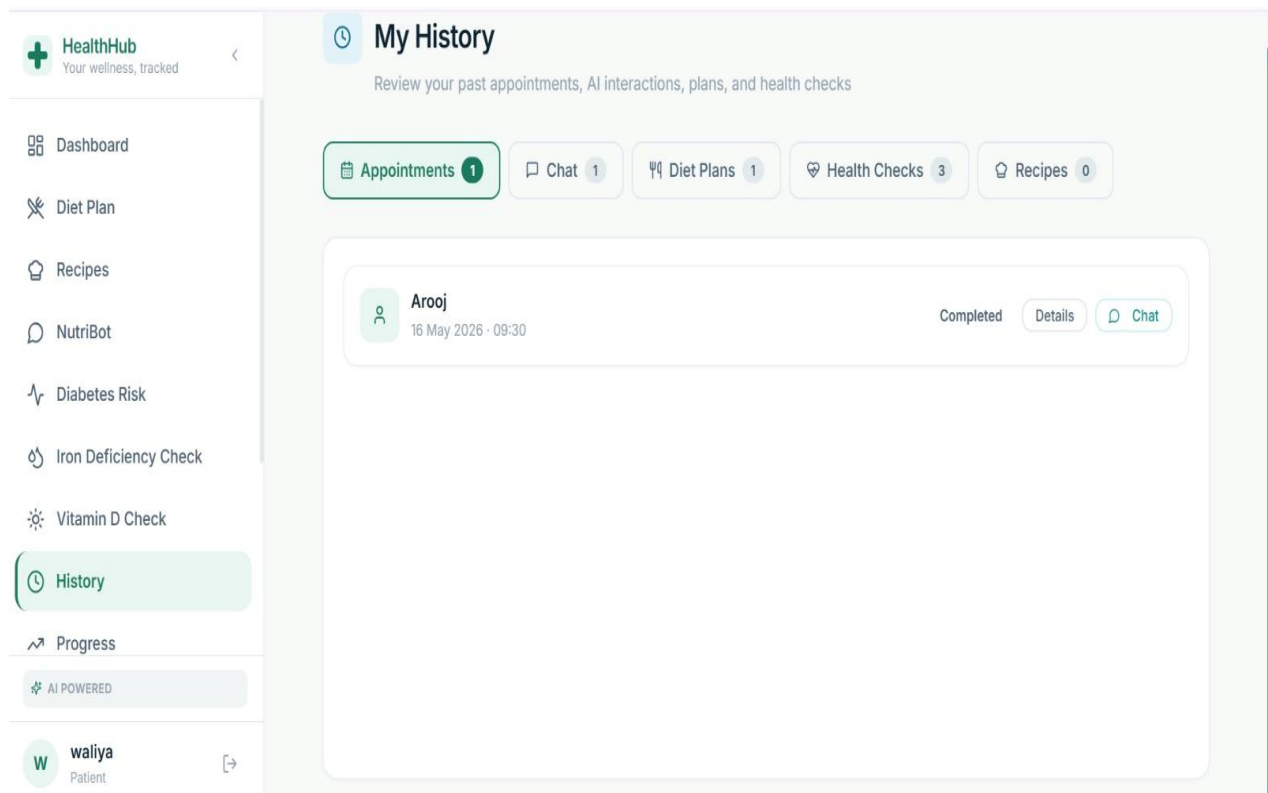


Figure 5.8: User History

Figure 5.9 presents the Progress interface of the Health Hub platform, designed to provide patients with an interactive dashboard to log and monitor their physical metrics over time. At the top of the interface, a sub-navigation bar allows users to toggle between two core entry metrics: Weight and Activity. The active workspace displays a "Log Weight" card component containing structured input fields, including an automated calendar date selector pre-filled with the current date, a numerical weight container mapping entries in kilograms, and an optional text block for adding descriptive personal notes. A high-contrast "+ Log Weight" action button commits the input parameters to the data store.

The lower section of the interface embeds a detailed data visualization panel titled "Weight Trend," which uses the Recharts library to render a dynamic line chart tracking long-term physical developments. The graphical area features a clear horizontal dashed reference line indicating the patient's specific health target threshold, allowing users to contrast their current trajectory against established wellness goals at a glance. The dashboard layout preserves the platform's standard design language, utilizing a persistent application sidebar for straightforward navigation, crisp contextual typography, and a cohesive green-and-white visual theme.

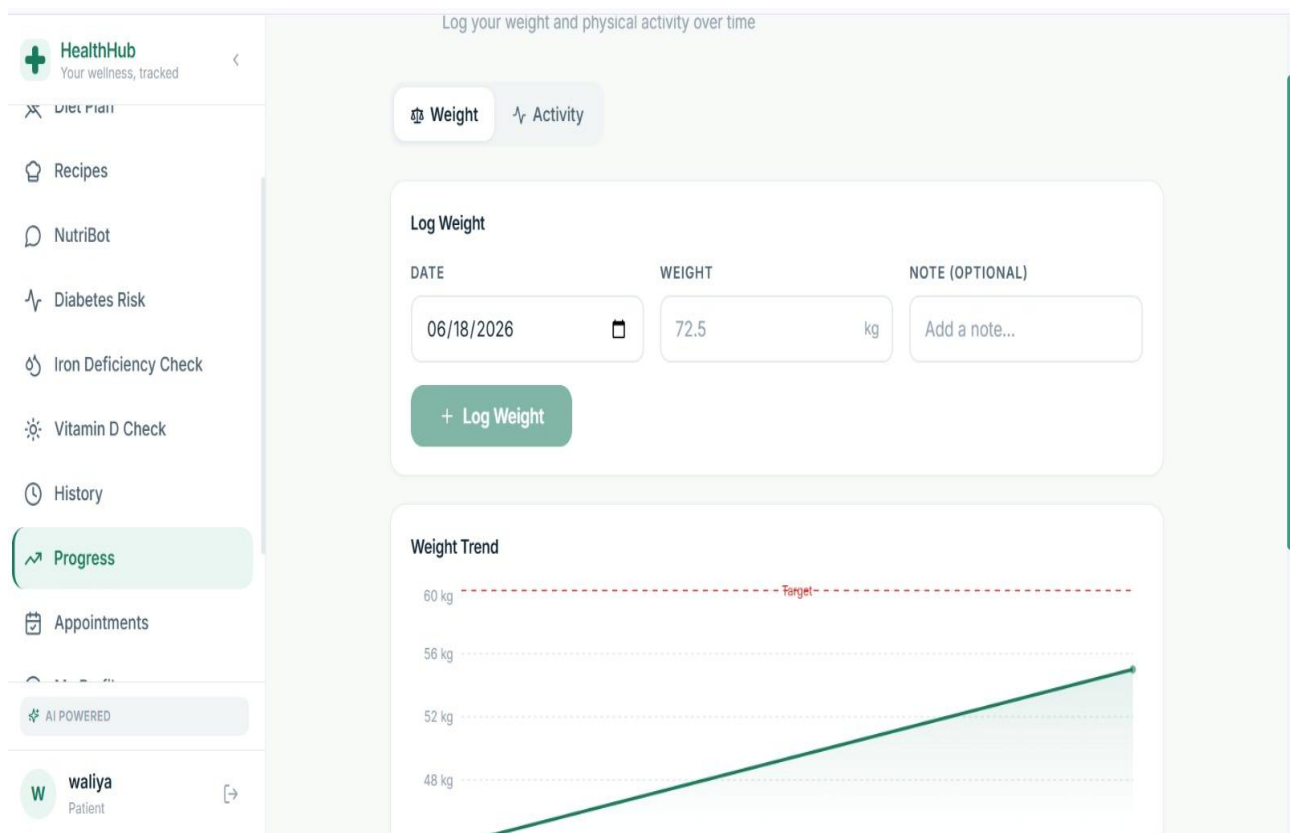


Figure 5.9: Progress Tracking

Figure 5.10 presents the Book an Appointment interface of the Health Hub platform, designed to provide patients with a direct gateway to browse verified nutritionists and book consultation sessions. Under a clean header block, the interface features a persistent, full-width search input field with the placeholder text "Search by name or specialization..." to allow real-time filter queries. Below the search filter, the UI organizes available practitioner profiles into a clean, horizontal card layout. Each profile card highlights key professional credentials, including a green verification badge for vetted experts, specialized diagnostic tags (e.g., Weight Management, Diabetes, Hypertension), consultation fee structures in local currency (PKR), and active weekly scheduling availability. High-contrast "Book Now" call-to-action buttons are embedded within each practitioner card to trigger booking transactions.

The lower section of the interface incorporates a separate "My Appointments" panel that acts as a localized transaction log for the user's booking history. This log summarizes past scheduled events, displaying individual record blocks that detail the assigned practitioner's name, precise calendar timestamps, and historical booking status flags such as "Completed". The entire module maintains absolute structural and visual consistency with the platform's overarching design philosophy, utilizing clean card boundaries, clear typographic hierarchies, and a persistent, green-accented navigation

sidebar to support a seamless human-in-the-loop consultation workflow.

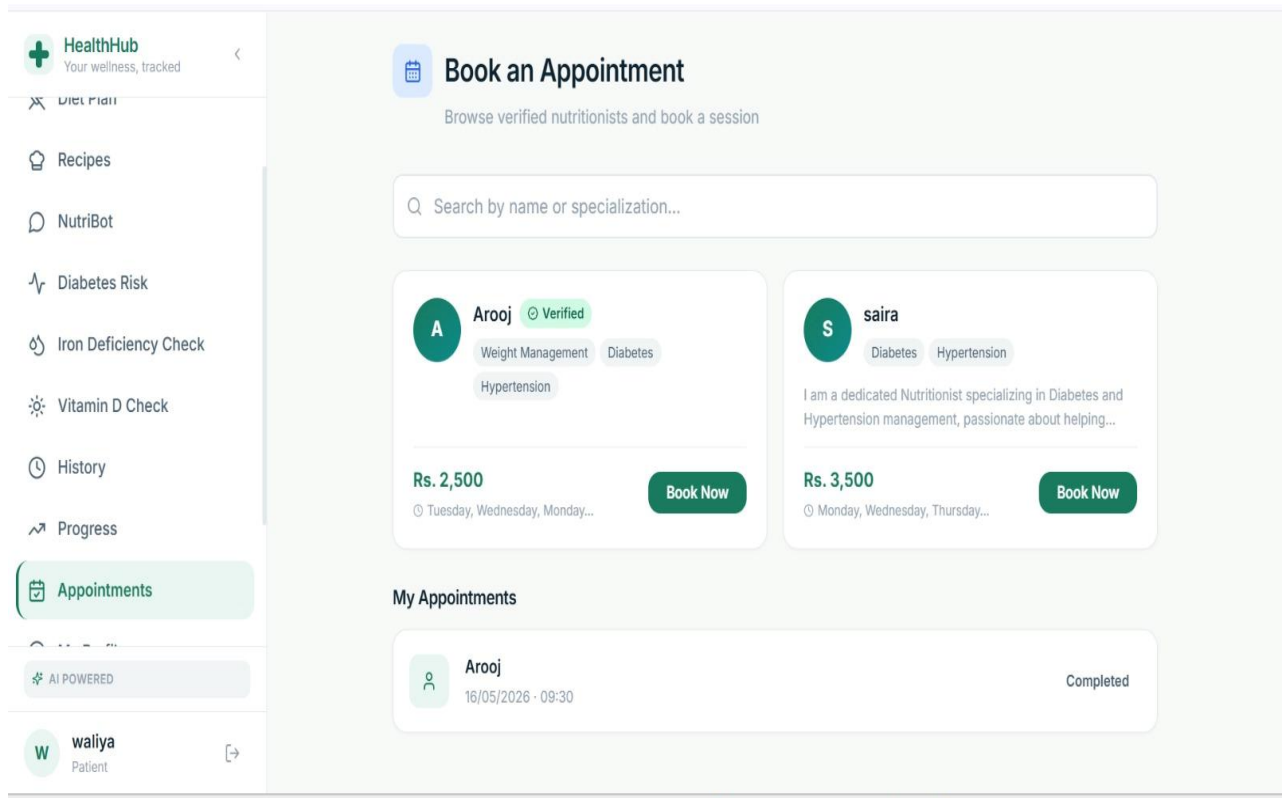


Figure 5.10: Appointment Booking

Figure 5.11 presents the My Profile interface of the Health Hub platform, designed to provide patients with a centralized configuration workspace to view and manage their personal biological data and nutritional settings. The interface utilizes a symmetric, dual-card layout that segregates core health parameters into distinct analytical domains. On the left, the "Body Metrics" card aggregates verified physiological values, including Age, Gender, Height, and Weight, while automatically displaying calculated derived outputs such as the user's specific BMI status tag (e.g., 43.1 Obese) and a precise baseline dynamic Calorie Target mapped in kilocalories per day.

To the right, the "Nutrition Settings" card tracks behavioral and medical constraints, detailing the primary wellness Goal, Diet Type preferences, financial Budget tiers, and active physical Activity Levels. At the bottom of this section, a dedicated "Medical Conditions" row showcases prominent, high-visibility tags such as Diabetes (Type 2), Hypertension, and High Cholesterol which act as system-wide feature flags to shape downstream machine learning risk screening and generative meal plan properties. Beneath these main data grids, an "Account Actions" management strip provides

clean, high-contrast action elements to seamlessly "Update Profile" or "Deactivate Account." The entire workspace maintains structural harmony with the application's minimalist framework, relying on crisp data hierarchies, sharp typography contrast, and a persistent navigation sidebar.

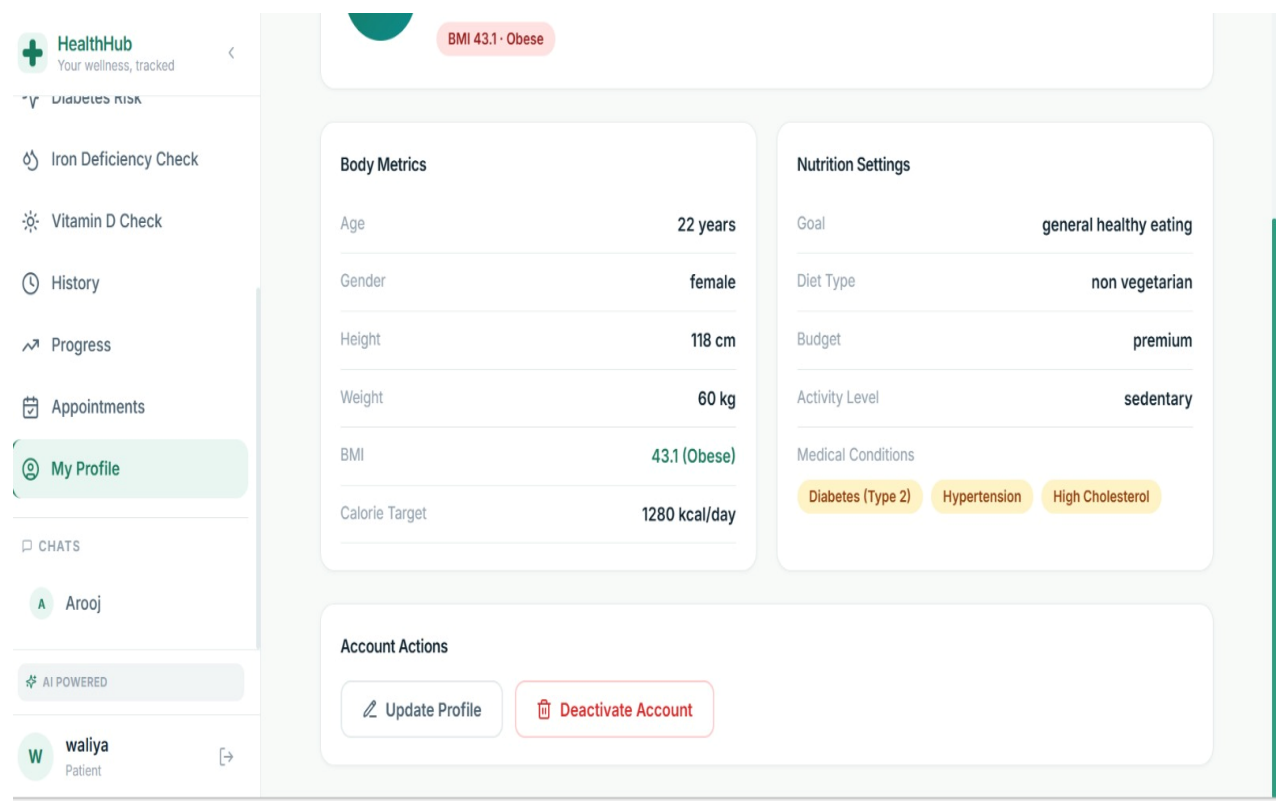


Figure 5.11: User Profile

Figure 5.12 presents the Welcome Back sign-in interface of the Health Hub platform, designed to provide users with a secure and intuitive entry point to access the system's core functionalities. The interface utilizes a distinct, split-screen layout that neatly balances platform branding with credential submission workflows. On the left, a vibrant green marketing panel reinforces the application's core identity as a "Nutrition & Wellness Platform," explicitly stating its value proposition of connecting users with certified nutritionists and generating personalized diet plans built specifically for Pakistan's rich food culture. This panel displays bulleted value markers with corresponding high-contrast icons to highlight secure professional consultations, AI-powered tracking workflows, and culturally tailored Pakistani food insights.

To the right, the interactive workspace contains a clean, centralized credential input block designed to process absolute user access. Under a prominent "Welcome back" heading, the form includes distinct text input containers for the user's "Email Address" and "Password," embedding subtle iconography (envelope and padlock symbols) alongside a visible toggle option to unmask password

entries. Directly above the password field, a "Forgot password?" hyperlink enables credential recovery routines, while a prominent, solid green "Sign In" call-to-action button is positioned at the base of the inputs to submit payload data to the Firebase Authentication pipeline. At the very bottom of the white workspace, an additional registration link ("Don't have an account? Create account") supports flexible user navigation, maintaining a clean, professional aesthetic that minimizes cognitive overhead during high-stakes account access.

HealthHub
Nutrition & Wellness Platform

Your health journey starts here

Connect with certified nutritionists and get personalised diet plans built for Pakistan's rich food culture.

- Secure, verified professional consultations
- AI-powered nutrition tracking & diet plans
- Culturally-tailored Pakistani food insights

Welcome back

Sign in to your account to continue

EMAIL ADDRESS

you@example.com

PASSWORD

Forgot password?

Sign In

Don't have an account? [Create account](#)

Figure 5.12: Welcome Back sign-in

Figure 5.13 presents the Nutritionist Dashboard interface of the Health Hub platform, designed to provide registered dietitians and medical experts with a centralized workspace to manage client consultation schedules and incoming requests. At the top of the main panel, the interface displays the nutritionist's profile name alongside a clear "Verified by PNC" medical verification badge, referencing their verified Pakistan Nursing Council (PNC) registration metrics. Below the credential header, a row of independent analytics cards aggregates key workspace metrics, providing instant counts for Upcoming Appointments, Pending Requests, and sessions Completed This Month.

The central area features a dedicated "Today's Appointments" schedule panel that dynamically

updates to list booked time slots for the day or displays a clean fallback illustration indicating that no sessions are currently scheduled. On the left side of the screen, the specialized workspace incorporates a distinct Nutritionist Portal sidebar menu, providing swift switching between the Dashboard, Appointments, and Profile pages. This sidebar also hosts a persistent "Patient Chats" messaging feed to support direct, real-time consultation communication loops with active clients.

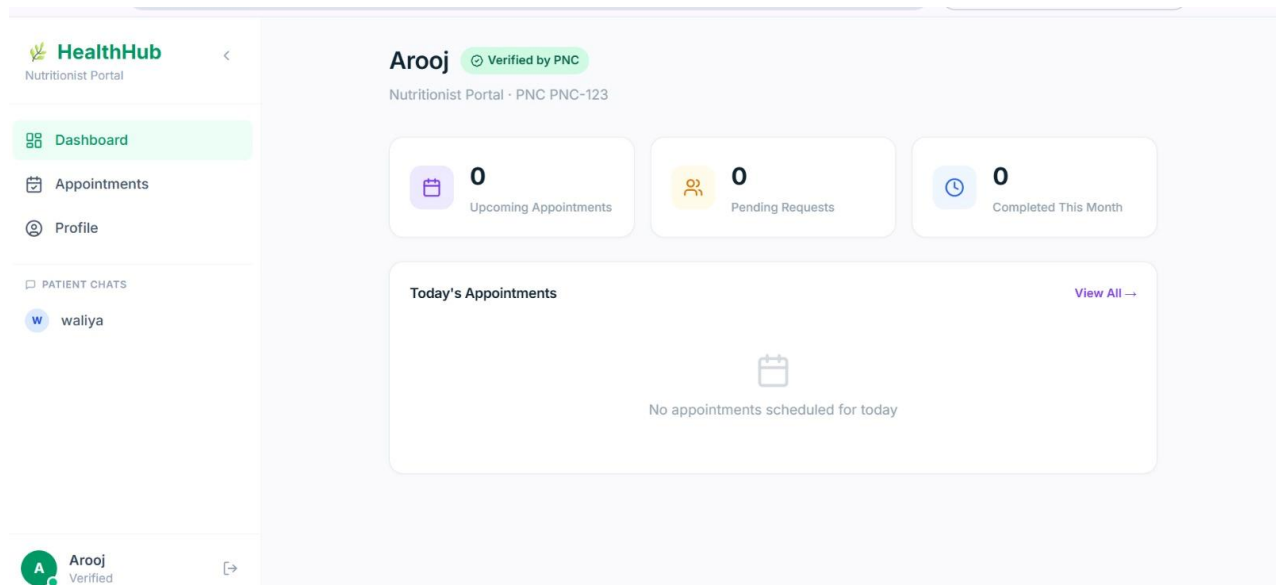


Figure 5.13: Nutritionist Dashboard

Figure 5.14 presents the Professional Profile interface of the Health Hub Nutritionist Portal, designed to provide registered dietitians with a structured management workspace to update their public clinical settings and weekly consultation availability. At the top of the interface, the "Consultation Details" card hosts secure input containers allowing professionals to establish their mandatory consultation fee in Pakistani Rupees (PKR) and verify their specific Pakistan Nursing Council (PNC) registration number.

The central section of the profile features an interactive, multi-select grid for specifying professional "Specializations." This container displays high-contrast, checked selection tags for active medical focus areas such as Weight Management, Diabetes, and Hypertension alongside unselected baseline options like General Nutrition, Child Nutrition, and Maternal Nutrition, allowing professionals to tailor their profiles to relevant patient health tracks. Directly underneath, the "Available Days" scheduler incorporates circular, high-contrast toggle chips representing the days of the week, enabling the practitioner to dynamically adjust their active booking availability window. The workspace maintains absolute visual alignment with the platform's portal guidelines, featuring a persistent sidebar menu, clear typography hierarchy, and an intuitive, scannable layout.

HealthHub
Nutritionist Portal

Professional Profile
Manage your public information and consultation availability

CONSULTATION DETAILS

CONSULTATION FEE (PKR) *
Rs 2500

PNC REGISTRATION NUMBER *
PNC-123

SPECIALIZATIONS *

☒ Weight Management ☒ Diabetes ☒ Hypertension

☐ General Nutrition ☐ Child Nutrition ☐ Maternal Nutrition

AVAILABLE DAYS *

☒ Mon ☒ Tue ☒ Wed ☒ Thu ☒ Fri ☒ Sat ☐ Sun

PATIENT CHATS

w waliya

Arooj
Verified

Figure 5.14: Nutritionist Portal

Figure 5.15 presents the Patient Chat interface within the Health Hub Nutritionist Portal, designed to facilitate a secure, bidirectional communication channel between registered practitioners and their active clients. At the top of the workspace, a dedicated header panel incorporates an intuitive "Back" navigation button alongside a stylized chat module indicator. The main chat canvas is engineered for synchronous messaging updates; when a conversation initializes or contains no historical log, it renders a clean, centered placeholder graphic and a friendly introductory prompt ("No messages yet. Say hello!") to encourage open engagement.

Directly below the message viewing canvas, a persistent input bar provides a spacious text box with explicit context prompts ("Type a message... (Enter to send)") alongside an integrated, high-contrast send action chip to submit dialogue payloads. On the left side of the screen, the interface maintains seamless platform continuity via the persistent Nutritionist Portal navigation sidebar, keeping the practitioner oriented within their active workspace while explicitly highlighting the currently selected client under the "Patient Chats" thread segment. The entire communication interface employs a minimalist layout, clear data boundaries, and a soothing green-and-white visual theme to foster an accessible digital environment for healthcare consulting.

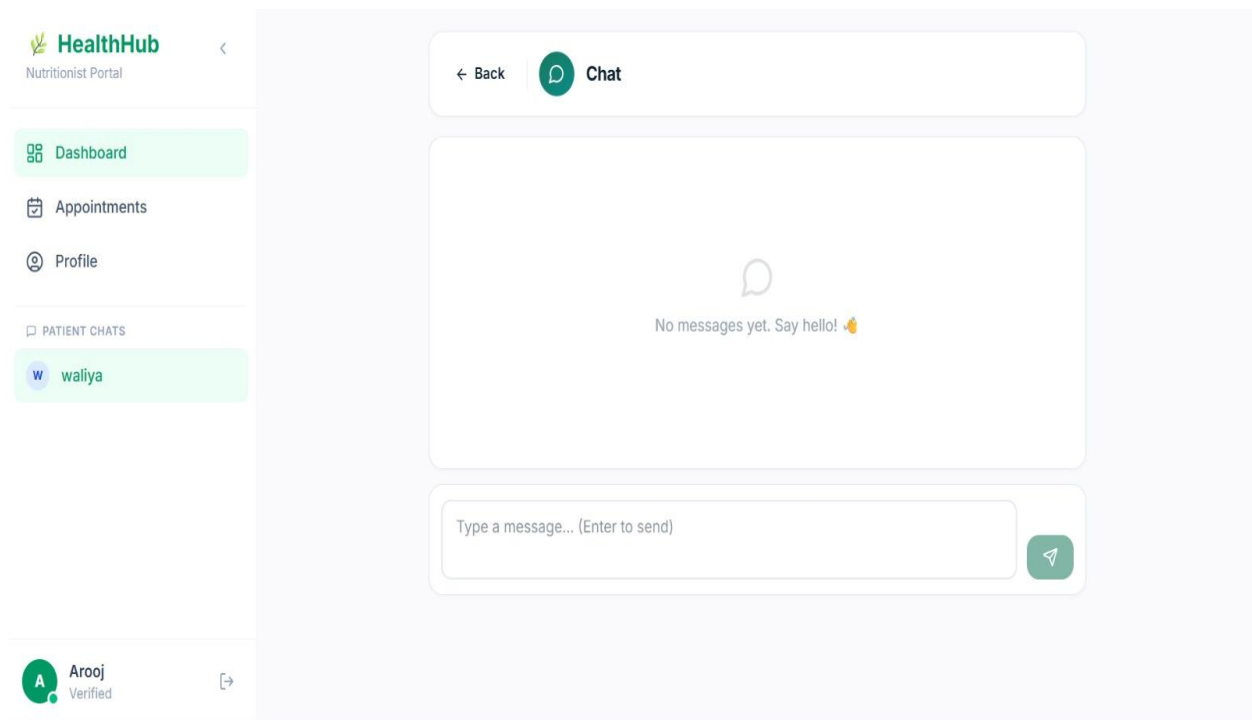


Figure 5.15: Patient Chat

Figure 5.16 presents the Admin Panel interface of the Health Hub platform, designed to provide system administrators with a comprehensive workspace to manage nutritionists and monitor platform overview metrics. At the top of the workspace, underneath a primary header icon, a horizontal segmentation row breaks backend management operations into active tab links: Overview, Pending, Verified, and PNC Verification. High-visibility orange badge indicators are embedded within the tabs to instantly call attention to actionable verification workloads, such as pending registrations and required credentials audits.

Directly below the sub-navigation row, a grid-based dashboard layout hosts independent micro-analytics cards that aggregate critical platform statistics. These elements display real-time counts tracking Total Users, active Nutritionists, metrics Pending Review, and Total Appointments processed through the infrastructure. The lower segment of the workspace displays a "Recent Appointments" container block acting as a live transaction log, which captures structural metadata such as patient names, assigned consultants, execution dates, and booking fulfillment status flags (e.g., Completed). On the left side of the screen, a dedicated "HealthHub Admin" sidebar provides centralized administrative navigation between platform metrics tracking and the "Verify Nutritionists" validation portal, rounded out by a persistent session management logout action button at the bottom.

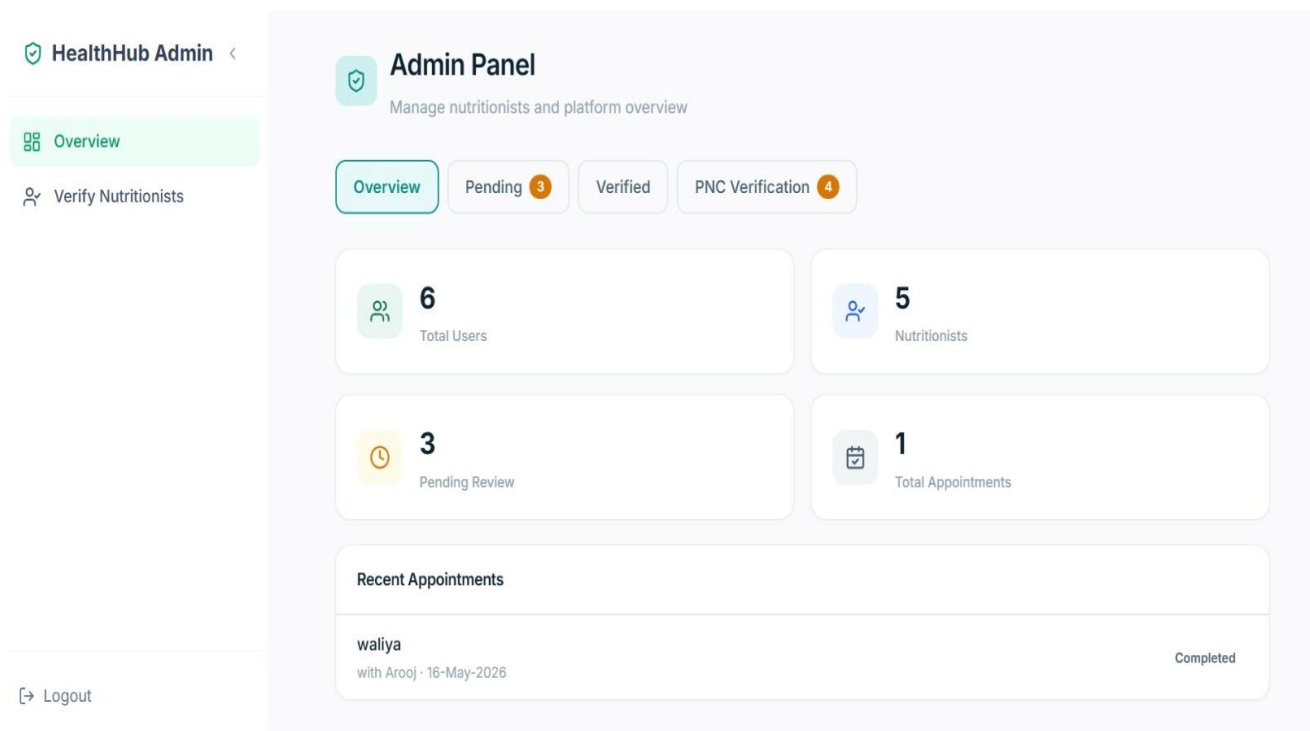


Figure 5.16: Admin Panel

Figure 5.17 presents the Pending Verification tab view within the Admin Panel of the Health Hub platform, designed to provide system administrators with a structured data grid to manage and audit new practitioner registration requests. At the top of the workspace, the horizontal sub-navigation row highlights the active "Pending" view status with a prominent numeric indicator badge showing three actionable requests. Below this dashboard header, a comprehensive verification table systematically arrays essential onboarding data across dedicated rows, mapping out critical registration parameters including Name, PNC Registration Number, Email address, registration date stamps, and a high-visibility, orange-highlighted "Pending" status flag.

To facilitate efficient, administrative decision-making workflows directly from the data grid, the "Actions" column embeds high-contrast, independent administrative interactive controllers, enabling the system administrator to instantly "Verify" or "Reject" the respective applicant profile payload. On the left side of the screen, the interface maintains overall design continuity via the persistent "HealthHub Admin" sidebar navigation structure, keeping the active view highlighted under the "Verify Nutritionists" portal section to reduce cognitive overhead during high-stakes credential checking.

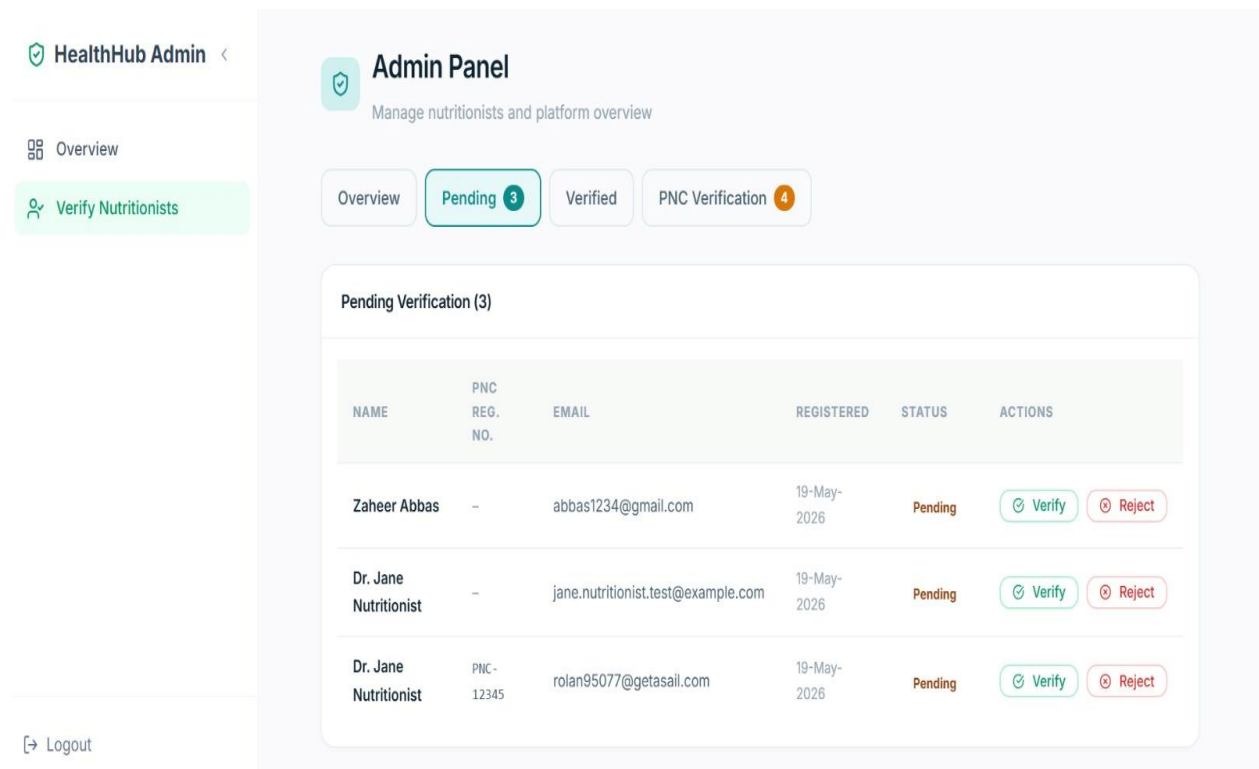


Figure 5.17: Pending Verification tab

Figure 5.18 presents the Verified tab view within the Admin Panel of the Health Hub platform, designed to provide system administrators with a structured data grid to monitor and manage approved healthcare professionals. At the top of the workspace, the horizontal sub-navigation row highlights the active "Verified" status tab, displaying the system overview alongside nearby actionable badges for pending queues. Below this dashboard header, a comprehensive "Verified Nutritionists" table systematically catalogs active practitioners who have successfully completed authentication and credential checking. The tabular view maps essential professional parameters across dedicated columns, including Name, PNC Registration Number, Email address, registration date stamps, and a high-visibility, green-highlighted "Verified" status flag.

To facilitate efficient system moderation and ongoing compliance audits directly from the data grid, the "Actions" column embeds an independent administrative interactive controller, enabling the system administrator to instantly "Revoke" practitioner credentials and alter their access tier if needed. On the left side of the screen, the interface maintains absolute design continuity via the persistent "HealthHub Admin" sidebar navigation structure, keeping the active view highlighted under the "Verify Nutritionists" portal section to ensure a clean, minimalist administrative workflow.

HealthHub Admin <

Overview

Verify Nutritionists

Logout

Admin Panel

Manage nutritionists and platform overview

Overview

Pending 3

Verified

PNC Verification 4

Verified Nutritionists (2)

NAME	PNC REG. NO.	EMAIL	REGISTERED	STATUS	ACTIONS
Arooj	PNC-123	aroojahmad35@gmail.com	04-May-2026	Verified	Revoke
saira	PNC-2026-45871	sairahuzai9@gmail.com	07-May-2026	Verified	Revoke

Figure 5.18: Verified tab

Chapter 6

Testing and Evaluation

6. Testing and Evaluation

This chapter details the rigorous testing and evaluation procedures conducted to ensure the reliability, accuracy, and security of the Health Hub platform. Given that the system provides nutritional advice based on clinical biomarkers and Machine Learning predictions, the testing phase focuses on both the functional performance of the web application and the clinical validity of the ML-generated diet plans. The goal is to verify that all system components, from the backend orchestration to the cloud database storage, operate seamlessly under various conditions while maintaining high data integrity and clinical safety.

6.1 Manual Testing

Manual testing is a process where a tester interacts directly with the Health Hub platform to verify that all AI-driven and manual functionalities are working correctly. Unlike automated testing, manual testing relies on human observation to identify nuanced errors in ML classifications, usability issues in the web interface, and unexpected behavior in the generative AI diet generation engine. It is a critical step in ensuring that the system meets its clinical and functional requirements while providing a smooth user experience. Manual testing allows testers to simulate real-world usage scenarios, such as entering edge-case biomarker values (e.g., extremely high glucose), making it easier to detect hidden logic errors that automated scripts might miss.

Manual testing in the Health Hub system is carried out in multiple stages, including unit testing, functional testing, integration testing, and system testing before deployment. This ensures that the ML modules work correctly in isolation and in combination with the backend services and cloud database.

6.1.1 System Testing

System testing is a high-level evaluation conducted after the complete development of the Health Hub platform. Its primary objective is to verify that the integrated system, from the client-side frontend to the hosted ML models, functions as intended and meets all specified medical and nutritional requirements. System testing focuses on the application as a whole, allowing the detection of logic inconsistencies in the diet generation workflow that might not be evident during the development of individual modules.

6.1.1.1 Purpose of System Testing

The purpose of system testing is to confirm that Health Hub meets functional requirements (like

TDEE calculation), performs reliably under variable network conditions, and is ready for clinical oversight. It helps in detecting hidden errors in the data flow, ensuring that the patient's clinical biomarkers are correctly retrieved from the cloud database and processed by the advanced machine learning models (such as the Stacking Ensemble and XGBoost classifiers). System testing also validates that the overall platform is stable, secure through robust authentication protocols, and provides a seamless experience for both patients and nutritionists.

6.1.1.2 Types of System Testing

To comprehensively evaluate the software ecosystem, several testing types and levels are incorporated into the verification lifecycle:

- **Unit Testing:** Focuses on individual logic components, such as the BMI formula, the tailored recipe filters, or the BMR calculation logic, ensuring they function correctly in isolation.
- **Functional Testing:** Evaluates core application features like Manual Biomarker Entry, ML-Diet Request generation, and the Real-time Chat workspace to ensure they perform according to design specifications.
- **Integration Testing:** Checks that multiple core modules, such as the application orchestrator and the cloud database, work together correctly, verifying the seamless flow of clinical markers between user profiles and the ML models.
- **Regression Testing:** Ensures that new updates to the LLM system prompts or underlying machine learning pipelines do not negatively impact existing functionalities, such as historical progress tracking.

6.1.1.3 Importance of Manual System Testing

Manual system testing is performed carefully to validate the complex AI-human interaction within Health Hub. It ensures that the system is reliable, secure, and user-friendly before it reaches the end-user. By manually verifying the accuracy of ML-predicted diet types against the patient's biomarkers, potential clinical errors or "hallucinations" are detected early. This human-centric approach improves the quality and reliability of the medical insights provided, giving confidence that the "Human-in-the-Loop" approval workflow functions effectively in real-world healthcare scenarios.

6.1.2 Unit Testing

Unit testing focuses on verifying the individual modules of the Health Hub platform to ensure that

each part functions correctly in isolation. It is the first step in our manual testing phase and forms the critical foundation for subsequent functional and integration tests.

Unit tests involve providing specific input values to a module such as a manual clinical marker or a Firebase authentication token and comparing the actual output with the expected result. Any discrepancies found during this stage, such as an error in an XGBoost risk screening prediction or a failed database transaction, are noted and fixed before moving on to higher-level system tests.

Table 6.1: Unit Testing Table

Test ID	Module Name	Test Case/Script	Description	Input Data	Expected Result	Actual Result	Status
UT01	Auth Module	User Login	Verify secure access via Firebase Auth.	Email: user@healthhub.com, Password: Password123	Successfully authenticate and redirect to Dashboard.	Authenticated via Firebase successfully.	Pass
UT02	Profile Module	BMI/BMR Calculation	Verify the accuracy of clinical formulas.	Weight: 60kg, Height: 118cm	BMI: 43.1, BMR & Calorie Target calculated.	Calculated 43.1 (Obese) correctly.	Pass
UT03	Clinical Engine	Marker Validation	Test rule-based validation for biomarkers.	Input: Fasting Glucose: 140 mg/dL	Flag as "High Risk" (Diabetic range).	Correctly flagged risk level.	Pass
UT04	ML Predictor	Risk Classification	Verify Stacking Ensemble / XGBoost multi-track classification.	Context: High Glucose, BMI: 43.1, Age: 22	Predict "Diabetes High" and "Iron Deficiency High" Risk.	Predicted correct clinical risk categories.	Pass
UT05	Generative AI	Diet Generation	Verify Llama 3.3 JSON arrays via Groq.	Goal: General Healthy Eating, Preference: Halal	Generate valid JSON with calorie-precise meal plan arrays.	Generated valid Halal-compliant JSON.	Pass
UT06	Database Layer	Firestore CRUD	Verify atomic transaction-safe operations for health records.	New Record: Weight 60kg	Data committed to Firestore with dynamic timestamp.	Record saved with transactional safety.	Pass
UT07	Analytics Module	Progress Rendering	Verify Recharts visualization update.	Data Point: [Date, Weight: 60]	Chart renders new data point in the weight trend line.	Visualized trend canvas successfully.	Pass

6.1.3 Functional Testing

Functional testing is conducted after unit testing and focuses on verifying that the Health Hub system's features work according to specified requirements. The main purpose of functional testing is to ensure that each high-level function performs correctly in real-world scenarios and meets both clinical design and user expectations through integrated module interaction.

A primary example of functional testing in Health Hub is the role-based redirection and end-to-end machine learning workflows, as described in Table 6.2. This test ensures that after a successful Firebase login, the system accurately redirects users to their specialized interfaces, such as the AI-powered Patient Dashboard or the verified Nutritionist Portal. Functional testing helps identify critical issues related to automated medical workflows, role-based data permissions (RBAC), and the overall reliability of the human-in-the-loop validation process.

Table 6.2: Functional Testing

Test ID	Module Name	Test Case/Script	Attribute & Value	Expected Result	Actual Result	Status
FT01	Auth & Role	Multi-Role Access	Patient vs. Nutritionist login via Firebase Auth	Correct role mapping to specific user/practitioner dashboards.	System correctly routed user based on Firestore role field.	Pass
FT02	Clinical Engine	End-to-End Biomarker Analysis	Input: Fasting Blood Glucose: 140, Hemoglobin: 11.5	Rule-based interpretation identifies high-risk paths and updates status.	Clinical status updated and stored instantly in Firestore.	Pass
FT03	ML Orchestration	Screening Classification	Trigger: Stacking Ensemble & XGBoost Local Models	Correct prediction of risk indicators (e.g., Diabetes High, Iron Deficiency High).	Local native models predicted clinically appropriate risks.	Pass
FT04	Diet Engine	Iterative Plan Generation	LLM Prompt via Groq Cloud / Llama 3.3	A Halal-compliant, structured, calorie-precise meal plan is built for 7 days.	Received valid array of nutritional distributions (~1330 kcal).	Pass
FT05	Validation Loop	Human-in-the-Loop Oversight	Nutritionist modifies and updates active dashboard plan	AI-generated plan is verified; configuration status shifts to custom approved.	Plan finalized with professional specialist oversight.	Pass
FT06	Health Analytics	Longitudinal Tracking	Update logged weight data	Recharts trend canvas updates	Visualization reflects weight	Pass

			from 60kg to 58kg	dynamically showing physical development.	trend metrics accurately.	
FT07	Consultation	Real-time Professional Sync	Firestore snapshot real-time text synchronization listener	Message sent by Nutritionist is reflected instantly on Patient UI chat panel.	Bidirectional text synchronization occurred with zero manual refresh.	Pass

6.1.4 Integration Testing

Integration testing ensures that multiple modules of the Health Hub platform work together correctly and that data flows seamlessly between them. This type of testing is conducted after unit and functional testing, focusing on verifying the interactions between complex components to identify errors that might occur when individual modules are combined.

For Health Hub, integration testing is vital because it validates the connection between the frontend client environment, the cloud-based Firebase infrastructure, local native ML inference pipes, and external AI services. Common integration scenarios include updating a biomarker metric and verifying that the database refreshes the risk badges, or triggering the diet plan engine and ensuring it is correctly stored in Firestore and displayed on the patient's dashboard. This ensures that the combined features operate as intended to provide a unified user experience.

Table 6.3: Integration Testing

Test ID	Modules Involved	Test Case/Test Script	Attribute & Value	Expected Result	Actual Result	Status
IT01	Auth & Dashboard	Login & Load Profile Data	Email: patient@test.com, PWD: ****	Login success, authenticated UID fetched, and Patient Dashboard initialized.	Dashboard canvas loaded with correct profile dataset.	Pass
IT02	Biomarker Form & Database	Update Metrics & Verify Sync	Input: Hemoglobin: 11.5 g/dL	Data updates in Firestore document and refreshes risk assessment flags.	Biomarker metrics updated in database and reflected on frontend.	Pass
IT03	Local ML & Groq AI	Model Context & LLM Pipeline	Flags: Diabetes High, Iron Deficiency High	LLM receives verified diagnostic context array and returns	Generative AI built precise plan matching backend	Pass

				tailored diet properties.	ML risk results.	
IT04	Diet Module & View	Plan Save & Layout Feed	Plan Duration: 7 Days, Target: 1330 kcal	Plan payload is committed to database and unlocks the accordion view layout.	Accordion cards populated with generated calorie records.	Pass
IT05	Consultation & Chat	Patient to Nutritionist Chat Sync	Message: "Review my biomarkers"	Message stored in Firestore 'Chats' collection and streams to Nutritionist feed.	Message synced across both UI panels seamlessly.	Pass
IT06	Tracking Form & Chart	Metric Update & Graphic Sync	Input: Weight 60kg, Date: 18/06	Firestore collection updates and Recharts module re-renders line plot.	Trend line re-rendered with new data point in real-time.	Pass

6.2 Automated Testing

Automated testing is integrated into the continuous verification workflow of the Health Hub platform to execute systematic validation suites, testing API endpoints, schema structures, and error resilience. This automation focuses on evaluating backend endpoint behavior within the FastAPI execution scope, schema safety constraints of the Firestore document layers, and latency patterns of external AI connection loops.

By executing script-driven integration assertions, automated testing verifies that user authentication states map reliably across routes, network timeout thresholds fail gracefully during traffic spikes, and analytical rendering components receive perfectly parsed payloads. This automated layer provides confidence that structural updates to the application code do not inadvertently disrupt established feature sets.

Table 6.4: Automated Testing

Test ID	Modules Involved	Test Case/Script	Attribute and Value	Expected Result	Actual Result	Status
AT01	Firebase Auth & FastAPI	Token Propagation & API Guard	Bearer Token -> UID Verification	Backend validates current signature and securely passes context parameters.	Session verification resolved successfully via middleware.	Pass

AT02	Clinical Input & Database	Document Injection & Safety	Input: Fasting Blood Glucose 140	Schema validator enforces numeric limits and writes document safely.	Document saved with absolute data integrity.	Pass
AT03	Local ML Model & Groq	Pipeline Flow & Payload Check	Risk Context: High Anemia Risk	ML model predictions structure the JSON context template dispatched to Groq.	System orchestration handled pipeline transitions correctly.	Pass
AT04	Account Action & Control	Profile State Transitions	Action Type: Deactivate Account	Account flag flips to inactive state; current active session token is invalidated.	Profile state change successfully committed and session killed.	Pass
AT05	Chat Service & Sync	Message Streaming Pipeline	Snapshot Listener Event Trigger	Data pipeline handles high-throughput messaging strings without dropping packets.	Event synchronization achieved zero-packet drop rates.	Pass
AT06	Chart Data & Recharts	Data Array Structural Check	Weight Trend Object Array	API returns chronological array sorting weights matching graph requirements.	Data array structure generated cleanly for visualization engine.	Pass
AT07	FastAPI & External Services	Gateway Timeout & Error Triggers	Gateway Timeout: Groq Cloud API	System traps the 504 error, falls back to locally cached data, or alerts user.	Exception caught cleanly with human-readable error response.	Pass

Chapter 7

CONCLUSION

7. Conclusion

The Health Hub system has been successfully designed and implemented as a comprehensive, Machine Learning-driven healthcare and nutritional management platform. Developed using a modern React frontend architecture seamlessly integrated with a cloud-based Firebase infrastructure, the platform provides advanced, personalized functionalities such as multi-track machine learning risk screening, rule-based clinical biomarker interpretation, dynamic BMI/TDEE tracking, and iterative, culturally tailored Pakistani recipe and meal plan generation via the Gemini API.

Through careful system design, a modular layered architecture, and rigorous testing including unit, functional, integration, and manual system evaluations the platform has proven to be reliable, clinically grounded, and technically robust. The orchestration of native machine learning models, such as the Stacking Ensemble and XGBoost classifiers, combined with real-time Firestore database synchronization, ensures high data integrity and minimal latency. Overall, the project successfully demonstrates how modern artificial intelligence, cloud-native backend services, and a human-in-the-loop approval workflow can be effectively leveraged to simplify complex medical data and provide actionable health insights while maintaining strict professional oversight.

7.1 Future Work

While the current version of Health Hub successfully covers core AI-driven nutritional and health screening functionalities, there are several opportunities for future enhancement to evolve the platform into a more holistic healthcare ecosystem:

7.1.1 Advanced Video Consultation

In future versions, the system can support real-time e-consultations through integrated WebRTC-based video calls directly within the application. This will allow patients to consult verified nutritionists remotely with a more personal touch, facilitating a face-to-face clinical review of the AI-generated insights and increasing accessibility for users across diverse geographical regions.

7.1.2 Multi-Language & Dialect AI Support

To reach a wider audience in Pakistan and globally, the AI and conversational text orchestration layer can be extended to support multiple languages, including Urdu and other regional dialects. This would allow users to input health markers and converse with NutriBot or receive diet plans in their native language, significantly reducing the barrier to digital healthcare literacy.

7.1.3 IoT and Wearable Ecosystem Integration

Health Hub could integrate directly with wearable devices and health APIs (such as Apple Health or Google Fit) to automatically record vital signs like heart rate, blood pressure, daily steps, and sleep patterns. This continuous data stream can be fed dynamically into the underlying machine learning models to provide even more precise, real-time diet and risk adjustments based on actual daily physical activity levels.

7.1.4 AI-Powered Computer Vision for Meal Logging

Future updates could include a computer vision module where users can take a photo of their meal. The system, using specialized image recognition models, would identify food items, estimate portion sizes, and automatically log the nutritional intake against their daily diet plan, replacing manual meal logging.

7.1.5 Pharmacy and Smart Grocery Integration

The system can be linked with e-commerce pharmacies and local grocery platforms. Once a personalized diet plan is verified and approved, the system could automatically compile a structured shopping list, allowing users to order the required healthy ingredients or nutritionist-recommended supplements with a single click.

7.1.6 Deep Learning for Predictive Health Analytics

By utilizing the longitudinal health datasets stored safely within the cloud database, future versions could implement deep sequential models to identify implicit physiological patterns over time. This would enable the system to alert users about potential long-term health anomalies such as early warning signs of chronic conditions well before they become critical, evolving Health Hub from a management tool into a proactive, preventive healthcare ecosystem.

8. References

- [1] "Pinch of Yum," [Online]. Available: <https://pinchofyum.com/>.
- [2] "Nutrition.gov," [Online]. Available: <https://www.nutrition.gov/>.
- [3] "Healthy Food Guid," [Online]. Available: <https://www.healthyfood.com/>.
- [4] [Online]. Available: <https://www.foodfusion.com/>.
- [5] "Healthy diet," [Online]. Available: <https://www.who.int/news-room/fact-sheets/detail/healthy-diet>.
- [6] "Food.," [Online]. Available: <https://www.food.com/>.
- [7] [Online]. Available:
https://www.nestlegoodnes.com/pk/?utm_source=google&utm_medium=cp&utm_campaign=22903247935&utm_term=cheap%20meals&utm_content=183955721916&gad_source=1&gad_campaignid=22903247935&gbraid=0AAAABAirTOkbdOqyxYv2c_u-WtZelD0Yg&gclid=CjwKCAjwobnGBhBNEiwAu2mpFNI.